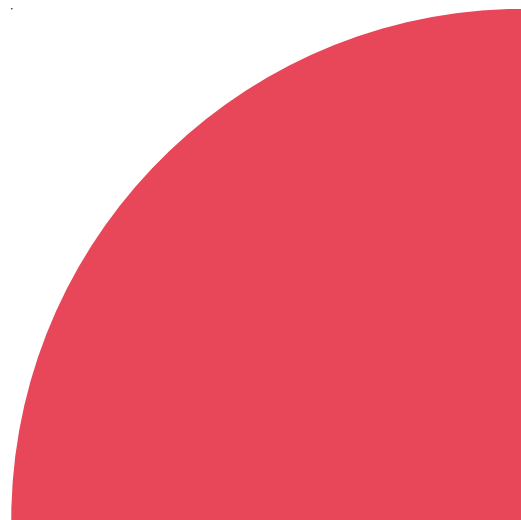
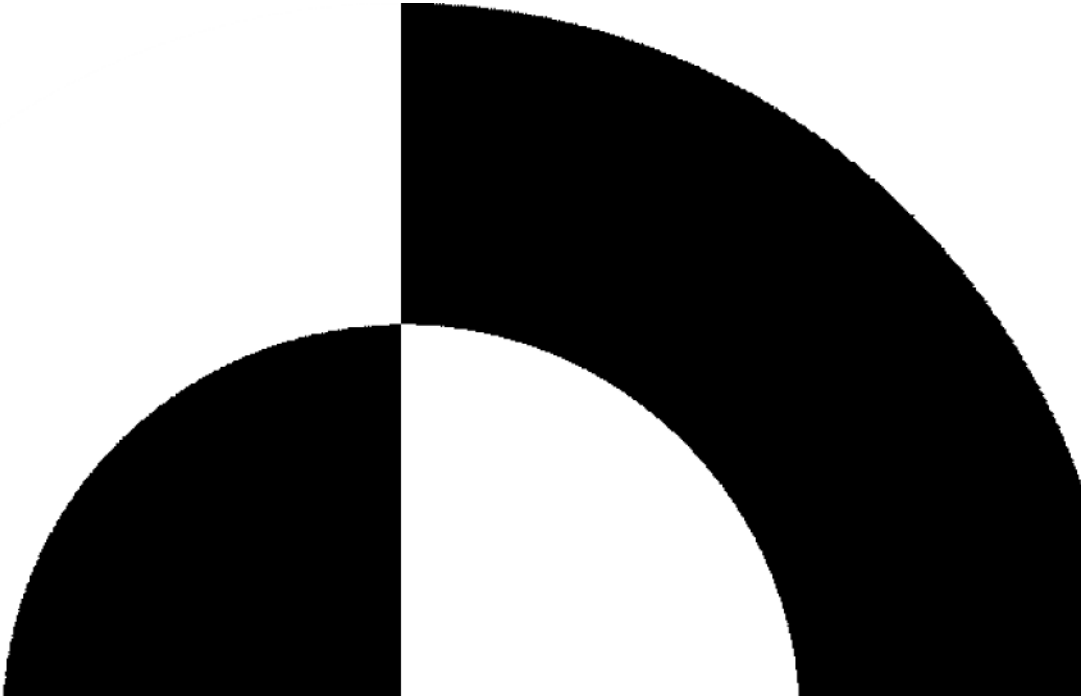


INORGANIC
PREVIOUS
YEARS
QUESTIONS

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[JEE MAIN]

- Q.1 The correct order of atomic radius is - [AIEEE 2002]
 (1) $Ce > Sm > Tb > Lu$ (2) $Lu > Tb > Sm > Ce$
 (3) $Tb > Lu > Sm > Ce$ (4) $Sm > Tb > Lu > Ce$
- Q.2 Ce^{3+} , La^{3+} , Pm^{3+} and Yb^{3+} have ionic radii in the increasing order as – [AIEEE 2002]
 (1) $La^{3+} < Ce^{3+} < Pm^{3+} < Yb^{3+}$ (2) $Yb^{3+} < Pm^{3+} < Ce^{3+} < La^{3+}$
 (3) $La^{3+} = Ce^{3+} < Pm^{3+} < Yb^{3+}$ (4) $Yb^{3+} < Pm^{3+} < La^{3+} < Ce^{3+}$
- Q.3 According to the Periodic Law of elements, the Variation in properties of elements is related to their ? [AIEEE 2003]
 (1) Nuclear masses (2) Atomic numbers
 (3) Nuclear neutron-proton number ratio (4) Atomic masses
- Q.4 The reduction in atomic size with increase in atomic number is a characteristic of elements of - [AIEEE 2003]
 (1) d-block (2) f-block (3) Radioactive series (4) High atomic masses
- Q.5 Which one of the following groups represent a collection of isoelectronic species ? [AIEEE 2003]
 (At. no. $Cs = 55$, $Br = 35$)
 (1) N^{3-} , F^- , Na^+ (2) Be , Al^{3+} , Cl^- (3) Ca^{2+} , Cs^+ , Br (4) Na^+ , Ca^{2+} , Mg^{2+}
- Q.6 The atomic numbers of vanadium (V), Chromium (Cr), manganese (Mn) and iron (Fe) respectively 23, 24, 25 and 26. Which one of these may be expected to have the higher second ionization enthalpy ? [AIEEE 2003]
 (1) Cr (2) Mn (3) Fe (4) V
- Q.7 Which one of the following sets of ions represents the collection of isoelectronic species ? [AIEEE 2004]
 (1) K^+ , Cl^- , Mg^{2+} , Sc^{3+} (2) Na^+ , Ca^{2+} , Sc^{3+} , F^-
 (3) K^+ , Ca^{2+} , Sc^{3+} , Cl^- (4) Na^+ , Mg^{2+} , Al^{3+} , Cl^-
- Q.8 Which one of the following ions has the highest value of ionic radius ? [AIEEE 2004]
 (1) O^{2-} (2) B^{3+} (3) Li^+ (4) F^-

- Q.9 Among Al_2O_3 , SiO_2 , P_2O_3 and SO_2 the correct order of acidic strength is : [AIEEE 2004]
 (1) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{SO}_2 < \text{P}_2\text{O}_3$ (2) $\text{SiO}_2 < \text{SO}_2 < \text{Al}_2\text{O}_3 < \text{P}_2\text{O}_3$
 (3) $\text{SO}_2 < \text{P}_2\text{O}_3 < \text{SiO}_2 < \text{Al}_2\text{O}_3$ (4) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_3 < \text{SO}_2$
- Q.10 The formation of the oxide ion $\text{O}_{(\text{g})}^{2-}$ requires first an exothermic and then an endothermic step as shown below. [AIEEE 2004]

$$\text{O}_{(\text{g})} + \text{e}^- = \text{O}_{(\text{g})}^- \quad \Delta H^\circ = -142 \text{ kJ mol}^{-1}$$

$$\text{O}_{(\text{g})}^- + \text{e}^- = \text{O}_{(\text{g})}^{2-} \quad \Delta H^\circ = 844 \text{ kJ mol}^{-1}$$
 This is because of :
 (1) O^- ion will tend to resist the addition of another electron
 (2) Oxygen has high electron affinity
 (3) Oxygen is more electronegative
 (4) O^- ion has comparatively larger size than oxygen atom
- Q.11 In which of the following arrangements the order is NOT according to the property indicated against it ? [AIEEE 2005]
 (1) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na} < \text{F}^-$ – increasing ionic size
 (2) $\text{B} < \text{C} < \text{N} < \text{O}$ – increasing first ionization enthalpy
 (3) $\text{I} < \text{Br} < \text{F} < \text{Cl}$ – increasing electron gain enthalpy (with negative sign)
 (4) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ – increasing metallic radius
- Q.12 Which of the following oxides is amphoteric in character ? [AIEEE 2005]
 (1) SnO_2 (2) SiO_2 (3) CO_2 (4) CaO
- Q.13 The lanthanide contraction is responsible for the fact that [AIEEE 2005]
 (1) Zr and Y have about the same radius (2) Zr and Nb have similar oxidation state
 (3) Zr and Hf have about the same radius (4) Zr and Zn have the same oxidation state
- Q.14 The increasing order of the first ionization enthalpies of the elements B, P, S and F (lowest first) is – [AIEEE 2006]
 (1) $\text{F} < \text{S} < \text{P} < \text{B}$ (2) $\text{P} < \text{S} < \text{B} < \text{F}$ (3) $\text{B} < \text{P} < \text{S} < \text{F}$ (4) $\text{B} < \text{S} < \text{P} < \text{F}$
- Q.15 Which one of the following sets of ions represents a collection of isoelectronic species ? [AIEEE 2006]
 (1) N^{3-} , O^{2-} , F^- , S^{2-} (2) Li^+ , Na^+ , Mg^{2+} , Ca^{2+}
 (3) K^+ , Cl^- , Ca^{2+} , Sc^{3+} (4) Ba^{2+} , Sr^{2+} , K^+ , Ca^{2+}
- Q.16 In which of the following arrangements, the sequence is not strictly according to the property written against it ? [AIEEE 2009]
 (1) $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$: Increasing acidic strength
 (2) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: Increasing acidic strength
 (3) $\text{NH}_3 > \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: Increasing basic strength
 (4) $\text{B} < \text{C} < \text{O} < \text{N}$: increasing first ionization enthalpy

- Q.17 The set representing the correct order of ionic radius is – [AIEEE 2009]
 (1) $\text{Li}^+ > \text{Be}^{2+} > \text{Na}^+ > \text{Mg}^{2+}$ (2) $\text{Na}^+ > \text{Li}^+ > \text{Mg}^{2+} > \text{Be}^{2+}$
 (3) $\text{Li}^{2+} > \text{Na}^+ > \text{Mg}^{2+} > \text{Be}^{2+}$ (4) $\text{Mg}^{2+} > \text{Be}^{2+} > \text{Li}^+ > \text{Na}^+$
- Q.18 The correct sequence which shows decreasing order of the ionic radii of the elements is [AIEEE 2010]
 (1) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+ > \text{F}^- > \text{O}^{2-}$ (2) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{O}^{2-} > \text{F}^-$
 (3) $\text{Na}^+ > \text{F}^- > \text{Mg}^{2+} > \text{O}^{2-} > \text{Al}^{3+}$ (4) $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
- Q.19 Which one of the following ordered presents the correct sequence of the increasing basic nature of the given oxides? [AIEEE 2011]
 (1) $\text{Al}_2\text{O}_3 < \text{MgO} < \text{Na}_2\text{O} < \text{K}_2\text{O}$ (2) $\text{MgO} < \text{K}_2\text{O} < \text{Al}_2\text{O}_3 < \text{Na}_2\text{O}$
 (3) $\text{Na}_2\text{O} < \text{K}_2\text{O} < \text{MgO} < \text{Al}_2\text{O}_3$ (4) $\text{K}_2\text{O} < \text{Na}_2\text{O} < \text{Al}_2\text{O}_3 < \text{MgO}$
- Q.20 The increasing order of the ionic radii of the given isoelectronic species is [AIEEE 2012]
 (1) $\text{Ca}^{2+}, \text{K}^+, \text{Cl}^-, \text{S}^{2-}$ (2) $\text{K}^+, \text{S}^{2-}, \text{Ca}^{2+}, \text{Cl}^-$
 (3) $\text{Cl}^-, \text{Ca}^{2+}, \text{K}^+, \text{S}^{2-}$ (4) $\text{S}^{2-}, \text{Cl}^-, \text{Ca}^{2+}, \text{K}^+$
- Q.21 The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be: [JEE (Main-Offline) 2013]
 (1) -5.1 eV (2) -10.2 eV (3) +2.55 eV (4) -2.55 eV
- Q.22 Which of the following represents the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar? [JEE (Main-Offline) 2013]
 (1) $\text{S} < \text{Se} < \text{Ca} < \text{Ba} < \text{Ar}$ (2) $\text{Ba} < \text{Ca} < \text{Se} < \text{S} < \text{Ar}$
 (3) $\text{Ca} < \text{Ba} < \text{S} < \text{Se} < \text{Ar}$ (4) $\text{Ca} < \text{S} < \text{Ba} < \text{Se} < \text{Ar}$
- Q.23 Electron gain enthalpy with negative sign of fluorine is less than that of chlorine due to: [JEE (Main-Online) 2013]
 (1) Bigger size of 2p orbital of fluorine (2) High ionization enthalpy of fluorine
 (3) Smaller size of chlorine atom (4) Smaller size of fluorine atom
- Q.24 Oxidation state of sulphur in anions SO_3^{2-} , $\text{S}_2\text{O}_4^{2-}$ and $\text{S}_2\text{O}_6^{2-}$ increases in the orders: [JEE (Main-Online) 2013]
 (1) $\text{S}_2\text{O}_6^{2-} < \text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-}$ (2) $\text{SO}_3^{2-} < \text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-}$
 (3) $\text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-} < \text{S}_2\text{O}_6^{2-}$ (4) $\text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-} < \text{SO}_3^{2-}$
- Q.25 The order of increasing sizes of atomic radii among the elements O, S, Se and As is: [JEE (Main-Online) 2013]
 (1) $\text{Se} < \text{S} < \text{As} < \text{O}$ (2) $\text{As} < \text{S} < \text{O} < \text{Se}$
 (3) $\text{O} < \text{S} < \text{Se} < \text{As}$ (4) $\text{O} < \text{S} < \text{As} < \text{Se}$

Q.26 Given

Reaction	Energy Change (in kJ)
$\text{Li(s)} \rightarrow \text{Li(g)}$	161
$\text{Li(g)} \rightarrow \text{Li}^+(\text{g})$	520
$\frac{1}{2}\text{F}_2(\text{g}) \rightarrow \text{F(g)}$	77
$\text{F(g)} + \text{e}^- \rightarrow \text{F}^-(\text{g})$	(Electron gain enthalpy)
$\text{Li}^+(\text{g}) + \text{F}^-(\text{g}) \rightarrow \text{LiF(s)}$	-1047
$\text{Li(s)} + \frac{1}{2}\text{F}_2(\text{g}) \rightarrow \text{LiF(s)}$	-617

Based on data provided, the value of electron gain enthalpy of fluorine would be :

- (1) -300 kJ mol^{-1} (2) -228 kJ mol^{-1} (3) -328 kJ mol^{-1} (4) -350 kJ mol^{-1}

[JEE (Main-Online) 2013]

Q.27 Which is the correct order of second ionization potential of C, N, O and F in the following ?

- (1) $\text{C} > \text{N} > \text{O} > \text{F}$ (2) $\text{F} > \text{O} > \text{N} > \text{C}$ (3) $\text{O} > \text{N} > \text{F} > \text{C}$ (4) $\text{O} > \text{F} > \text{N} > \text{C}$

[JEE (Main-Online) 2013]

Q.28 Which of the following arrangements represents the increasing order (smallest to largest) of ionic radii of the given species O^{2-} , S^{2-} , N^{3-} , P^{3-} ?

- (1) $\text{N}^{3-} < \text{O}^{2-} < \text{P}^{3-} < \text{S}^{2-}$ (2) $\text{O}^{2-} < \text{P}^{3-} < \text{N}^{3-} < \text{S}^{2-}$
 (3) $\text{N}^{3-} < \text{S}^{2-} < \text{O}^{2-} < \text{P}^{3-}$ (4) $\text{O}^{2-} < \text{N}^{3-} < \text{S}^{2-} < \text{P}^{3-}$

[JEE (Main-Online) 2014]

Q.29 Similarity in chemical properties of the atoms of elements in a group of the Periodic table is most closely related to :

- (1) number of valence electrons (2) number of principal energy levels
 (3) atomic masses (4) atomic numbers

[JEE (Main-Online) 2014]

Q.30 Which one of the following has largest ionic radius ?

- (1) Li^+ (2) B^{3+} (3) O^{2-} (4) F^-

[JEE (Main-Online) 2014]

Q.31 The ionic radii (in Å) of N^{3-} , O^{2-} and F^- are respectively ?

- (1) 1.71, 1.40 and 1.36 (2) 1.71, 1.36 and 1.40
 (3) 1.36, 1.40 and 1.71 (4) 1.36, 1.71 and 1.40

[JEE (Main-Offline) 2015]

Q.32 If the principal quantum number $n = 6$, the correct sequence of filling of electrons will be :

- (1) $ns \rightarrow (n-2)f \rightarrow (n-1)d \rightarrow np$ (2) $ns \rightarrow (n-1)d \rightarrow (n-2)f \rightarrow np$
 (3) $ns \rightarrow (n-2)f \rightarrow np \rightarrow (n-1)d$ (4) $ns \rightarrow np \rightarrow (n-1)d \rightarrow (n-2)f$

[JEE (Main-Online) 2015]

Q.33 In the long form of the periodic table, the valence shell electronic configuration of $5s^25p^4$ corresponds to the element present in :

- (1) Group 16 and period 5 (2) Group 16 and period 6
 (3) Group 17 and period 6 (4) Group 17 and period 5

[JEE (Main-Online) 2015]

Q.34 Which of the following atoms has the highest first ionization energy ? [JEE (Main-Offline) 2016]
 (1) Sc (2) Rb (3) Na (4) K

Q.35 The following statements concern elements in the periodic table. Which of the following is true ?
 (1) All the elements in Group 17 are gases.
 (2) The Group 13 elements are all metals.
 (3) Elements of Group 16 have lower ionization enthalpy values compared to those of Group 15 in the corresponding periods.
 (4) For Group 15 elements, the stability of +5 oxidation state increases down the group.
 [JEE (Main-Online) 2016]

Q.36 The group having isoelectronic species is : [JEE (Main-Offline) 2017]
 (1) O^- , F^- , Na, Mg^+ (2) O^{2-} , F^- , Na, Mg^{2+}
 (3) O^- , F^- , Na^+ , Mg^{2+} (4) O^{2-} , F^- , Na^+ , Mg^{2+}

Q.37 Consider the following ionization enthalpies of two elements 'A' and 'B'.

Element	Ionisation enthalpy (kJ/mol)		
	1 st	2 nd	3 rd
A	899	1757	14847
B	737	1450	7731

Which of the following statements is correct ?

- (1) Both 'A' and 'B' belong to group-1 where 'B' comes below 'A'.
 (2) Both 'A' and 'B' belong to group-1 where 'A' comes below 'B'.
 (3) Both 'A' and 'B' belong to group-2 where 'B' comes below 'A'.
 (4) Both 'A' and 'B' belong to group-2 where 'A' comes below 'B'.

[JEE (Main-Online) 2017]

Q.38 The electronic configuration with the highest ionization enthalpy is :
 (1) [Ne] $3s^2 3p^1$ (2) [Ne] $3s^2 3p^2$ (3) [Ne] $3s^2 3p^3$ (4) [Ar] $3d^{10} 4s^2 4p^3$
 [JEE (Main-Online) 2017]

Q.39 The **correct** order of electron affinity is :
 (1) $Cl > F > O$ (2) $O > F > Cl$ (3) $F > Cl > O$ (4) $F > O > Cl$
 [JEE (Main-Online) 2018]

Q.40 For Na^+ , Mg^{2+} , F^- and O^{2-} ; the **correct** order of increasing ionic radii is :
 (1) $Mg^{2+} < O^{2-} < Na^+ < F^-$ (2) $Mg^{2+} < Na^+ < F^- < O^{2-}$
 (3) $Na^+ < Mg^{2+} < F^- < O^{2-}$ (4) $O^{2-} < F^- < Na^+ < Mg^{2+}$
 [JEE (Main-Online) 2018]

Q.41 In general, the properties that decrease and increase down a group in the periodic table, respectively are
 (1) Electron gain enthalpy and electronegativity
 (2) Electronegativity and electron gain enthalpy
 (3) Atomic radius and electronegativity
 (4) Electronegativity and atomic radius

[JEE (Main-Online) 2019]

- Q.42 When the first electron gain enthalpy ($\Delta_{\text{eg}}H$) of oxygen is -141 kJ/mol , its second electron gain enthalpy is :
(1) a more negative value than the first (2) negative, but less negative than the first
(3) a positive value (4) almost the same as that of the first.
[JEE (Main-Online) 2019]
- Q.43 The electronegativity of aluminium is similar to :
(1) Beryllium (2) Boron (3) Lithium (4) Carbon
[JEE (Main-Online) 2019]
- Q.44 The 71st electron of an element X with an atomic number of 71 enters into the orbital :
(1) 6s (2) 6p (3) 5d (4) 4f
[JEE (Main-Online) 2019]
- Q.45 The amphoteric hydroxide is
(1) $\text{Be}(\text{OH})_2$ (2) $\text{Ca}(\text{OH})_2$ (3) $\text{Mg}(\text{OH})_2$ (4) $\text{Sr}(\text{OH})_2$
[JEE (Main-Online) 2019]
- Q.46 The correct order of the atomic radii of C, Cs, Al and S is
(1) $\text{C} < \text{S} < \text{Al} < \text{Cs}$ (2) $\text{S} < \text{C} < \text{Cs} < \text{Al}$ (3) $\text{S} < \text{C} < \text{Al} < \text{Cs}$ (4) $\text{C} < \text{S} < \text{Cs} < \text{Al}$
[JEE (Main-Online) 2019]
- Q.47 The correct option with respect to the Pauling electronegativity values of the elements is:
(1) $\text{Te} > \text{Se}$ (2) $\text{Ga} < \text{Ge}$ (3) $\text{Si} < \text{Al}$ (4) $\text{P} > \text{S}$
[JEE (Main-Online) 2019]
- Q.48 The relative stability of +1 oxidation state of group 13 elements follows the order:
(1) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$ (2) $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$ (3) $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$ (4) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
[JEE (Main-Online) 2019]
- Q.49 The element with $Z = 120$ (not yet discovered) will be an / a :
(1) Inner-transition metal (2) Transition metal
(3) Alkaline earth metal (4) Alkali metal
[JEE (Main-Online) 2019]
- Q.50 The correct order of atomic radii is:
(1) $\text{Ho} > \text{N} > \text{Eu} > \text{Ce}$ (2) $\text{Ce} > \text{Eu} > \text{Ho} > \text{N}$
(3) $\text{Eu} > \text{Ce} > \text{Ho} > \text{N}$ (4) $\text{N} > \text{Ce} > \text{Eu} > \text{Ho}$
[JEE (Main-Online) 2019]
- Q.51 In general, the property (magnitudes only) that shows an opposite trend in comparison to other properties across a period is
(1) Electronegativity (2) Electron gain enthalpy
(3) Ionization enthalpy (4) Atomic radius
[JEE (Main-Online) 2020]

- Q.52 Three elements X, Y and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is:
 (1) $Z < Y < X$ (2) $X < Z < Y$ (3) $X < Y < Z$ (4) $Y < X < Z$
[JEE (Main-Online) 2020]
- Q.53 The atomic number of the element unnilennium is :
 (1) 119 (2) 108 (3) 102 (4) 109
[JEE (Main-Online) 2020]
- Q.54 The five successive ionization enthalpies of an elements are 800, 2427, 3658, 25024 and 32824 kJ mol⁻¹. The number of valence electrons in the element is :
 (1) 2 (2) 3 (3) 4 (4) 5
[JEE (Main-Online) 2020]
- Q.55 The elements with atomic numbers 101 and 104 belong to, respectively:
 (1) Group 11 and Group 4 (2) Actinoids and Group 4
 (3) Actinoids and Group 6 (4) Group 6 and Actinoids
[JEE (Main-Online) 2020]
- Q.56 The ionic radii of O²⁻, F⁻, Na⁺ and Mg²⁺ are in the order :
 (1) $F^- > O^{2-} > Na^+ > Mg^{2+}$ (2) $Mg^{2+} > Na^+ > F^- > O^{2-}$
 (3) $O^{2-} > F^- > Mg^{2+} > Na^+$ (4) $O^{2-} > F^- > Na^+ > Mg^{2+}$
[JEE (Main-Online) 2020]
- Q.57 The process that is NOT endothermic in nature is
 (1) $Ar_{(g)} + e^- \rightarrow Ar_{(g)}^-$ (2) $H_{(g)} + e^- \rightarrow H_{(g)}^-$
 (3) $Na_{(g)} \rightarrow Na_{(g)}^+ + e^-$ (4) $O_{(g)}^- + e^- \rightarrow O_{(g)}^{2-}$
[JEE (Main-Online) 2020]
- Q.58 In the sixth period, the orbitals that are filled are
 (1) 6s, 5f, 6d, 6p (2) 6s, 6p, 6d, 6f (3) 6s, 5d, 5f, 6p (4) 6s, 4f, 5d, 6p
[JEE (Main-Online) 2020]
- Q.59 The correct order of the ionic radii of O²⁻, N³⁻, F⁻, Mg²⁺, Na⁺ and Al³⁺ is :
 (1) $Al^{3+} < Na^+ < Mg^{2+} < O^{2-} < F^- < N^{3-}$ (2) $N^{3-} < O^{2-} < F^- < Na^+ < Mg^{2+} < Al^{3+}$
 (3) $Al^{3+} < Mg^{2+} < Na^+ < F^- < O^{2-} < N^{3-}$ (4) $N^{3-} < F^- < O^{2-} < Mg^{2+} < Na^+ < Al^{3+}$
[JEE (Main-Online) 2020]
- Q.60 The atomic number of Unnilunium is _____.
[JEE (Main-Online) 2020]

- Q.61 The electron gain enthalpy (in kJ/mol) of fluorine, chlorine, bromine and iodine, respectively, are
(1) $-296, -325, -333$ and -349 (2) $-333, -349, -325$ and -296
(3) $-349, -333, -325$ and -296 (4) $-333, -325, -349$ and -296
[JEE (Main-Online) 2020]
- Q.62 Within each pair of element F & Cl, S & Se, and Li & Na, respectively, the elements that release more energy upon an electron gain are
(1) Cl, Se and Na (2) Cl, S and Li (3) F, S and Li (4) F, Se and Na
[JEE (Main-Online) 2020]
- Q.63 The increasing order of the atomic radii of the following elements is
(a) C (b) O (c) F
(d) Cl (e) Br
(1) $(a) < (b) < (c) < (d) < (e)$ (2) $(c) < (b) < (a) < (d) < (e)$
(3) $(d) < (c) < (b) < (a) < (e)$ (4) $(b) < (c) < (d) < (a) < (e)$
[JEE (Main-Online) 2020]
- Q.64 B has a smaller first ionization enthalpy than Be. Consider the following statements:
(I) it is easier to remove 2p electron than 2s electron.
(II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be.
(III) 2s electron has more penetration power than 2p electron
(IV) atomic radius of B is more than Be
(atomic number B = 5, Be = 4)
The correct statements are
(1) (I), (II) and (IV) (2) (II), (III) and (IV) (3) (I), (II) and (III) (4) (I), (III) and (IV)
[JEE (Main-Online) 2020]
- Q.65 The acidic basic and amphoteric oxides, respectively, are
(1) Cl_2O , CaO , P_4O_{10} (2) Na_2O , SO_3 , Al_2O_3
(3) N_2O_3 , Li_2O , Al_2O_3 (4) MgO , Cl_2O , Al_2O_3
[JEE (Main-Online) 2020]
- Q.66 The oxidation states of transition metal atoms in $\text{K}_2\text{Cr}_2\text{O}_7$, KMnO_4 and K_2FeO_4 , respectively, are x, y and z. The sum of x, y and z is _____.
[JEE (Main-Online) 2020]

[JEE ADVANCED]

- Q.1 Moving from right to left in a periodic table, the atomic size is: **[JEE 1995]**
(A) increased (B) decreased (C) remains constant (D) none of these
- Q.2 The increasing order of electronegativity in the following elements: **[JEE 1995]**
(A) C, N, Si, P (B) N, Si, C, P (C) Si, P, C, N (D) P, Si, N, C
- Q.3 One element has atomic weight 39. Its electronic configuration is $1s^2, 2s^2 2p^6, 3s^2 3p^6 4s^1$. The true statement for that element is: **[JEE 1995]**
(A) Highest value of IE (B) Transition element
(C) Isotone with ${}_{18}\text{Ar}^{38}$ (D) None
- Q.4 The number of paired electrons in oxygen atom is: **[JEE 1995]**
(A) 6 (B) 16 (C) 8 (D) 32
- Q.5 The decreasing size of K^+ , Ca^{2+} , Cl^- & S^{2-} follows the order: **[REE 1995]**
(A) $\text{K}^+ > \text{Ca}^{2+} > \text{S}^{2-} > \text{Cl}^-$ (B) $\text{K}^+ > \text{Ca}^{2+} > \text{Cl}^- > \text{S}^{2-}$
(C) $\text{Ca}^{2+} > \text{K}^+ > \text{Cl}^- > \text{S}^{2-}$ (D) $\text{S}^{2-} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{2+}$
- Q.6 Which of the following oxide is neutral? **[JEE 1996]**
(A) CO (B) SnO_2 (C) ZnO (D) SiO_2
- Q.7 Which of the following has the maximum number of unpaired electrons **[JEE 1996]**
(A) Mg^{2+} (B) Ti^{3+} (C) V^{3+} (D) Fe^{2+}
- Q.8 The following acids have been arranged in the order of decreasing acid strength. Identify the correct order **[JEE 1996]**
 ClOH(I) BrOH(II) IOH(III)
(A) $\text{I} > \text{II} > \text{III}$ (B) $\text{II} > \text{I} > \text{III}$ (C) $\text{III} > \text{II} > \text{I}$ (D) $\text{I} > \text{III} > \text{II}$
- Q.9 The incorrect statement among the following is: **[JEE 1997]**
(A) the first ionisation potential of Al is less than the first ionisation potential of Mg
(B) the second ionisation potential of Mg is greater than the second ionisation potential of Na
(C) the first ionisation potential of Na is less than the first ionisation potential of Mg
(D) the third ionisation potential of Mg is greater than the third ionisation potential of Al
- Q.10 Which of the following are amphoteric? **[REE 1997]**
(A) Be(OH)_2 (B) Sr(OH)_2 (C) Ca(OH)_2 (D) Al(OH)_3
- Q.11 Li^+ , Mg^{2+} , K^+ , Al^{3+} (Arrange in increasing order of radii) **[JEE 1997]**

- Q.12 Which one of the following statement (s) is (are) correct? [JEE 1998]
 (A) The electronic configuration of Cr is $[\text{Ar}] 3d^5 4s^1$. (Atomic No. of Cr = 24)
 (B) The magnetic quantum number may have a negative value
 (C) In silver atom, 23 electrons have a spin of one type and 24 of the opposite type. (Atomic No. of Ag = 47)
 (D) The oxidation state of nitrogen in HN_3 is -3 .
- Q.13 **Statement-1:** F atom has a less negative electron gain enthalpy than Cl atom. [JEE 2000]
Statement-2: Additional electron is repelled more efficiently by 3p electron in Cl atom than by 2p electron in F atom.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- Q.14 **Statement-1:** $\text{Al}(\text{OH})_3$ is amphoteric in nature. [JEE 2000]
Statement-2: $\text{Al}-\text{O}$ and $\text{O}-\text{H}$ bonds can be broken with equal ease in $\text{Al}(\text{OH})_3$.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- Q.15 The correct order of radii is: [JEE 2000]
 (A) $\text{N} < \text{Be} < \text{B}$ (B) $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$ (C) $\text{Na} < \text{Li} < \text{K}$ (D) $\text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}^{4+}$
- Q.16 The correct order of acidic strength is: [JEE 2000]
 (A) $\text{Cl}_2\text{O}_7 > \text{SO}_3 > \text{P}_4\text{O}_{10}$ (B) $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$
 (C) $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$ (D) $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$
- Q.17 The IE_1 of Be is greater than that of B. [T/F] [JEE 2001]
- Q.18 The set representing correct order of IP_1 is [JEE 2001]
 (A) $\text{K} > \text{Na} > \text{Li}$ (B) $\text{Be} > \text{Mg} > \text{Ca}$ (C) $\text{B} > \text{C} > \text{N}$ (D) $\text{Fe} > \text{Si} > \text{C}$
- Q.19 Identify the least stable ion amongst the following: [JEE 2002]
 (A) Li^- (B) Be^- (C) B^- (D) C^-
- Q.20 Identify the correct order of acidic strengths of CO_2 , CuO , CaO , H_2O : [JEE 2002]
 (A) $\text{CaO} < \text{CuO} < \text{H}_2\text{O} < \text{CO}_2$ (B) $\text{H}_2\text{O} < \text{CuO} < \text{CaO} < \text{CO}_2$
 (C) $\text{CaO} < \text{H}_2\text{O} < \text{CuO} < \text{CO}_2$ (D) $\text{H}_2\text{O} < \text{CO}_2 < \text{CaO} < \text{CuO}$
- Q.21 Among the following, the number of elements showing only one non-zero oxidation state is [JEE 2010]
 O, Cl, F, N, P, Sn, Tl, Na, Ti

- Q.22 The increasing order of atomic radii of the following Group 13 elements is [JEE (Advanced) 2016]
(A) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$ (B) $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$
(C) $\text{Al} < \text{In} < \text{Ga} < \text{Tl}$ (D) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$
- Q.23 The option(s) with only amphoteric oxide is(are) [JEE (Advanced) 2017]
(A) Cr_2O_3 , BeO , SnO , SnO_2 (B) Cr_2O_3 , CrO , SnO , PbO
(C) ZnO , Al_2O_3 , PbO , PbO_2 (D) NO , B_2O_3 , PbO , SnO_2

pmssir

[JEE MAIN]

- Q.1 In which of the following species is the underlined carbon having sp^3 -hybridisation?
 (1) $CH_3-\underline{C}OOH$ (2) $CH_3-\underline{C}H_2OH$ (3) $CH_3-\underline{C}OCH_3$ (4) $CH_2=\underline{C}H-CH_3$ [AIEEE 2002]
- Q.2 Which of the following statements is true?
 (1) HF is less polar than HBr [AIEEE 2002]
 (2) Water does not contain any ions
 (3) Chemical bond formation takes place when forces of attraction overcome the forces of repulsion
 (4) In covalent cond, transfer of electrons takes place
- Q.3 A square planar complex is formed by hybridisation of which atomic orbital? [AIEEE 2002]
 (1) s, p_x, p_y, d_{yz} (2) $s, p_x, p_y, d_{x^2-y^2}$ (3) s, p_x, p_y, d_{z^2} (4) s, p_x, p_y, d_{xy}
- Q.4 Which one of the following pairs of molecules will have permanent dipole moments for both members:
 (1) NO_2 and CO_2 (2) NO_2 and O_3 [AIEEE 2003]
 (3) SiF_4 and CO_2 (4) SiF_4 and NO_2
- Q.5 The pair of species having identical shapes for molecules of both species is : [AIEEE 2003]
 (1) XeF_2, CO_2 (2) BF_3, PCl_3 (3) PF_5, IF_5 (4) CF_4, SF_4
- Q.6 The correct order of bond angles (smallest first) in H_2S, NH_3, BF_3 and SiH_4 is : [AIEEE 2004]
 (1) $H_2S < NH_3 < SiH_4 < BF_3$ (2) $NH_3 < H_2S < SiH_4 < BF_3$
 (3) $H_2S < SiH_4 < NH_3 < BF_3$ (4) $H_2S < NH_3 < BF_3 < SiH_4$
- Q.7 The maximum number of 90° angles between bond pair-bond pair of electrons is observed in : [AIEEE 2004]
 (1) dsp^2 (2) sp^3d hybridization
 (3) dsp^3 hybridization (4) sp^3d^2 hybridization
- Q.8 The molecular shapes of SF_4, CF_4 and XeF_4 are: [AIEEE 2005]
 (1) the same with 2, 0 and 1 lone pair of electrons on the central atom, respectively
 (2) the same with 1, 1 and 1 lone pair of electrons on the central atoms, respectively
 (3) different with 0, 1 and 2 lone pair of electrons on the central atoms, respectively
 (4) different with 1, 0 and 2 lone pair of electrons on the central atoms, respectively
- Q.9 Of the following sets which one does not contain isoelectronic species? [AIEEE 2005]
 (1) $PO_4^{3-}, SO_4^{2-}, ClO_4^-$ (2) CN^-, N_2, C_2^{2-}
 (3) $SO_3^{2-}, CO_3^{2-}, NO_3^-$ (4) $BO_3^{3-}, CO_3^{2-}, NO_3^-$

- Q.10 In which of the following molecules/ions are all the bonds not equal? [AIEEE 2006]
 (1) XeF_4 (2) BF_4^- (3) SF_4 (4) SiF_4
- Q.11 The decreasing value of bond angles from NH_3 (106°) to SbH_3 (101°) down group-15 of the periodic table is due to: [AIEEE 2006]
 (1) decreasing ℓp – bp repulsion (2) increasing electronegativity
 (3) increasing bp – bp repulsion (4) increasing p-orbital character in sp^3
- Q.12 The hybridisation of orbitals of N atom in NO_3^- , NO_2^+ and NH_4^+ are respectively [AIEEE 2011]
 (1) sp , sp^2 , sp^3 (2) sp^2 , sp , sp^3 (3) sp , sp^3 , sp^2 (4) sp^2 , sp^3 , sp
- Q.13 The structure of IF_7 is [AIEEE 2011]
 (1) square pyramid (2) trigonal bipyramid
 (3) octahedral (4) pentagonal bipyramid
- Q.14 The molecule having smallest bond angle is [AIEEE 2012]
 (1) SbCl_3 (2) PCl_3 (3) NCl_3 (4) AsCl_3
- Q.15 In which of the following pairs the two species are not isostructural? [AIEEE 2012]
 (1) PF_5 and BrF_5 (2) AlF_6^{3-} and SF_6 (3) CO_3^{2-} and NO_3^- (4) PCl_4^+ and SiCl_4
- Q.16 In which of the following ionization processes the bond energy has increased and also the magnetic behaviour has changed from paramagnetic to diamagnetic? [JEE (Main-Online) 2013]
 (1) $\text{N}_2 \longrightarrow \text{N}_2^+$ (2) $\text{O}_2 \longrightarrow \text{O}_2^+$ (3) $\text{C}_2 \longrightarrow \text{C}_2^+$ (4) $\text{NO} \longrightarrow \text{NO}^+$
- Q.17 Which one of the following molecules is polar? [JEE (Main-Online) 2013]
 (1) CF_4 (2) IF_5 (3) XeF_4 (4) SbF_5
- Q.18 Trigonal bipyramid geometry is shown by: [JEE (Main-Online) 2013]
 (1) $[\text{XeF}_8]^{2-}$ (2) XeOF_2 (3) XeO_3F_2 (4) FXeOSO_2F
- Q.19 XeO_4 molecule is tetrahedral having: [JEE (Main-Online) 2013]
 (1) Four $\text{p}\pi$ – $\text{d}\pi$ bonds (2) One $\text{p}\pi$ – $\text{d}\pi$ bond
 (3) Three $\text{p}\pi$ – $\text{d}\pi$ bonds (4) Two $\text{p}\pi$ – $\text{d}\pi$ bonds
- Q.20 The shape of IF_6^- is: [JEE (Main-Online) 2013]
 (1) Pyramidal (2) Octahedral
 (3) Trigonal distorted octahedron (4) Square antiprism
- Q.21 Which has trigonal bipyramidal shape? [JEE (Main-Online) 2013]
 (1) XeOF_2 (2) XeO_3F_2 (3) XeO_3 (4) XeOF_4

Q.22 In which of the following sets, all the given species are isostructural?

- (1) BF_3 , NF_3 , PF_3 , AlF_3 (2) BF_4^- , CCl_4 , NH_4^+ , PCl_4^+
 (3) CO_2 , NO_2 , ClO_2 , SiO_2 (4) PCl_3 , AlCl_3 , BCl_3 , SbCl_3

[JEE (Main-Online) 2013]

Q.23 Which one of the following arrangement represents the correct order of the proton affinity of the given species:

- (1) $\text{NH}_2^- < \text{HS}^- < \text{I}^- < \text{F}^-$ (2) $\text{I}^- < \text{F}^- < \text{HS}^- < \text{NH}_2^-$
 (3) $\text{HS}^- < \text{NH}_2^- < \text{F}^- < \text{I}^-$ (4) $\text{F}^- < \text{I}^- < \text{NH}_2^- < \text{HS}^-$

[JEE (Main-Online) 2013]

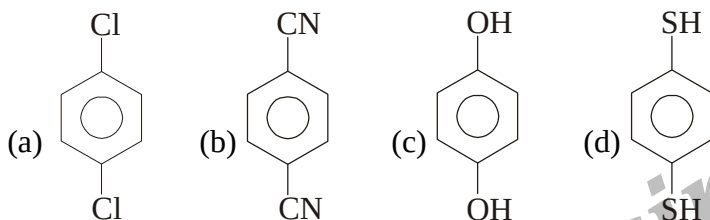
Q.24 The correct statement for the molecule, CsI_3 is :

[JEE (Main-Offline) 2014]

- (1) it contains Cs^+ and I_3^- ions. (2) it contains Cs^{3+} and I^- ions.
 (3) it contains Cs^+ , I^- and lattice I_2 molecule (4) it is a covalent molecule

Q.25 For which of the following molecule significant $\mu \neq 0$?

[JEE (Main-Offline) 2014]



- (1) (a) and (b) (2) only (c) (3) (c) and (d) (4) only (a)

Q.26 For the compounds

CH_3Cl , CH_3Br , CH_3I and CH_3F , the correct order of increasing C-halogen bond length is :

- (1) $\text{CH}_3\text{F} < \text{CH}_3\text{Cl} < \text{CH}_3\text{Br} < \text{CH}_3\text{I}$ (2) $\text{CH}_3\text{F} < \text{CH}_3\text{I} < \text{CH}_3\text{Br} < \text{CH}_3\text{Cl}$
 (3) $\text{CH}_3\text{Cl} < \text{CH}_3\text{Br} < \text{CH}_3\text{F} < \text{CH}_3\text{I}$ (4) $\text{CH}_3\text{F} < \text{CH}_3\text{Br} < \text{CH}_3\text{Cl} < \text{CH}_3\text{I}$

Q.27 In allene (C_3H_4), the type(s) of hybridization of the carbon atoms is (are):

- (1) sp^2 and sp^3 (2) only sp^2 (3) sp^2 and sp (4) sp and sp^3

[JEE (Main-Online) 2014]

Q.28 Shapes of certain interhalogen compounds are stated below. Which one of them is not correctly stated?

- (1) ICl_3 : planar dimeric (2) IF_7 : pentagonal bipyramid
 (3) BrF_3 Planar T-shaped (4) BrF_5 : trigonal bipyramid

[JEE (Main-Online) 2014]

Q.29 Permanent hardness in water cannot be cured by :

- (1) Calgon's method (2) Ion exchange method
 (3) Treatment with washing soda (4) Boiling

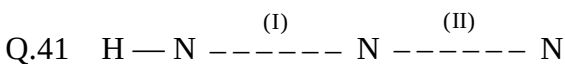
[JEE (Main-Online) 2015]

Q.30 The geometry of XeOF_4 by VSEPR theory is :

- (1) square pyramidal (2) octahedral (3) pentagonal planar (4) trigonal bipyramidal

[JEE (Main-Online) 2015]

- Q.31 Choose the incorrect formula out of the four compounds for an element X below :
 (1) $X_2(SO_4)_3$ (2) X_2O_3 (3) X_2Cl_3 (4) XPO_4
[JEE (Main-Online) 2015]
- Q.32 The group of molecules having identical shape is:
 (1) SF_4 , XeF_4 , CCl_4 (2) ClF_3 , $XeOF_2$, XeF_3^+
 (3) BF_3 , PCl_3 , XeO_3 (4) PCl_5 , IF_5 , XeO_2F_2
[JEE (Main-Online) 2016]
- Q.33 The bond angle H-X-H is the greatest in the compound :
 (1) CH_4 (2) NH_3 (3) H_2O (4) PH_3
[JEE (Main-Online) 2016]
- Q.34 The bond angle H-X-H is the greatest in the compound :
 (1) CH_4 (2) NH_3 (3) H_2O (4) PH_3
[JEE (Main-Online) 2016]
- Q.35 sp^3d^2 hybridization is not displayed by :
 (1) BrF_5 (2) SF_6 (3) $[CrF_6]^{3-}$ (4) PF_5
[JEE (Main-Online) 2017]
- Q.36 The group having triangular planar structures is :
 (1) BF_3 , NF_3 , CO_3^{2-} (2) CO_3^{2-} , NO_3^- , SO_3
 (3) NH_3 , SO_3 , CO_3^{2-} (4) NCl_3 , BCl_3 , SO_3
[JEE (Main-Online) 2017]
- Q.37 Total number of lone pair of electrons in I_3^- ion is :
 (1) 9 (2) 12 (3) 3 (4) 6
[JEE (Main-Offline) 2018]
- Q.38 Which of the following compounds contain(s) no covalent bond(s) ?
 KCl , PH_3 , O_2 , B_2H_6 , H_2SO_4
 (1) KCl (2) KCl , B_2H_6 (3) KCl , B_2H_6 , PH_3 (4) KCl , H_2SO_4
[JEE (Main-Offline) 2018]
- Q.39 In XeO_3F_2 , the number of bond pair(s), π -bond(s) and lone pair(s) on Xe atom respectively are:
 (1) 5, 2, 0 (2) 4, 2, 2 (3) 4, 4, 0 (4) 5, 3, 0
[JEE (Main-Online) 2018]
- Q.40 Which of the following is a Lewis acid?
 (1) NF_3 (2) PH_3 (3) NaH (4) $B(CH_3)_3$
[JEE (Main-Online) 2018]



In hydrogen azide (above) the bond orders of bonds (I) and (II) are

- | | (I) | (II) |
|-----|-----|------|
| (1) | > 2 | > 2 |
| (2) | < 2 | > 2 |
| (3) | < 2 | < 2 |
| (4) | > 2 | < 2 |

[JEE (Main-Online) 2018]

Q.42 The decreasing order of bond angles in BF_3 , NH_3 , PF_3 and I_3^- is :

- | | |
|--|--|
| (1) $\text{BF}_3 > \text{NH}_3 > \text{PF}_3 > \text{I}_3^-$ | (2) $\text{BF}_3 > \text{I}_3^- > \text{PF}_3 > \text{NH}_3$ |
| (3) $\text{I}_3^- > \text{BF}_3 > \text{NH}_3 > \text{PF}_3$ | (4) $\text{I}_3^- > \text{NH}_3 > \text{PF}_3 > \text{BF}_3$ |

[JEE (Main-Online) 2018]

Q.43 Identify the pair in which the geometry of the species is a T-shape and square pyramidal, respectively :

- | | |
|---|---|
| (1) IO_3^- and IO_2F_2^- | (2) ClF_3 and IO_4^- |
| (3) XeOF_2 and XeOF_4 | (4) ICl_2^- and ICl_5 |

[JEE (Main-Online) 2018]

Q.44 Which of the following conversions involves change in both shape and hybridisation ?

- | | | | |
|---|--|---|---|
| (1) $\text{NH}_3 \rightarrow \text{NH}_4^+$ | (2) $\text{CH}_4 \rightarrow \text{C}_2\text{H}_6$ | (3) $\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$ | (4) $\text{BF}_3 \rightarrow \text{BF}_4^-$ |
|---|--|---|---|

[JEE (Main-Online) 2018]

Q.45 The **incorrect** geometry is represented by :

- | | |
|-------------------------------------|---|
| (1) BF_3 - trigonal planar | (2) H_2O - bent |
| (3) NF_3 - trigonal planar | (4) AsF_5 - trigonal bipyramidal |

[JEE (Main-Online) 2018]

Q.46 The type of hybridisation and number of lone pair(s) of electrons of Xe in XeOF_4 respectively, are :

- | | | | |
|---------------------------------|-----------------------------------|-----------------------------------|---------------------------------|
| (1) sp^3d and 1 | (2) sp^3d^2 and 1 | (3) sp^3d^2 and 2 | (4) sp^3d and 2 |
|---------------------------------|-----------------------------------|-----------------------------------|---------------------------------|

[JEE (Main-Online) 2019]

Q.47 The element that shows greater ability to form $\text{p}\pi - \text{p}\pi$ multiple bonds, is :

- | | | | |
|--------|--------|--------|-------|
| (1) Sn | (2) Si | (3) Ge | (4) C |
|--------|--------|--------|-------|

[JEE (Main-Online) 2019]

Q.48 The dipole moments of CCl_4 , CHCl_3 and CH_4 are in the order

- | | |
|--|--|
| (1) $\text{CHCl}_3 < \text{CH}_4 = \text{CCl}_4$ | (2) $\text{CH}_4 = \text{CCl}_4 < \text{CHCl}_3$ |
| (3) $\text{CH}_4 < \text{CCl}_4 < \text{CHCl}_3$ | (4) $\text{CCl}_4 < \text{CH}_4 < \text{CHCl}_3$ |

[JEE (Main-Online) 2020]

Q.49 Arrange the following bonds according to their average bond energies in descending order

$\text{C} - \text{Cl}$, $\text{C} - \text{Br}$, $\text{C} - \text{F}$, $\text{C} - \text{I}$

- | | |
|---|---|
| (1) $\text{C} - \text{F} > \text{C} - \text{Cl} > \text{C} - \text{Br} > \text{C} - \text{I}$ | (2) $\text{C} - \text{Br} > \text{C} - \text{I} > \text{C} - \text{Cl} > \text{C} - \text{F}$ |
| (3) $\text{C} - \text{I} > \text{C} - \text{Br} > \text{C} - \text{Cl} > \text{C} - \text{F}$ | (4) $\text{C} - \text{Cl} > \text{C} - \text{Br} > \text{C} - \text{I} > \text{C} - \text{F}$ |

[JEE (Main-Online) 2020]

Q.50 The number of sp^2 hybrid orbitals in a molecule of benzene is

- (1) 24 (2) 6 (3) 12 (4) 18

[JEE (Main-Online) 2020]

Q.51 If AB_4 molecule is a polar molecule, a possible geometry of AB_4 is :

- (1) Square pyramidal (2) Tetrahedral
(3) Square planar (4) Rectangular planar

[JEE (Main-Online) 2020]

Q.52 The shape/structure of $[XeF_5]^-$ and XeO_3F_2 , respectively, are :

- (1) pentagonal planar and trigonal bipyramidal
(2) trigonal bipyramidal and pentagonal planar
(3) octahedral and square pyramidal
(4) trigonal bipyramidal and trigonal bipyramidal

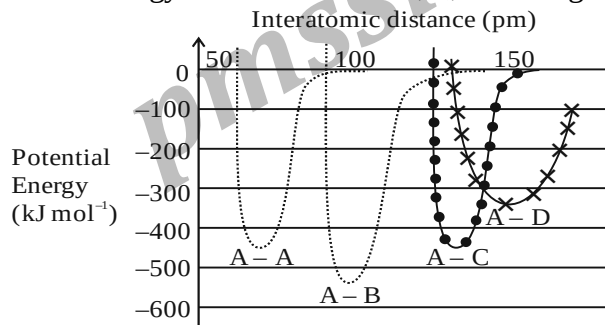
[JEE (Main-Online) 2020]

Q.53 The molecular geometry of SF_6 is octahedral. What is the geometry of SF_4 (including lone pair(s) of electrons, if any) ?

- (1) Trigonal bipyramidal (2) Square planar
(3) Tetrahedral (4) Pyramidal

[JEE (Main-Online) 2020]

Q.54 The intermolecular potential energy for the molecules A, B, C and D given below suggests that :



- (1) D is more electronegative than other atoms (2) A-D has the shortest bond length
(3) A-B has the stiffest bond (4) A-A has the largest bond enthalpy

[JEE (Main-Online) 2020]

Q.55 The structure of PCl_5 in the solid state is

- (1) square pyramidal
(2) tetrahedral $[PCl_4]^+$ and octahedral $[PCl_6]^-$
(3) square planar $[PCl_4]^+$ and octahedral $[PCl_6]^-$
(4) trigonal bipyramidal

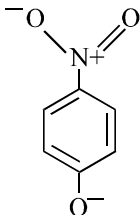
[JEE (Main-Online) 2020]

Q.56 The compound that has the largest H-M-H bond angle (M=N, O, S, C), is :

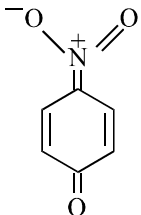
- (1) H_2O (2) CH_4 (3) NH_3 (4) H_2S

[JEE (Main-Online) 2020]

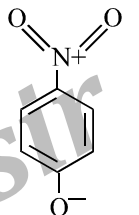
[JEE ADVANCED]

- Q.1 The geometry & the type of hybrid orbitals present about the central atom in BF_3 is : **[JEE 1998]**
 (A) linear, sp (B) trigonal planar, sp^2 (C) tetrahedral, sp^3 (D) pyramidal, sp^3
- Q.2 The correct order of increasing C - O bond length of, CO , CO_3^{2-} , CO_2 is **[JEE 1999]**
 (A) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$ (B) $\text{CO}_2 < \text{CO}_3^{2-} < \text{CO}$
 (C) $\text{CO} < \text{CO}_3^{2-} < \text{CO}_2$ (D) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
- Q.3 The geometry of H_2S and its dipole moment are **[JEE 1999]**
 (A) angular & non zero (B) angular & zero
 (C) linear & non zero (D) linear & zero
- Q.4 In compounds type ECl_3 , where $\text{E} = \text{B}, \text{P}, \text{As}$ or Bi , the angles $\text{Cl} - \text{E} - \text{Cl}$ for different E are in the order **[JEE 1999]**
 (A) $\text{B} > \text{P} = \text{As} = \text{Bi}$ (B) $\text{B} > \text{P} > \text{As} > \text{Bi}$ (C) $\text{B} < \text{P} = \text{As} = \text{Bi}$ (D) $\text{B} < \text{P} < \text{As} < \text{Bi}$
- Q.5 The most likely representation of resonance structure of p-nitrophenoxide is: **[JEE 1999]**
- 

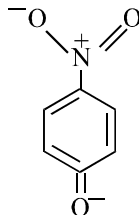
(A)



(B)

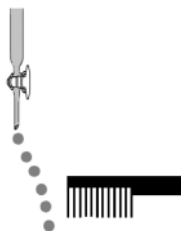


(C)



(D)
- Q.6 The hybridization of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ are **[JEE 2000]**
 (A) sp^2 , sp^3 and sp^2 respectively (B) sp , sp^2 and sp^3 respectively
 (C) sp^2 , sp and sp^3 respectively (D) sp^2 , sp^3 and sp respectively
- Q.7 Specify hybridization of N and B atoms in a 1 : 1 complex of BF_3 and NH_3 **[JEE 2002]**
 (A) N : tetrahedral, sp^3 ; B : tetrahedral, sp^3 (B) N : pyramidal, sp^3 ; B : pyramidal, sp^3
 (C) N : pyramidal, sp^3 ; B : planar, sp^2 (D) N : pyramidal, sp^3 ; B : tetrahedral, sp^3
- Q.8 The nodal plane in the π -bond of ethene is located in **[JEE 2002]**
 (A) the molecular plane
 (B) a plane parallel to the molecular plane
 (C) a plane perpendicular to the molecular plane which bisects, the carbon-carbon σ bond at right angle.
 (D) a plane perpendicular to the molecular plane which contains, the carbon-carbon bond.
- Q.9 Which of the following are isoelectronic and isostructural ? NO_3^- , CO_3^{2-} , ClO_3^- , SO_3 **[JEE 2003]**
 (A) NO_3^- , CO_3^{2-} (B) SO_3 , NO_3^- (C) ClO_3^- , CO_3^{2-} (D) CO_3^{2-} , SO_3

- Q.10 Which species has the maximum number of lone pair of electrons on the central atom? [JEE 2005]
 (A) ClO_3^- (B) XeF_4 (C) SF_4 (D) I_3^-
- Q.11 The percentage of p-character in the orbitals forming P–P bonds in P_4 is [JEE 2007]
 (A) 25 (B) 33 (C) 50 (D) 75
- Q.12 The structure of XeO_3 is [JEE 2007]
 (A) linear (B) planar (C) pyramidal (D) T-shaped
- Q.13 The species having pyramidal shape is [JEE 2010]
 (A) SO_3 (B) BrF_3 (C) SiO_3^{2-} (D) OSF_2
- Q.14 Based on VSEPR theory, the number of 90 degree F–Br–F angles in BrF_5 is [JEE 2010]
- Q.15 In allene (C_3H_4), the type(s) of hybridisation of the carbon atoms is (are) [JEE (Advanced) 2012]
 (A) sp and sp^3 (B) sp and sp^2 (C) only sp^2 (D) sp^2 and sp^3
- Q.16 The shape of XeO_2F_2 molecule is : [JEE (Advanced) 2012]
 (A) trigonal bipyramidal (B) square planar
 (C) tetrahedral (D) see-saw
- Q.17 Among the triatomic molecules / ions, BeCl_2 , N_3^- , N_2O , NO_2^+ , O_3 , SCl_2 , ICl_2^- , I_3^- and XeF_2 the total number of linear molecule(s) / ion(s) where the hybridization of the central atom does not have contribution from the d-orbital(s) is :
 [Atomic number : S = 16, Cl = 17, I = 53 and Xe = 54] [JEE (Advanced) 2015]
- Q.18 The sum of the number of lone pairs of electrons on each central atom in the following species is
 $[\text{TeBr}_6]^{2-}$, $[\text{BrF}_2]^+$, SNF_3 , and $[\text{XeF}_3]^-$
 (Atomic numbers: N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54) [JEE (Advanced) 2017]
- Q.19 Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature.
 (1) BF_3 , O_3 , SF_6 , XeF_6 (2) BeCl_2 , CO_2 , BCl_3 , CHCl_3
 (3) NO_2 , NH_3 , POCl_3 , CH_3Cl (4) SO_2 , $\text{C}_6\text{H}_5\text{Cl}$, H_2Se , BrF_5
 [JEE (Advanced) 2019]
- Q.20 Consider the following compounds in the liquid form :
 O_2 , HF , H_2O , NH_3 , H_2O_2 , CCl_4 , CHCl_3 , C_6H_6 , $\text{C}_6\text{H}_5\text{Cl}$.
 When a charged comb is brought near their flowing stream, how many of them show deflection as per the following figure?



[JEE (Advanced) 2020]

[JEE MAIN]

- Q.1 The reason for double helical structure of DNA is operation of: [AIEEE 2003]
 (1) dipole-dipole interaction (2) hydrogen bonding
 (3) electrostatic attraction (4) vander Wall's forces
- Q.2 The bond order in NO is 2.5 while that in NO^+ is 3. Which of the following statements is true for these two species? [AIEEE 2004]
 (1) Bond length in NO^+ is equal to that in NO (2) Bond length in NO is greater than in NO^+
 (3) Bond length in NO^+ is greater than in NO (4) Bond length is unpredictable
- Q.3 The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively: [AIEEE 2004]
 (1) sp^3 and sp^2 (2) sp^2 and sp^3
 (3) sp^2 and sp^2 (4) sp^3 and sp^3
- Q.4 Beryllium and aluminium exhibit many properties which are similar. But, the two elements differ in : [AIEEE 2004]
 (1) Forming covalent halides
 (2) Forming polymeric hydrides
 (3) Exhibiting maximum covalency in compound
 (4) Exhibiting amphoteric nature in their oxides
- Q.5 Which one of the following species is diamagnetic in nature? [AIEEE 2005]
 (1) He_2^+ (2) H_2 (3) H_2^+ (4) H_2^-
- Q.6 The number and type of bond between two carbon atom in calcium carbide are : [AIEEE 2005]
 (1) One sigma, one pi bond (2) One sigma, two pi bond
 (3) Two sigma, one pi bond (4) Two sigma, two pi bond
- Q.7 Which of the following molecules/ions does not contain unpaired electrons? [AIEEE 2006]
 (1) N_2^+ (2) O_2 (3) O_2^{2-} (4) B_2
- Q.8 Among the following mixtures, dipole-dipole as the major interaction, is present in : [AIEEE 2006]
 (1) KCl and water (2) benzene and carbon tetrachloride
 (3) benzene and ethanol (4) acetonitrile and acetone
- Q.9 A metal, M forms chlorides in its +2 and +4 oxidation states. Which of the following statements about these chlorides is correct? [AIEEE 2006]
 (1) MCl_2 is more ionic than MCl_4
 (2) MCl_2 is more easily hydrolysed than MCl_4
 (3) MCl_2 is more volatile than MCl_4
 (4) MCl_2 is more soluble in anhydrous ethanol than MCl_4

- Q.10 In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed [AIEEE 2007]
 (1) $\text{NO} \rightarrow \text{NO}^+$ (2) $\text{O}_2 \rightarrow \text{O}_2^+$ (3) $\text{N}_2 \rightarrow \text{N}_2^+$ (4) $\text{C}_2 \rightarrow \text{C}_2^+$
- Q.11 Which of the following hydrogen bonds is the strongest [AIEEE 2007]
 (1) $\text{F} - \text{H} \cdots \text{F}$ (2) $\text{O} - \text{H} \cdots \text{O}$ (3) $\text{O} - \text{H} \cdots \text{F}$ (4) $\text{O} - \text{H} \cdots \text{N}$
- Q.12 Which of the following species exhibits the diamagnetic behaviour: [AIEEE 2007]
 (1) O_2^+ (2) O_2 (3) NO (4) O_2^{2-}
- Q.13 The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+ , Ca^{+2} , Mg^{+2} , Be^{+2} [AIEEE 2007]
 (1) $\text{Be}^{+2} < \text{K}^+ < \text{Ca}^{+2} < \text{Mg}^{+2}$ (2) $\text{K}^+ < \text{Ca}^{+2} < \text{Mg}^{+2} < \text{Be}^{+2}$
 (3) $\text{Ca}^{+2} < \text{Mg}^{+2} < \text{Be}^{+2} < \text{K}^+$ (4) $\text{Mg}^{+2} < \text{Be}^{+2} < \text{K}^+ < \text{Ca}^{+2}$
- Q.14 Which one of the following pairs of species have the same bond order : [AIEEE 2008]
 (1) CN^- and NO^+ (2) CN^- and CN^+
 (3) O_2^- and CN^- (4) NO^+ and CN^+
- Q.15 Using molecule orbital theory predict which of the following species has the shortest bond length- [AIEEE 2009]
 (1) O_2^{2+} (2) O_2^+ (3) O_2^- (4) O_2^{2-}
- Q.16 Ortho-Nitrophenol is less soluble in water than p- and m-Nitrophenols because : [AIEEE 2012]
 (1) o-Nitrophenol shows intermolecular H-bonding
 (2) Melting point of o-Nitrophenol is lower than those of m- and p-isomers.
 (3) o-Nitrophenol is more volatile in steam than those of m- and p-isomers.
 (4) o-Nitrophenol shows intramolecular H-bonding
- Q.17 Stability of the species Li_2 , Li_2^- and Li_2^+ increases in the order of: [JEE (Main-Offline) 2013]
 (1) $\text{Li}_2^- < \text{Li}_2^+ < \text{Li}_2$ (2) $\text{Li}_2 < \text{Li}_2^- < \text{Li}_2^+$ (3) $\text{Li}_2^- < \text{Li}_2 < \text{Li}_2^+$ (4) $\text{Li}_2 < \text{Li}_2^+ < \text{Li}_2^-$
- Q.18 Which one of the following molecules is expected to exhibit diamagnetic behaviour? [JEE (Main-Offline) 2013]
 (1) N_2 (2) O_2 (3) S_2 (4) C_2
- Q.19 In which of the following pairs of molecules / ions, both the species are not likely to exist? [JEE (Main-Offline) 2013]
 (1) H_2^- , He_2^{2-} (2) H_2^{2+} , He_2 (3) H_2^- , He_2^{2+} (4) H_2^+ , He_2^{2-}
- Q.20 The solubility order for alkali metal fluoride in water is : [JEE (Main-Online) 2013]
 (1) $\text{RbF} < \text{KF} < \text{NaF} < \text{LiF}$ (2) $\text{LiF} < \text{RbF} < \text{KF} < \text{NaF}$
 (3) $\text{LiF} > \text{NaF} > \text{KF} > \text{RbF}$ (4) $\text{LiF} < \text{NaF} < \text{KF} < \text{RbF}$

- Q.21 Bond order normally gives idea of stability of a molecular species. All the molecules viz. H_2 , Li_2 and B_2 have the same bond order yet they are not equally stable. Their stability order is :
 (1) $Li_2 > H_2 > Br_2$ (2) $Li_2 > B_2 > H_2$ (3) $H_2 > B_2 > Li_2$ (4) $B_2 > H_2 > Li_2$
[JEE (Main-Online) 2013]
- Q.22 The internuclear distances in O—O bonds for O_2^+ , O_2 , O_2^- and O_2^{2-} respectively are :
 (1) 1.30 Å, 1.49 Å, 1.12 Å, 1.21 Å (2) 1.12 Å, 1.21 Å, 1.30 Å, 1.49 Å
 (3) 1.21 Å, 1.12 Å, 1.49 Å, 1.30 Å (4) 1.49 Å, 1.21 Å, 1.12 Å, 1.30 Å
[JEE (Main-Online) 2013]
- Q.23 Which one of the following properties is not shown by NO? **[JEE (Main-Offline) 2014]**
 (1) It is a neutral oxide
 (2) It combines with oxygen to form nitrogen dioxide
 (3) It's bond order is 2.5
 (4) It is diamagnetic in gaseous state
- Q.24 The number and type of bonds in C_2^{2-} ion in CaC_2 are :
 (1) Two σ bonds and one π -bond (2) Two σ bonds and two π -bonds
 (3) One σ bond and one π -bond (4) One σ bond and two π -bonds
[JEE (Main-Online) 2014]
- Q.25 Which of the following has unpaired electron(s) ?
 (1) N_2 (2) O_2^- (3) O_2^{2-} (4) N_2^{2+}
[JEE (Main-Online) 2014]
- Q.26 Which one of the following does not have a pyramidal shape ?
 (1) $P(CH_3)_3$ (2) $(SiH_3)_3N$ (3) $P(SiH_3)_3$ (4) $(CH_3)_3N$
[JEE (Main-Online) 2014]
- Q.27 The correct order of bond dissociation energy among N_2 , O_2 , O_2^- is shown in which of the following arrangements ?
 (1) $O_2^- > O_2 > N_2$ (2) $N_2 > O_2 > O_2^-$ (3) $N_2 > O_2^- > O_2$ (4) $O_2 > O_2^- > N_2$
[JEE (Main-Online) 2014]
- Q.28 Amongst $LiCl$, $RbCl$, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are :
 (1) $MgCl_2$ and $BeCl_2$ (2) $LiCl$ and $RbCl$ (3) $RbCl$ and $MgCl_2$ (4) $RbCl$ and $BeCl_2$
[JEE (Main-Online) 2014]
- Q.29 The reason for double helical structure of DNA is the operation of :
 (1) Hydrogen bonding (2) Electrostatic attractions
 (3) Dipole-Dipole interactions (4) van der Waals forces
[JEE (Main-Online) 2014]

- Q.30 Which one of the following molecules is paramagnetic ?
 (1) CO (2) N₂ (3) O₃ (4) NO
 [JEE (Main-Online) 2014]
- Q.31 The intermolecular interaction that is dependent on the inverse cube of distance between the molecules is:
 (1) London force (2) Hydrogen bond
 (3) Ion-ion interaction (4) Ion-dipole interaction
 [JEE (Main-Offline) 2015]
- Q.32 After understanding the assertion and reason, choose the correct option.
Assertion : In the bonding molecular orbital (MO) of H₂, electron density is increased between the nuclei.
Reason : The bonding MO is $\Psi_A + \Psi_B$, which shows destructive interference of the combining electron waves.
 (1) Assertion is correct, reason is incorrect.
 (2) Assertion and reason are correct, but reason is not the correct explanation for the assertion.
 (3) Assertion and reason are correct and reason is the correct explanation for the assertion.
 (4) Assertion is incorrect, reason is correct.
 [JEE (Main-Online) 2015]
- Q.33 Which of the alkaline earth metal halides given below is essentially covalent in nature ?
 (1) CaCl₂ (2) BeCl₂ (3) SrCl₂ (4) MgCl₂
 [JEE (Main-Online) 2015]
- Q.34 Which of the following compounds has a P-P bond ?
 (1) H₄P₂O₇ (2) H₄P₂O₆ (3) (HPO₃)₃ (4) H₄P₂O₆
 [JEE (Main-Online) 2015]
- Q.35 Which intermolecular force is most responsible in allowing xenon gas to liquefy?
 (1) Dipole-dipole (2) Ion-dipole
 (3) Instantaneous dipole-induced dipole (4) Ionic
 [JEE (Main-Online) 2016]
- Q.36 The correct order of the solubility of alkaline-earth metal sulphates in water is:
 (1) Mg < Ca < Sr < Ba (2) Mg < Sr < Ca < Ba
 (3) Mg > Sr > Ca > Ba (4) Mg > Ca > Sr > Ba
 [JEE (Main-Online) 2016]
- Q.37 Which of the following species is not paramagnetic ?
 (1) CO (B) O₂ (C) B₂ (D) NO
 [JEE (Main-Offline) 2017]
- Q.38 Which of the following is paramagnetic ?
 (1) NO⁺ (2) CO (3) O₂²⁻ (4) B₂
 [JEE (Main-Online) 2017]

- Q.39 The number of S = O and S – OH bonds present in peroxodisulphuric acid and pyrosulphuric acid respectively are :
 (1) (2 and 2) and (2 and 2) (2) (2 and 4) and (2 and 4)
 (3) (4 and 2) and (2 and 4) (4) (4 and 2) and (4 and 2)
[JEE (Main-Online) 2017]

- Q.40 Which of the following salts is the most basic in aqueous solution? **[JEE (Main-Offline) 2018]**
 (1) FeCl_3 (2) $\text{Pb}(\text{CH}_3\text{COO})_2$ (3) $\text{Al}(\text{CN})_3$ (4) CH_3COOK

- Q.41 Which of the following are Lewis acids? **[JEE (Main-Offline) 2018]**
 (1) PH_3 and SiCl_4 (2) BCl_3 and AlCl_3 (3) PH_3 and BCl_3 (4) AlCl_3 and SiCl_4

- Q.42 According to molecular orbital theory, which of the following will **not** be a viable molecule?
 (1) H_2^- (2) H_2^{2-} (3) He_2^{2+} (4) He_2^+
[JEE (Main-Offline) 2018]

- Q.43 Which of the following best describes the diagram below of a molecular orbital ?



- (1) An antibonding σ orbital (2) A non-bonding orbital
 (3) A bonding π orbital (4) An antibonding π orbital

[JEE (Main-Online) 2018]

- Q.44 The number of P – O bonds in P_4O_6 is :
 (1) 12 (2) 6 (3) 9 (4) 18
[JEE (Main-Online) 2018]

- Q.45 In graphite and diamond, the percentage of p-characters of the hybrid orbitals in hybridisation are respectively:
 (1) 50 and 75 (2) 33 and 25 (3) 67 and 75 (4) 3 and 75
[JEE (Main-Online) 2018]

- Q.46 In the molecular orbital diagram for the molecular ion, N_2^+ , the number of electrons in the σ_{2p} molecular orbital is :
 (1) 2 (2) 3 (3) 1 (4) 0
[JEE (Main-Online) 2018]

- Q.47 Among the oxides of nitrogen :
 N_2O_3 , N_2O_4 and N_2O_5 ; the molecule(s) having nitrogen-nitrogen bond is / are :
 (1) Only N_2O_5 (2) N_2O_3 and N_2O_5 (3) N_2O_4 and N_2O_5 (4) N_2O_3 and N_2O_4
[JEE (Main-Online) 2018]

- Q.48 According to molecular orbital theory, which of the following is true with respect to Li_2^+ and Li_2^- ?
 (1) Both are stable (2) Li_2^+ is unstable and Li_2^- is stable
 (3) Both are unstable (4) Li_2^+ is stable and Li_2^- is unstable
[JEE (Main-Online) 2019]
- Q.49 Which amongst the following is the strongest acid ?
 (1) CHI_3 (2) $\text{CH}(\text{CN})_3$ (3) CHCl_3 (4) CHBr_3
[JEE (Main-Online) 2019]
- Q.50 Good reducing nature of H_3PO_2 is attributed to the presence of :
 (1) Two P–OH bonds (2) Two P–H bonds (3) One P–OH bond (4) One P–H bond
[JEE (Main-Online) 2019]
- Q.51 In which of the following processes, the bond order has increased and paramagnetic character has changed to diamagnetic?
 (1) $\text{O}_2 \rightarrow \text{O}_2^{2-}$ (2) $\text{O}_2 \rightarrow \text{O}_2^+$ (3) $\text{N}_2 \rightarrow \text{N}_2^+$ (4) $\text{NO} \rightarrow \text{NO}^+$
[JEE (Main-Online) 2019]
- Q.52 Two pi and half sigma bonds are present in :
 (1) N_2^+ (2) O_2^+ (3) O_2 (4) N_2
[JEE (Main-Online) 2019]
- Q.53 The number of 2-centre-2-electron and 3-centre-2-electron bonds in B_2H_6 , respectively, are :
 (1) 2 and 2 (2) 4 and 2 (3) 2 and 4 (4) 2 and 1
[JEE (Main-Online) 2019]
- Q.54 The hydride that is NOT electron deficient is:
 (1) SiH_4 (2) B_2H_6 (3) GaH_3 (4) AlH_3
[JEE (Main-Online) 2019]
- Q.55 Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature. **[JEE (Main-Offline) 2019]**
 (1) BF_3 , O_3 , SF_6 , XeF_6 (2) BeCl_2 , CO_2 , BCl_3 , CHCl_3
 (3) NO_2 , NH_3 , POCl_3 , CH_3Cl (4) SO_2 , $\text{C}_6\text{H}_5\text{Cl}$, H_2Se , BrF_5
- Q.56 Among B_2H_6 , $\text{B}_3\text{N}_3\text{H}_6$, N_2O , N_2O_4 , $\text{H}_2\text{S}_2\text{O}_3$ and $\text{H}_2\text{S}_2\text{O}_8$, the total number of molecules containing covalent bond between two atoms of the same kind is _____ **[JEE (Main-Offline) 2019]**
- Q.57 According to molecular orbital theory, which of the following is true with respect to Li_2^+ and Li_2^- ?
 (1) Both are stable (2) Li_2^+ is unstable and Li_2^- is stable
 (3) Both are unstable (4) Li_2^+ is stable and Li_2^- is unstable
[JEE (Main-Online) 2019]
- Q.58 Which amongst the following is the strongest acid ?
 (1) CHI_3 (2) $\text{CH}(\text{CN})_3$ (3) CHCl_3 (4) CHBr_3
[JEE (Main-Online) 2019]

- Q.59 Good reducing nature of H_3PO_2 is attributed to the presence of :
 (1) Two P–OH bonds (2) Two P–H bonds (3) One P–OH bond (4) One P–H bond
[JEE (Main-Online) 2019]
- Q.60 In which of the following processes, the bond order has increased and paramagnetic character has changed to diamagnetic?
 (1) $\text{O}_2 \rightarrow \text{O}_2^{2-}$ (2) $\text{O}_2 \rightarrow \text{O}_2^+$ (3) $\text{N}_2 \rightarrow \text{N}_2^+$ (4) $\text{NO} \rightarrow \text{NO}^+$
[JEE (Main-Online) 2019]
- Q.61 Two pi and half sigma bonds are present in :
 (1) N_2^+ (2) O_2^+ (3) O_2 (4) N_2
[JEE (Main-Online) 2019]
- Q.62 The number of 2-centre-2-electron and 3-centre-2-electron bonds in B_2H_6 , respectively, are :
 (1) 2 and 2 (2) 4 and 2 (3) 2 and 4 (4) 2 and 1
[JEE (Main-Online) 2019]
- Q.63 The hydride that is NOT electron deficient is:
 (1) SiH_4 (2) B_2H_6 (3) GaH_3 (4) AlH_3
[JEE (Main-Online) 2019]
- Q.64 Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature. **[JEE (Main-Offline) 2019]**
 (1) BF_3 , O_3 , SF_6 , XeF_6 (2) BeCl_2 , CO_2 , BCl_3 , CHCl_3
 (3) NO_2 , NH_3 , POCl_3 , CH_3Cl (4) SO_2 , $\text{C}_6\text{H}_5\text{Cl}$, H_2Se , BrF_5
- Q.65 Among B_2H_6 , $\text{B}_3\text{N}_3\text{H}_6$, N_2O , N_2O_4 , $\text{H}_2\text{S}_2\text{O}_3$ and $\text{H}_2\text{S}_2\text{O}_8$, the total number of molecules containing covalent bond between two atoms of the same kind is _____ **[JEE (Main-Offline) 2019]**
- Q.66 Match the type of interaction in Column A with the distance dependence of their interaction energy in Column B :
- | A | B |
|----------------------------------|----------------------------------|
| (I) iron - ion | (a) $\frac{1}{r}$ |
| (II) dipole - dipole | (b) $\frac{1}{r^2}$ |
| (III) London dispersion | (c) $\frac{1}{r^3}$ |
| | (d) $\frac{1}{r^6}$ |
| (1) (I)-(a), (II)-(b), (III)-(c) | (2) (I)-(a), (II)-(c), (III)-(d) |
| (3) (I)-(a), (II)-(b), (III)-(d) | (4) (I)-(b), (II)-(d), (III)-(c) |
- [JEE (Main-Online) 2020]**

- Q.67 If the boiling point of H_2O is 373 K, the boiling point of H_2S will be :
 (1) greater than 300 K but less than 373 K (2) less than 300 K
 (3) equal to 373 K (4) more than 373 K
[JEE (Main-Online) 2020]
- Q.68 In a molecule of pyrophosphoric acid, the number of P–OH, P = O and P–O–P bonds/moiety(ies) respectively are :
 (1) 3, 3 and 3 (2) 2, 4 and 1 (3) 4, 2 and 0 (4) 4, 2 and 1
[JEE (Main-Online) 2020]
- Q.69 Of the species, NO , NO^+ , NO^{2+} and NO^- , the one with minimum bond strength is :
 (1) NO^{2+} (2) NO^+ (3) NO (4) NO^-
[JEE (Main-Online) 2020]
- Q.70 An alkaline earth metal 'M' readily forms water soluble sulphate and water insoluble hydroxide. Its oxide MO is very stable to heat and does not have rock-salt structure. M is :-
 (1) Ca (2) Be (3) Mg (4) Sr
[JEE (Main-Online) 2020]
- Q.71 Among the sulphates of alkaline earth metals, the solubilities of BeSO_4 and MgSO_4 in water, respectively, are:
 (1) high and high (2) poor and poor (3) high and poor (4) poor and high
[JEE (Main-Online) 2020]
- Q.72 The number of Cl = O bonds in perchloric acid is, " _____ "
[JEE (Main-Online) 2020]
- Q.73 The relative strength of interionic/intermolecular forces in decreasing order is
 (1) ion-dipole > ion-ion > dipole-dipole (2) ion-dipole > dipole-dipole > ion-ion
 (3) ion-ion > ion-dipole > dipole-dipole (4) dipole-dipole > ion-dipole > ion-ion
[JEE (Main-Online) 2020]
- Q.74 The bond order and the magnetic characteristics and CN^- are
 (1) 3, paramagnetic (2) $2\frac{1}{2}$, diamagnetic
 (3) 3, diamagnetic (4) $2\frac{1}{2}$, paramagnetic
[JEE (Main-Online) 2020]

- Q.75 The predominant intermolecular forces present in ethyl acetate, a liquid, are
- (1) London dispersion, dipole-dipole and hydrogen bonding
 - (2) hydrogen bonding and London dispersion
 - (3) dipole-dipole and hydrogen bonding
 - (4) London dispersion and dipole-dipole

[JEE (Main-Online) 2020]

- Q.76 If the magnetic moment of a dioxygen species is 1.73 B.M, it may be

- (1) O_2 or O_2^+ (2) O_2^- or O_2^+ (3) O_2 or O_2^- (4) O_2 , O_2^- , O_2^+

[JEE (Main-Online) 2020]

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[JEE ADVANCED]

- Q.1 In the dichromate anion [JEE 1999]
 (A) 4 Cr - O bonds are equivalent (B) 6 Cr - O bonds are equivalent
 (C) all Cr - O bonds are equivalent (D) all Cr - O bonds are non equivalent
- Q.2 Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is [JEE 2000]
 (A) H_2O because of hydrogen bonding (B) H_2Te because of higher molecular weight
 (C) H_2S because of hydrogen bonding (D) H_2Se because of lower molecular weight
- Q.3 The common features among the species CN^- , CO and NO^+ are [JEE 2001]
 (A) Bond order three and isoelectronic (B) Bond order three and weak field ligands
 (C) Bond order two and π - acceptors (D) Isoelectronic and weak field ligands
- Q.4 Which of the following molecular species has unpaired electron(s)? [JEE 2002]
 (A) N_2 (B) F_2 (C) O_2^- (D) O_2^{2-}
- Q.5 According to molecular orbital theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ [JEE 2004]
 (A) Paramagnetic and Bond order $< \text{O}_2$ (B) Paramagnetic and Bond order $> \text{O}_2$
 (C) Diamagnetic and Bond order $< \text{O}_2$ (D) Diamagnetic and Bond order $> \text{O}_2$
- Q.6 Decreasing order of the O-O bond length present in them [JEE 2004]
 O_2 , KO_2 and $\text{O}_2 [\text{AsF}_4]$
- Q.7 Among the following, the paramagnetic compound is [JEE 2007]
 (A) Na_2O_2 (B) O_3 (C) N_2O (D) KO_2
- Q.8 The species having bond order different from that in CO is [JEE 2007]
 (A) NO^- (B) NO^+ (C) CN^- (D) N_2
- Q.9 **Statement-1** : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid. because
Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding. [JEE 2007]
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.
- Q.10 The number of water molecule(s) directly bonded to the metal centre in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is [JEE 2009]

Q.11 Match each of the diatomic molecules in Column I with its property / properties in Column II.

Column I**Column II****[JEE 2009]**(A) B_2

(P) Paramagnetic

(B) N_2

(Q) undergoes oxidation

(C) O_2^-

(R) Undergoes reduction

(D) O_2 (S) Bond order ≥ 2

(T) Mixing of 's' and 'p' orbitals

Q.12 The nitrogen oxide(s) that contain(s) N–N bond(s) is (are)

[JEE 2009](A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5

Q.13 Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is **[JEE 2010]**

(A) 1 and diamagnetic

(B) 0 and diamagnetic

(C) 1 and paramagnetic

(D) 0 and paramagnetic

Q.14 The total number of diprotic acids among the following is **[JEE 2010]**

 H_3PO_4 H_2SO_4 H_3PO_3 H_2CO_3 $H_2S_2O_7$ H_3BO_3 H_3PO_2 H_2CrO_4 H_2SO_3

Q.15 The difference in the oxidation numbers of the two types of sulphur atoms in $Na_2S_4O_6$ is **[JEE 2011]**

Q.16 With respect to graphite and diamond, which of the statement(s) given below is (are) correct?

[JEE (Advanced) 2012]

(A) Graphite is harder than diamond

(B) Graphite has higher electrical conductivity than diamond.

(C) Graphite has higher thermal conductivity than diamond.

(D) Graphite has higher C–C bond order than diamond.

Q.17 Hydrogen bonding plays a central role in the following phenomena :

[JEE (Advanced) 2014]

(A) Ice floats in water.

(B) Higher Lewis basicity of primary amines than tertiary amines in aqueous solutions

(C) Formic acid is more acidic than acetic acid.

(D) Dimerisation of acetic acid in benzene.

Q.18 Assuming 2s-2p mixing in NOT operative, the paramagnetic species among the following is

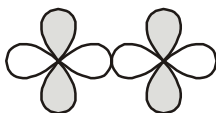
[JEE (Advanced) 2014](A) Be_2 (B) B_2 (C) C_2 (D) N_2

Q.19 Match the orbital overlap figures shown in List-I with the description given in List-II and select the correct answer using the code given below the lists. [JEE (Advanced) 2014]

List-I

List-II

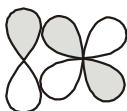
P.



1.

p – d π antibonding

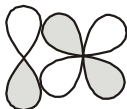
Q.



2.

d – d σ bonding

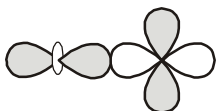
R.



3.

p – d π bonding

S.



4.

d – d σ antibonding

Code:

	P	Q	R	S
(A)	2	1	3	4
(B)	4	3	1	2
(C)	2	3	1	4
(D)	4	1	3	2

Q.20 The correct statement(s) regarding, (i) HClO , (ii) HClO_2 , (iii) HClO_3 and (iv) HClO_4 , is(are)

- (A) The number of $\text{Cl}=\text{O}$ bonds in (ii) and (iii) together is two
 (B) The number of lone pairs of electrons on Cl in (ii) and (iii) together is three
 (C) The hybridization of Cl in (iv) is sp^3
 (D) Amongst (i) to (iv), the strongest acid is (i)

[JEE (Advanced) 2015]

Q.21 The total number of lone pairs of electrons in N_2O_3 is :

[JEE (Advanced) 2015]

Q.22 According to Molecular Orbital Theory,

[JEE (Advanced) 2016]

- (A) C_2^{2-} is expected to be diamagnetic
 (B) O_2^{2+} is expected to have a longer bond length than O_2
 (C) N_2^+ and N_2^- have the same bond order
 (D) He_2^+ has the same energy as two isolated He atoms

Q.23 The colour of the X_2 molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to [JEE (Advanced) 2017]

- (A) the physical state of X_2 at room temperature changes from gas to solid down the group
 (B) decrease in HOMO-LUMO gap down the group
 (C) decrease in $\pi^*-\sigma^*$ gap down the group
 (D) decrease in ionization energy down the group

- Q.24 Among H_2 , He_2^+ , Li_2 , Be_2 , B_2 , C_2 , N_2 , O_2^- , and F_2 , the number of diamagnetic species is (Atomic numbers : H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9) **[JEE (Advanced) 2017]**
- Q.25 The order of the oxidation state of the phosphorus atom in H_3PO_2 , H_3PO_4 , H_3PO_3 and $\text{H}_4\text{P}_2\text{O}_6$ is **[JEE (Advanced) 2017]**
- (A) $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6$ (B) $\text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6$
(C) $\text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_4$ (D) $\text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2$
- Q.26 Based on the compounds of group 15 elements, the correct statement(s) is (are)
- (A) Bi_2O_5 is more basic than N_2O_5
(B) NF_3 is more covalent than BiF_3
(C) PH_3 boils at lower temperature than NH_3
(D) The N-N single bond is stronger than the P-P single bond **[JEE (Advanced) 2018]**
- Q.27 The total number of compounds having at least one bridging oxo group among the molecules given below is ____.
- N_2O_3 , N_2O_5 , P_4O_6 , P_4O_7 , $\text{H}_4\text{P}_2\text{O}_5$, $\text{H}_5\text{P}_3\text{O}_{10}$, $\text{H}_2\text{S}_2\text{O}_3$, $\text{H}_2\text{S}_2\text{O}_5$ **[JEE (Advanced) 2018]**
- Q.28 Among B_2H_6 , $\text{B}_3\text{N}_3\text{H}_6$, N_2O , N_2O_4 , $\text{H}_2\text{S}_2\text{O}_3$ and $\text{H}_2\text{S}_2\text{O}_8$, the total number of molecules containing covalent bond between two atoms of the same kind is _____ **[JEE (Advanced) 2019]**

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[JEE MAIN]

- Q.1 In $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$, the isomerism shown is – [AIEEE 2002]
 (1) Ligand (2) Optical (3) Geometrical (4) Ionization
- Q.2 In the complexes $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{CN})_6]^{3+}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{FeCl}_6]^{3-}$, more stability is shown by – [AIEEE 2002]
 (1) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (2) $[\text{Fe}(\text{CN})_6]^{3+}$ (3) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (4) $[\text{FeCl}_6]^{3-}$
- Q.3 One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of $\text{AgCl}(\text{s})$. The structure of the complex is – [AIEEE 2003]
 (1) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$ (2) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2] \text{Cl} \cdot \text{NH}_3$
 (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}] \text{Cl}_2 \cdot \text{NH}_3$ (4) $[\text{Co}(\text{NH}_3)_5\text{Cl}] \text{Cl}_2$
- Q.4 In the coordination compound $\text{K}_4[\text{Ni}(\text{CN})_4]$, the oxidation state of nickel is – [AIEEE 2003]
 (1) 0 (2) +1 (3) +2 (4) -1
- Q.5 The number of 3d-electrons remained in Fe^{2+} (At. no. of Fe = 26) is – [AIEEE 2003]
 (1) 4 (2) 5 (3) 6 (4) 3
- Q.6 Ammonia forms the complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ion with copper ions in alkaline solutions but not in acidic solution. What is the reason for it – [AIEEE 2003]
 (1) In acidic solution hydration protects copper ions
 (2) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available
 (3) In alkaline solutions insoluble $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of any alkali
 (4) Copper hydroxide is an amphoteric substance
- Q.7 Among the properties (a) reducing (b) oxidising (c) complexing, the set of properties shown by CN^- ion towards metal species is – [AIEEE 2004]
 (1) c, a (2) b, c (3) a, b (4) a, b, c
- Q.8 The coordination number of a central metal atom in a complex is determined by – [AIEEE 2004]
 (1) The number of ligands around a metal ion bonded by sigma and pi-bonds both
 (2) The number of ligands around a metal ion bonded by pi-bonds
 (3) The number of ligands around a metal ion bonded by coordinate bonds
 (4) The number of only anionic ligands bonded to the metal ion
- Q.9 Which one of the following complexes is an outer orbitals complex [AIEEE 2004]
 (1) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (2) $[\text{Mn}(\text{CN})_6]^{4-}$ (3) $[\text{Fe}(\text{CN})_6]^{4-}$ (4) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 (Atomic nos. : Mn = 25; Fe = 26; Co = 27; Ni = 28)

- Q.10 Coordination compounds have great importance in biological systems. In this context which of the following statements is incorrect ? **[AIEEE 2004]**
 (1) Cyanocobalamin is vitamin B₁₂ and contains cobalt
 (2) Haemoglobin is the red pigment of blood and contains iron
 (3) Chlorophylls are green pigments in plants and contain calcium
 (4) Carboxypeptidase-A is an enzyme and contains zinc
- Q.11 The correct order of magnetic moments (spin only values in B.M.) among is – **[AIEEE 2004]**
 (1) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$ (2) $[\text{MnCl}_4]^{4-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
 (3) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$ (4) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
 (Atomic nos : Mn = 25 ; Fe = 26 ; Co = 27 ; Ni = 28)
- Q.12 The value of the 'spin only' magnetic moment for one of the following configurations is 2.84 BM . The correct one is – **[AIEEE 2005]**
 (1) d⁴ (in strong ligand field) (2) d⁴ (in weak ligand field)
 (3) d³ (in weak as well as in strong field) (4) d⁵ (in strong ligand field)
- Q.13 The IUPAC name for the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5] \text{Cl}_2$ is – **[AIEEE 2006]**
 (1) pentaammine nitrito-N-cobalt (II) chloride (2) pentaammine nitrito-N-cobalt (III) chloride
 (3) nitrito-N-pentaamminecobalt (III) chloride (4) nitrito-N-pentaamminecobalt (II) chloride
- Q.14 Nickel (Z = 28) combines with a negative monodentate ligand X⁻ to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron in the nickel and geometry of this complex ion are respectively **[AIEEE 2006]**
 (1) one, square planar (2) two, square planar (3) one, tetrahedral (4) two, tetrahedral
- Q.15 In $\text{Fe}(\text{CO})_5$, the Fe–C bond possesses – **[AIEEE 2006]**
 (1) ionic character (2) σ-character only (3) π-character (4) both σ and π character
- Q.16 How many EDTA (ethylenediaminetetraacetate ion) molecules are required to make an octahedral complex with a Ca²⁺ ion ? **[AIEEE 2006]**
 (1) One (2) Two (3) Six (4) Three
- Q.17 The "spin-only" magnetic moment [in units of Bohr magneton] of Ni²⁺ in aqueous solution would be (At. No. Ni = 28) – **[AIEEE 2006]**
 (1) 0 (2) 1.73 (3) 2.84 (4) 4.90
- Q.18 Which one of the following has a square planar geometry - (Co = 27, Ni = 28, Fe = 26, Pt = 78) – **[AIEEE 2007]**
 (1) $[\text{CoCl}_4]^{2-}$ (2) $[\text{FeCl}_4]^{2-}$ (3) $[\text{NiCl}_4]^{2-}$ (4) $[\text{PtCl}_4]^{2-}$
- Q.19 The coordination number and the oxidation state of the element 'E' in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4)] \text{NO}_2$ (where (en) is ethylene diamine) are, respectively – **[AIEEE 2008]**
 (1) 6 and +2 (2) 4 and +2 (3) 4 and +3 (4) 6 and +3

- Q.20 In which of the following octahedral complexes of Co (at. no. 27), will the magnitude of Δ_0 be the highest ? [AIEEE 2008]
 (1) $[\text{Co}(\text{CN})_6]^{3-}$ (2) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (3) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (4) $[\text{Co}(\text{NH}_3)_6]^{3+}$
- Q.21 Which of the following has an optical isomer [AIEEE 2009]
 (1) $[\text{Co}(\text{en})(\text{NH}_3)_2]^{2+}$ (2) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$
 (3) $[\text{Co}(\text{en})_2(\text{NH}_3)_2]^{3+}$ (4) $[\text{Co}(\text{NH}_3)_3\text{Cl}]^+$
- Q.22 Which of the following pairs represents linkage isomers ? [AIEEE 2009]
 (1) $[\text{Pd}(\text{PPh}_3)_2(\text{NCS})_2]$ and $[\text{Pd}(\text{PPh}_3)_2(\text{SCN})_2]$
 (2) $[\text{Co}(\text{NH}_3)_5\text{NO}_3]$ SO_4 and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]$ NO_3
 (3) $[\text{PtCl}_2(\text{NH}_3)_4]$ Br_2 and $[\text{PtBr}_2(\text{NH}_3)_4]$ Cl_2
 (4) $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$
- Q.23 A solution containing 2.675 g of $\text{CoCl}_3 \cdot 6\text{NH}_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of AgNO_3 to give 4.78 g of AgCl (molar mass = 143.5 g mol^{-1}). The formula of the complex is : [AIEEE 2010]
 (Atomic mass of Ag = 108 u)
 (1) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (2) $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$
 (3) $[\text{CoCl}_3(\text{NH}_3)_3]$ (4) $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$
- Q.24 Which one of the following has an optical isomer ? (en = ethylenediamine) [AIEEE 2010]
 (1) $[\text{Zn}(\text{en})(\text{NH}_3)_2]^{2+}$ (2) $[\text{Co}(\text{en})_3]^{3+}$
 (3) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$ (4) $[\text{Zn}(\text{en})_2]^{2+}$
- Q.25 Which of the following facts about the complex $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is wrong ? [AIEEE 2011]
 (1) The complex involves d^2sp^3 hybridisation and is octahedral in shape.
 (2) The complex is paramagnetic
 (3) The complex is an outer orbital complex.
 (4) The complex gives white precipitate with silver nitrate solution.
- Q.26 The magnetic moment (spin only) of $[\text{NiCl}_4]^{2-}$ is [AIEEE 2011]
 (1) 1.82 BM (2) 5.46 BM (3) 2.82 BM (4) 1.41 BM
- Q.27 Which among the following will be named as dibromidobis (ethylene diamine) chromium (III) bromide ? [AIEEE 2012]
 (1) $[\text{Cr}(\text{en})\text{Br}_4]^-$ (2) $[\text{Cr}(\text{en})\text{Br}_2]^-$ (3) $[\text{Cr}(\text{en})_3]\text{Br}_3$ (4) $[\text{Cr}(\text{en})_2\text{Br}_2]\text{Br}$
- Q.28 Which of the following complex species is not expected to exhibit optical isomerism? [JEE (Main-Offline) 2013]
 (1) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (2) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
 (3) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$ (4) $[\text{Co}(\text{en})_3]^{3+}$
- Q.29 Type of isomerism which exists between $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{SCN})_2]$ and $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{NCS})_2]$ is :
 (1) Solvate isomerism (2) Linkage isomerism
 (3) Ionisation isomerism (4) Coordination isomerism [JEE (Main-Online) 2013]

Q.30 Which of the following is diamagnetic ?

- (1) $[\text{Co(ox)}]^{3-}$ (2) $[\text{FeF}_6]^{3-}$
 (3) $[\text{CoF}_6]^{3-}$ (4) $[\text{Fe(CN)}_6]^{3-}$

[JEE (Main-Online) 2013]

Q.31 The magnetic moment of the complex anion $[\text{Cr(NO)}(\text{NH}_3)(\text{CN})_4]^{2-}$ is :

- (1) 1.73 BM (2) 3.87 BM (3) 2.82 BM (4) 5.91 BM

[JEE (Main-Online) 2013]

Q.32 The structure of which of the following chloro species can be explained on the basis of dsp^2 hybridisation?

- (1) CoCl_4^{2-} (2) NiCl_4^{2-} (3) PdCl_4^{2-} (4) FeCl_4^{2-}

[JEE (Main-Online) 2013]

Q.33 In which of the following octahedral complex species the magnitude of Δ_o will be maximum?

- (1) $[\text{Co(H}_2\text{O)}_6]^{2+}$ (2) $[\text{Co(NH}_3)_6]^{3+}$ (3) $[\text{Co(CN)}_6]^{3-}$ (4) $[\text{Co(C}_2\text{O}_4)_3]^{3-}$

[JEE (Main-Online) 2013]

Q.34 The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1 , L_2 , L_3 and L_4 absorb wavelengths in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is :

- (1) $\text{L}_1 < \text{L}_3 < \text{L}_2 < \text{L}_4$ (2) $\text{L}_3 < \text{L}_2 < \text{L}_4 < \text{L}_1$
 (3) $\text{L}_1 < \text{L}_2 < \text{L}_4 < \text{L}_3$ (4) $\text{L}_4 < \text{L}_3 < \text{L}_2 < \text{L}_1$

[JEE (Main-Offline) 2014]

Q.35 The equation which is balanced and represents the correct product(s) is : [JEE (Main-Offline) 2014]

- (1) $[\text{CoCl(NH}_3)_5]^+ + 5\text{H}^+ \rightarrow \text{Co}^{2+} + 5\text{NH}_4^+ + \text{Cl}^-$
 (2) $[\text{Mg(H}_2\text{O)}_6]^{2+} + (\text{EDTA})^{4-} \xrightarrow{\text{excess NaOH}} [\text{Mg(EDTA)}]^{2+} + 6\text{H}_2\text{O}$
 (3) $\text{CuSO}_4 + 4\text{KCN} \rightarrow \text{K}_2[\text{Cu(CN)}_4] + \text{K}_2\text{SO}_4$
 (4) $\text{Li}_2\text{O} + 2\text{KCl} \rightarrow 2\text{LiCl} + \text{K}_2\text{O}$

Q.36 The correct statement about the magnetic properties of $[\text{Fe(CN)}_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ is : ($Z = 26$)

- (1) $[\text{Fe(CN)}_6]^{3-}$ is paramagnetic, $[\text{FeF}_6]^{3-}$ is diamagnetic.
 (2) $[\text{Fe(CN)}_6]^{3-}$ is diamagnetic, $[\text{FeF}_6]^{3-}$ is paramagnetic.
 (3) both are paramagnetic.
 (4) both are diamagnetic

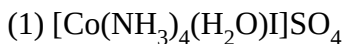
[JEE (Main-Online) 2014]

Q.37 An octahedral complex of Co^{3+} is diamagnetic. The hybridisation involved in the formation of the complex is :

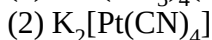
- (1) sp^3d^2 (2) dsp^2 (3) d^2sp^3 (4) dsp^3d

[JEE (Main-Online) 2014]

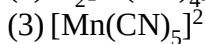
Q.38 Which of the following name formula combinations is not correct ?

Formula**Name**

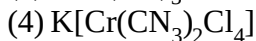
Tetraammine aquaiodo cobalt(III) sulphate



Potassium tetracyanoplatinate (II)



Pentacyanomagnate(II) ion

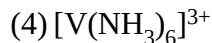
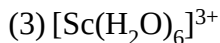
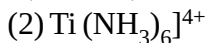
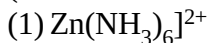


Potassium diammine tetrachlorochromate(III)

[JEE (Main-Online) 2014]

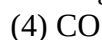
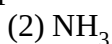
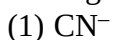
Q.39 Which one of the following complexes will most likely absorb visible light ?

(At nos. Sc = 21, Ti = 22, V = 23, Zn = 30)



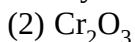
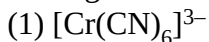
[JEE (Main-Online) 2014]

Q.40 Among the following species the one which causes the highest CFSE, Δ_o as a ligand is :



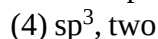
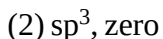
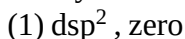
[JEE (Main-Online) 2014]

Q.41 Amongst the following identify the species with an atom in +6 oxidation state :



[JEE (Main-Online) 2014]

Q.42 Nickel ($Z = 28$) combines with a uninegative monodentate ligand to form a diamagnetic complex $[\text{NiL}_4]^{2-}$. The hybridisation involved and the number of unpaired electrons present in the complex are respectively



[JEE (Main-Online) 2014]

Q.43 An octahedral complex with molecular composition $\text{M}.5\text{NH}_3.\text{Cl}.\text{SO}_4$ has two isomers, A and B. The solution of A gives a white precipitate with AgNO_3 solution and the solution of B gives white precipitate with BaCl_2 solution. the type of isomerism exhibited by the complex is :

(1) Linkage isomerism

(2) Ionisation isomerism

(3) Coordinate isomerism

(4) Geometrical isomerism

[JEE (Main-Online) 2014]

Q.44 The number of geometric isomers that can exist for square planar $[\text{Pt}(\text{Cl})(\text{py})(\text{NH}_3)(\text{NH}_2\text{OH})]^+$ is (py = pyridine) :

(1) 4

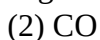
(2) 6

(3) 2

(4) 3

[JEE (Main-Offline) 2015]

Q.45 Which molecule/ion among the following cannot act as a ligand in complex compounds ?



[JEE (Main-Online) 2015]

- Q.46 The correct statement on the isomerism associated with the following complex ions,
 (a) $[\text{Ni}(\text{H}_2\text{O})_5\text{NH}_3]^{2+}$
 (b) $[\text{Ni}(\text{H}_2\text{O})_4(\text{NH}_3)_2]^{2+}$ and
 (c) $[\text{Ni}(\text{H}_2\text{O})_3(\text{NH}_3)_3]^{2+}$ is
 (1) (a) and (b) show geometrical and optical isomerism
 (2) (a) and (b) show only geometrical isomerism
 (3) (b) and (c) show geometrical and optical isomerism
 (4) (b) and (c) show only geometrical isomerism
[JEE (Main-Online) 2015]
- Q.47 Which of the following complex ions has electrons that are symmetrically filled in both t_{2g} and e_g orbitals?
 (1) $[\text{FeF}_6]^{3-}$ (2) $[\text{Mn}(\text{CN})_6]^{4-}$ (3) $[\text{Co}(\text{NH}_3)_6]^{2+}$ (4) $[\text{CoF}_6]^{3-}$
[JEE (Main-Online) 2015]
- Q.48 When concentrated HCl is added to an aqueous solution of CoCl_2 , its colour changes from reddish pink to deep blue. Which complex ion gives blue colour in this reaction?
 (1) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{CoCl}_6]^{3-}$ (3) $[\text{CoCl}_4]^{2-}$ (4) $[\text{CoCl}_6]^{4-}$
[JEE (Main-Online) 2015]
- Q.49 Which one of the following complexes shows optical isomerism?
 (1) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (2) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
 (3) $\text{cis}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (4) $\text{trans}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
 (en = ethylenediamine)
[JEE (Main-Offline) 2016]
- Q.50 The pair having the same magnetic moment is:
 [At. No.: Cr=24, Mn=25, Fe =26, Co =27]
 (1) $[\text{CoCl}_4]^{2-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$
 (3) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$
[JEE (Main-Offline) 2016]
- Q.51 Which one of the following complexes will consume more equivalents of aqueous solution of $\text{Ag}(\text{NO}_3)$?
 (1) $\text{Na}_3[\text{CrCl}_6]$ (2) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$ (3) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (4) $\text{Na}_2[\text{CrCl}_5(\text{H}_2\text{O})]$
[JEE (Main-Online) 2016]
- Q.52 Identify the correct trend given below:
[Atomic number : Ti = 22, Cr = 24 and Mo = 42]
 (1) Δ_o of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_o of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 (2) Δ_o of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_o of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 (3) Δ_o of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_o of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 (4) Δ_o of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_o of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
[JEE (Main-Online) 2016]
- Q.53 Which of the following is an example of homoleptic complex?
 (1) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (2) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
 (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ (4) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
[JEE (Main-Online) 2016]

- Q.54 The transition metal ions responsible for color in ruby and emerald are, respectively :
 (1) Cr^{3+} and Co^{3+} (2) Co^{3+} and Cr^{3+} (3) Co^{3+} and Co^{3+} (4) Cr^{3+} and Cr^{3+}
[JEE (Main-Online) 2016]
- Q.55 On treatment of 100 mL of 0.1 M solution of $\text{CoCl}_3 \cdot 6\text{H}_2\text{O}$ with excess AgNO_3 ; 1.2×10^{22} ions are precipitated. The complex is :
 (1) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$ (2) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
 (3) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (4) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
[JEE (Main-Offline) 2017]
- Q.56 $[\text{Co}_2(\text{CO})_8]$ displays :
 (1) one Co-Co bond, six terminal CO and two bridging CO
 (2) one Co-Co bond, four terminal CO and four bridging CO
 (3) no Co-Co bond, six terminal CO and two bridging CO
 (4) no Co-Co bond, four terminal CO and four bridging CO
[JEE (Main-Online) 2017]
- Q.57 Consider the following reaction and statements:

$$[\text{Co}(\text{NH}_3)_4\text{Br}_2]^+ + \text{Br}^- \longrightarrow [\text{Co}(\text{NH}_3)_3\text{Br}_3] + \text{NH}_3$$

 (I) Two isomers are produced if the reactant complex ion is a cis-isomer.
 (II) Two isomers are produced if the reactant complex ion is a trans-isomer.
 (III) Only one isomer is produced if the reactant complex ion is a trans-isomer.
 (IV) Only one isomer is produced if the reactant complex ion is a cis-isomer.
 The correct statements are:
 (1) (III) and (IV) (2) (II) and (IV) (3) (I) and (II) (4) (I) and (III)
[JEE (Main-Offline) 2018]
- Q.58 The oxidation states of Cr in $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$, $[\text{Cr}(\text{C}_6\text{H}_6)_2]$, and $\text{K}_2[\text{Cr}(\text{CN})_2(\text{O})_2(\text{O}_2)(\text{NH}_3)]$ respectively are :
 (1) +3, 0 and +6 (2) +3, 0 and +4 (3) +3, +4 and +6 (4) +3, +2 and +4
[JEE (Main-Offline) 2018]
- Q.59 The correct order of spin-only magnetic moments among the following is :
 [Atomic number : Mn = 25, Co = 27, Ni = 28, Zn = 30]
 (1) $[\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
 (2) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
 (3) $[\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
 (4) $[\text{ZnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
[JEE (Main-Online) 2018]
- Q.60 The total number of possible isomers for square-planar $[\text{Pt}(\text{Cl})(\text{NO}_2)(\text{NO}_3)(\text{SCN})]^{2-}$ is :
 (1) 24 (2) 12 (3) 8 (4) 16
[JEE (Main-Online) 2018]
- Q.61 The **correct** combination is
 (1) $[\text{NiCl}_4]^{2-}$ paramagnetic ; $[\text{Ni}(\text{CO})_4]$ tetrahedral
 (2) $[\text{NiCl}_4]^{2-}$ square planar ; $[\text{Ni}(\text{CN})_4]^{2-}$ paramagnetic
 (3) $[\text{Ni}(\text{CN})_4]^{2-}$ tetrahedral ; $[\text{Ni}(\text{CO})_4]$ paramagnetic
 (4) $[\text{NiCl}_4]^{2-}$ diamagnetic ; $[\text{Ni}(\text{CO})_4]$ square planar
[JEE (Main-Online) 2018]

Q.62 In Wilkinson's catalyst, the hybridization of central metal ion and its shape are respectively :

- (1) sp^3d , trigonal bipyramidal (2) sp^3 , tetrahedral
(3) dsp^2 , square planar (4) d^2sp^3 , octahedral

[JEE (Main-Online) 2018]

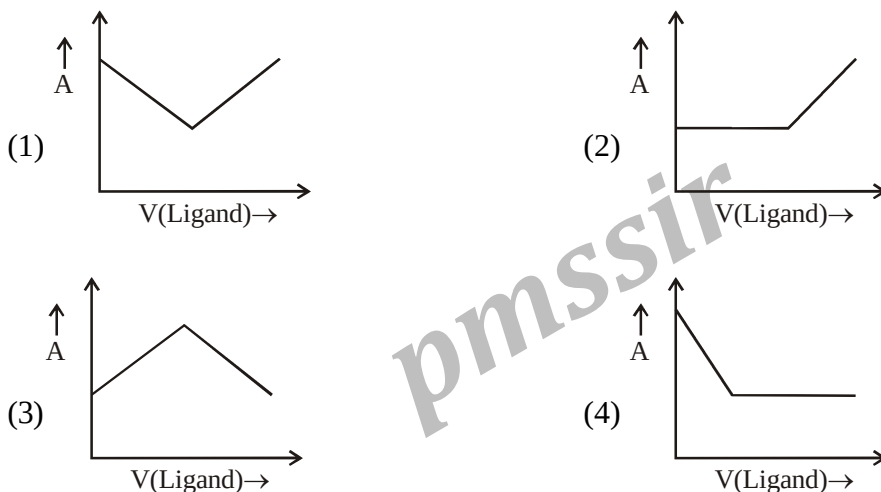
Q.63 Which of the following complexes will show geometrical isomerism ?

- (1) aquachlorobis(ethylenediamine)cobalt(II) chloride
(2) pentaquachlorochromium(III)chloride
(3) potassium amminetrichloroplatinate(II)
(4) potassium tris(oxalato)chromate(III)

[JEE (Main-Online) 2018]

Q.64 In a complexometric titration of metal ion with ligand

$M(\text{Metal ion}) + L(\text{Ligand}) \rightarrow C(\text{Complex})$ end point is estimated spectrophotometrically (through light absorption). If 'M' and 'C' do not absorb light and only 'L' absorbs, then the titration plot between absorbed light (A) versus volume of ligand 'L' (V) would look like :



[JEE (Main-Online) 2018]

Q.65 Two complexes $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (A) and $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (B) are violet and yellow coloured, respectively. The incorrect statement regarding them is :

- (1) Both are paramagnetic with three unpaired electrons.
(2) Both absorb energies corresponding to their complementary colors.
(3) Δ_0 value for (A) is less than that of (B).
(4) Δ_0 values of (A) and (B) are calculated from the energies of violet and yellow light, respectively.

[JEE (Main-Online) 2019]

Q.66 The complex that has highest crystal field splitting energy (Δ), is :

- (1) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ (2) $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]\text{Cl}_3$
(3) $\text{K}_3[\text{Co}(\text{CN})_6]$ (4) $\text{K}_2[\text{CoCl}_4]$

[JEE (Main-Online) 2019]

- Q.67 Homoleptic octahedral complexes of a metal ion M^{3+} with three monodentate ligands L_1 , L_2 and L_3 absorb wavelengths in the region of green, blue and red respectively. The increasing order of the ligand strength is :
 (1) $L_3 < L_1 < L_2$ (2) $L_2 < L_1 < L_3$ (3) $L_3 < L_2 < L_1$ (4) $L_1 < L_2 < L_3$
[JEE (Main-Online) 2019]
- Q.68 The total number of isomers for a square planar complex $[M(F)(Cl)(SCN)(NO_2)]$ is :
 (1) 16 (2) 4 (3) 8 (4) 12
[JEE (Main-Online) 2019]
- Q.69 Wilkinson catalyst is :
 (1) $[(Et_3P)_3RhCl]$ (2) $[(Ph_3P)_3RhCl]$ ($Et = C_2H_5$)
 (3) $[(Ph_3P)_3IrCl]$ (4) $[(Et_3P)_3IrCl]$
[JEE (Main-Online) 2019]
- Q.70 A reaction of cobalt (III) chloride and ethylenediamine in a 1 : 2 mole ratio generates two isomeric products A (violet coloured) and B (green coloured). A can show optical activity, but, B is optically inactive. What type of isomers does A and B represent?
 (1) Linkage isomers (2) Geometrical isomers
 (3) Ionisation isomers (4) Coordination isomers
[JEE (Main-Online) 2019]
- Q.71 The difference in the number of unpaired electrons of a metal ion in its high-spin and low-spin octahedral complexes is two. the metal ion is :
 (1) Co^{2+} (2) Fe^{2+} (3) Ni^{2+} (4) Mn^{2+}
[JEE (Main-Online) 2019]
- Q.72 The coordination number of Th in $K_4[Th(C_2O_4)_4(OH_2)_2]$ is: ($C_2O_4^{2-} = \text{Oxalato}$)
 (1) 14 (2) 6 (3) 8 (4) 10
[JEE (Main-Online) 2019]
- Q.73 The metal d-orbitals that are directly facing the ligands in $K_3[Co(CN)_6]$ are:
 (1) d_{xy} , d_{xz} and d_{yz} (2) d_{xz} , d_{yz} and d_{z^2} (3) d_{xy} and $d_{x^2-y^2}$ (4) $d_{x^2-y^2}$ and d_{z^2}
[JEE (Main-Online) 2019]
- Q.74 The pair of metal ions that can give a spin only magnetic moment of 3.9 BM for the complex $[M(H_2O)_6]Cl_2$ is:
 (1) V^{2+} and Fe^{2+} (2) Co^{2+} and Fe^{2+} (3) V^{2+} and Co^{2+} (4) Cr^{2+} and Mn^{2+}
[JEE (Main-Online) 2019]
- Q.75 $Mn_2(CO)_{10}$ is an organometallic compound due to the presence of :
 (1) Mn – C bond (2) Mn – O bond (3) Mn – Mn bond (4) C – O bond
[JEE (Main-Online) 2019]

Q.76 The magnetic moment of an octahedral homoleptic Mn(II) complex is 5.9 BM. The suitable ligand for this complex is :

- (1) CO (2) ethylenediamine (3) NCS^- (4) CN^-

[JEE (Main-Online) 2019]

Q.77 For octahedral Mn(II) and tetrahedral Ni(II) complexes, consider the following statements :

- (I) both the complexes can be high spin
 (II) Ni(II) complex can very rarely be low spin.
 (III) with strong field ligands, Mn(II) complexes can be low spin.
 (IV) aqueous solution of Mn(II) ions is yellow in color.

The correct statements are :

- (1) (I), (III) and (IV) only (2) (II), (III) and (IV) only
 (3) (I), (II) and (III) only (4) (I) and (II) only

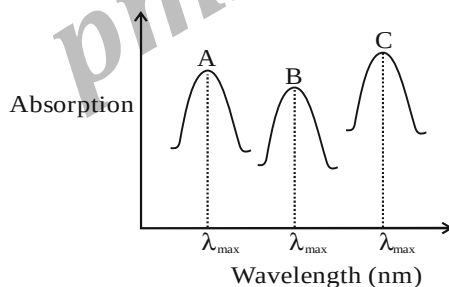
[JEE (Main-Online) 2020]

Q.78 Consider that a d^6 metal ion (M^{2+}) forms a complex with aqua ligands, and the spin only magnetic moment of the complex is 4.90 BM. The geometry and the crystal field stabilization energy of the complex is :

- (1) tetrahedral and $-1.6 \Delta_t + 1P$ (2) tetrahedral and $-0.6 \Delta_t$
 (3) octahedral and $-1.6 \Delta_0$ (4) octahedral and $-2.4 \Delta_0 + 2P$

[JEE (Main-Online) 2020]

Q.79 Simplified absorption spectra of three complexes (i), (ii) and (iii) of M^{n+} ion are provided below; their $\lambda_{\max}$ values are marked as A, B and C respectively. The correct match between the complexes and their $\lambda_{\max}$ values is :



- (i) $[\text{M}(\text{NCS})_6]^{(-6+n)}$ (ii) $[\text{MF}_6]^{(-6+n)}$ (iii) $[\text{M}(\text{NH}_3)_6]^{n+}$
 (1) A-(ii), B-(i), C-(iii) (2) A-(iii), B-(i), C-(ii)
 (3) A-(ii), B-(iii), C-(i) (4) A-(i), B-(ii), C-(iii)

[JEE (Main-Online) 2020]

Q.80 The one that is not expected to show isomerism is :

- (1) $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ (2) $[\text{Ni}(\text{NH}_3)_2\text{Cl}_2]$
 (3) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (4) $[\text{Ni}(\text{en})_3]^{2+}$

[JEE (Main-Online) 2020]

Q.81 The complex that can show optical activity is :

- (1) $\text{trans-}[\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^-$ (2) $\text{cis-}[\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^-$
 (3) $\text{cis-}[\text{CrCl}_2(\text{ox})_2]^{3-}$ (ox = oxalate) (4) $\text{trans-}[\text{Cr}(\text{Cl}_2)(\text{ox})_2]^{3-}$

[JEE (Main-Online) 2020]

- Q.82 The electronic spectrum of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ shows a single broad peak with a maximum at $20,300 \text{ cm}^{-1}$. The crystal field stabilization energy (CFSE) of the complex ion, in kJ mol^{-1} , is :
 $(1 \text{ kJ mol}^{-1} = 83.7 \text{ cm}^{-1})$
 (1) 242.5 (2) 83.7 (3) 145.5 (4) 97
[JEE (Main-Online) 2020]
- Q.83 The d-electron configuration of $[\text{Ru}(\text{en})_3]\text{Cl}_2$ and $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_2$, respectively are :
 (1) $t_{2g}^4 e_g^2$ and $t_{2g}^6 e_g^0$ (2) $t_{2g}^6 e_g^0$ and $t_{2g}^6 e_g^0$ (3) $t_{2g}^6 e_g^0$ and $t_{2g}^4 e_g^2$ (4) $t_{2g}^4 e_g^2$ and $t_{2g}^4 e_g^2$
[JEE (Main-Online) 2020]
- Q.84 Complex A has a composition of $\text{H}_{12}\text{O}_6\text{Cl}_3\text{Cr}$. If the complex on treatment with conc. H_2SO_4 loses 13.5% of its original mass, the correct molecular formula of A is :
 [Given : atomic mass of Cr = 52 amu and Cl = 35 amu]
 (1) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (2) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
 (3) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ (4) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
[JEE (Main-Online) 2020]
- Q.85 The number of isomers possible for $[\text{Pt}(\text{en})(\text{NO}_2)_2]$ is:
 (1) 3 (2) 2 (3) 1 (4) 4
[JEE (Main-Online) 2020]
- Q.86 The pair in which both the species have the same magnetic moment (spin only) is :
 (1) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})]^{2+}$ (2) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$
 (3) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Co}(\text{OH})_4]^{2-}$ and $[\text{Fe}(\text{NH}_3)_6]^{2+}$
[JEE (Main-Online) 2020]
- Q.87 The Crystal Field Stabilization Energy (CFSE) of $[\text{CoF}_3(\text{H}_2\text{O})_3](\Delta_0 < P)$ is :-
 (1) $-0.8 \Delta_0$ (2) $-0.4 \Delta_0 + P$ (3) $-0.8 \Delta_0 + 2P$ (4) $-0.4 \Delta_0$
[JEE (Main-Online) 2020]
- Q.88 The one that can exhibit highest paramagnetic behaviour among the following is :-
 gly = glycinato; bpy = 2, 2'-bipyridine
 (1) $[\text{Pd}(\text{gly})_2]$ (2) $[\text{Ti}(\text{NH}_3)_6]^{3+}$
 (3) $[\text{Co}(\text{OX})_2(\text{OH})_2]^{-}(\Delta_0 > P)$ (4) $[\text{Fe}(\text{en})(\text{bpy})(\text{NH}_3)_2]^{2+}$
[JEE (Main-Online) 2020]
- Q.89 The molecule in which hybrid MOs involve only one d-orbital of the central atom is :-
 (1) $[\text{Ni}(\text{CN})_4]^{2-}$ (2) $[\text{CrF}_6]^{3-}$ (3) BrF_5 (4) XeF_4
[JEE (Main-Online) 2020]
- Q.90 The values of the crystal field stabilization energies for a high spin d^6 metal ion in octahedral and tetrahedral fields, respectively, are :
 (1) $-0.4 \Delta_0$ and $-0.27 \Delta_t$ (2) $-1.6 \Delta_0$ and $-0.4 \Delta_t$
 (3) $-0.4 \Delta_0$ and $-0.6 \Delta_t$ (4) $-2.4 \Delta_0$ and $-0.6 \Delta_t$
[JEE (Main-Online) 2020]

- Q.91 The total number of coordination sites in ethylenediaminetetraacetate (EDTA^{4-}) is
[JEE (Main-Online) 2020]
- Q.92 Consider the complex ions,
 $\text{trans-}[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (A) and $\text{cis-}[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (B). The correct statement regarding them is :
(1) both (A) and (B) can be optically active
(2) both (A) and (B) cannot be optically active
(3) (A) can be optically active, but (B) cannot be optically active
(4) (A) cannot be optically active, but (B) can be optically active
[JEE (Main-Online) 2020]
- Q.93 Considering that $\Delta_0 > P$, the magnetic moment (in BM) of $[\text{Ru}(\text{H}_2\text{O})_6]^{2+}$ would be _____.
[JEE (Main-Online) 2020]
- Q.94 The species that has a spin only magnetic moment of 5.9 BM, is -
(1) $\text{Ni}(\text{CO})_4(\text{T}_d)$ (2) $[\text{MnBr}_4]^{2-}(\text{T}_d)$
(3) $[\text{NiCl}_4]^{2-}(\text{T}_d)$ (4) $[\text{Ni}(\text{CN})_4]^{2-}$ (square planar)
[JEE (Main-Online) 2020]
- Q.95 For a d^4 metal ion in an octahedral field, the correct electronic configuration is :
(1) $t_{2g}^4 e_g^0$ when $\Delta_0 < P$ (2) $e_g^2 t_{2g}^2$ when $\Delta_0 < P$
(3) $t_{2g}^3 e_g^1$ when $\Delta_0 < P$ (4) $t_{2g}^3 e_g^1$ when $\Delta_0 > P$
[JEE (Main-Online) 2020]
- Q.96 The correct order of the calculated spin-only magnetic moments of complexes (A) to (D) is
(A) $\text{Ni}(\text{CO})_4$
(B) $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$
(C) $\text{Na}_2[\text{Ni}(\text{CN})_4]$
(D) $\text{PdCl}_2(\text{PPh}_3)_2$
(1) (A) $\approx$ (C) < (B) $\approx$ (D) (2) (C) $\approx$ (D) < (B) < (A)
(3) (C) < (D) < (B) < (A) (4) (A) $\approx$ (C) $\approx$ (D) < (B)
[JEE (Main-Online) 2020]
- Q.97 The theory that can completely/properly explain the nature of bonding in $[\text{Ni}(\text{CO})_4]$ is
(1) Crystal field theory (2) Werner's theory
(3) Molecular orbital theory (4) Valence bond theory
[JEE (Main-Online) 2020]
- Q.98 The IUPAC name of the complex $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)] \text{Cl}$ is
(1) Bisammine (methanamine) chlorido platinum (II) chloride
(2) Diammine (methanamine) chlorido platinum (II) chloride
(3) Diammine chlorido (methanamine) platinum (II) chloride
(4) Diammine chlorido (aminomethane) platinum (II) chloride
[JEE (Main-Online) 2020]

Q.99 The number of possible optical isomers for the complexes MA_2B_2 with sp^3 and dsp^2 hybridized metal atom, respectively, is

Note: A and B are unidentate neutral and unidentate monoanionic ligands, respectively.

- (1) 0 and 1 (2) 0 and 0 (3) 2 and 2 (4) 0 and 2

[JEE (Main-Online) 2020]

Q.100 Among the statements (a)-(d), the incorrect ones as

- (a) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments
 (b) When $\Delta_0 < P$, the d-electron configuration of Co(III) in an octahedral complex is $t_{eg}^4 e_g^2$
 (c) Wavelength of light absorbed by $[Co(en)_3]^{3+}$ is lower than that of $[CoF_6]^{3-}$.
 (d) If the D_0 for an octahedral complex of Co(III) is $18,000\text{ cm}^{-1}$, the Δ_t for its tetrahedral complex with the same ligand will be $16,000\text{ cm}^{-1}$.

- (1) (b) and (c) only (2) (a) and (b) only
 (3) (a) and (d) only (4) (c) and (d) only

[JEE (Main-Online) 2020]

Q.101 The complex that can show fac- and mer- isomers is

- (1) $[Co(NH_3)_3(NO_2)_3]$ (2) $[Pt(NH_3)_2Cl_2]$
 (3) $[Co(NH_3)_4Cl_2]^+$ (4) $[CoCl_2(en)_2]$

[JEE (Main-Online) 2020]

Q.102 Among (a) - (d), the complexes that can display geometrical isomerism are

- (a) $[Pt(NH_3)_3Cl]^+$
 (b) $[Pt(NH_3)Cl_5]^-$
 (c) $[Pt(NH_3)_4Cl(NO_2)]$
 (d) $[Pt(NH_3)_4ClBr]^{2+}$
 (1) (d) and (a) (2) (c) and (d) (3) (a) and (b) (4) (b) and (c)

[JEE (Main-Online) 2020]

Q.103 Complexes (ML_5) of metals Ni and Fe have ideal square pyramidal and bipyramidal geometries, respectively. The sum of the 90° , 120° and 180° L-M-L angles in the two complexes is _____.

[JEE (Main-Online) 2020]

Q.104 Complex X of composition $Cr(H_2O)_6Cl_n$ has a spin only magnetic moment of 3.83 BM. It reacts with $AgNO_3$ and shows geometrical isomerism. The IUPAC nomenclature of X is

- (1) Dichloridotetraaqua chromium(IV) chloride dihydrate
 (2) Tetraaquadichlorido chromium(IV) chloride dihydrate
 (3) Tetraaquadichlorido chromimum(III) chloride dihydrate
 (4) Hexaaqua chromimum(III) chloride

[JEE (Main-Online) 2020]

Q.105 $[\text{Pd}(\text{F})(\text{Cl})(\text{Br})(\text{I})]^{2-}$ has n number of geometrical isomers. Then, the spin-only magnetic moment and crystal field stabilisation energy [CFSE] of $[\text{Fe}(\text{CN})_6]^{n-6}$, respectively, are

[Note: Ignore the pairing energy]

(1) 5.92 BM and 0

(2) 0 BM and $-2.4 \Delta_0$

(3) 1.73 BM and $-2.0 \Delta_0$

(4) 2.84 BM and $-1.6 \Delta_0$

[JEE (Main-Online) 2020]

Q.106 The correct order of the spin-only magnetic moments of the following complexes is

(I) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Br}_2$

(II) $\text{Na}_4[\text{Fe}(\text{CN})]$

(III) $\text{Na}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ ($\Delta_0 > P$)

(IV) $(\text{Et}_4\text{N}_2)[\text{CoCl}_4]$

(1) (III) > (I) > (II) > (IV)

(2) (I) > (IV) > (III) > (II)

(3) (II) $\approx$ (I) > (IV) > (III)

(4) (III) > (I) > (IV) > (II)

[JEE (Main-Online) 2020]

Q.107 The isomer(s) of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ that has/have a Cl–Co–Cl angle of 90° , is/are

(1) Meridional and trans

(2) cis and trans

(3) trans only

(4) cis only

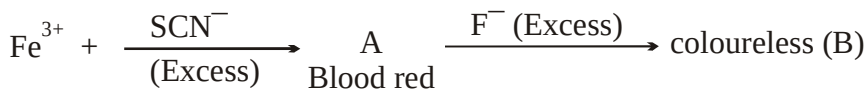
[JEE (Main-Online) 2020]

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[JEE ADVANCED]

- Q.1 Draw the structures of $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $\text{Ni}(\text{CO})_4$. Write the hybridisation of atomic orbitals of the transition metal in each case. **[JEE 2000]**
- Q.2 The complex ion which has no 'd' electrons in the central metal atom is : **[JEE 2001]**
 [At No. Cr = 24, Mn = 25, Fe = 26, Co = 27]
 (A) $[\text{MnO}_4]^-$ (B) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (C) $[\text{Fe}(\text{CN})_6]^{3-}$ (D) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
- Q.3 The correct order of hybridisation of the central atom in the following species. **[JEE 2001]**
 NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 is [At No. Pt = 78]
 (A) $\text{dsp}^2, \text{sp}^3\text{d}, \text{sp}^2$ and sp^3 (B) $\text{sp}^3, \text{dsp}^2$, sp^3d , sp^2
 (C) $\text{dsp}^2, \text{sp}^2, \text{sp}^3$ and sp^3d (D) dsp^2 , sp^3, sp^2 and sp^3d
- Q.4 Deduce the structures of $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ considering the hybridisation. Calculate the magnetic moment (spin only) of the species. **[JEE 2002]**
- Q.5 Write the IUPAC name of the given complex along with its a hybridisation and structure. **[JEE 2003]**
 $\text{K}_2 [\text{Cr}(\text{NO}) (\text{NH}_3) (\text{CN})_4]$, $\mu = 1.73$
- Q.6 When dimethyl glyoxime is added to the aqueous solution of nickel (II) chloride in presence of dilute ammonia solution, a rosy red coloured ppt is obtained. **[JEE 2004]**
 (a) Draw the structure of rosy red substance.
 (b) Write the oxidation state of nickel in the substance and hybridisation.
 (c) State whether the substance is paramagnetic or diamagnetic.
- Q.7 The species having tetrahedral shape is : **[JEE 2004]**
 (A) $[\text{PdCl}_4]^{2-}$ (B) $[\text{Ni}(\text{CN})_4]^{2-}$ (C) $[\text{Pd}(\text{CN})_4]^{2-}$ (D) $[\text{NiCl}_4]^{2-}$
- Q.8 The pair of compounds having metals in their highest oxidation state is **[JEE 2004]**
 (A) MnO_2 , FeCl_3 (B) $[\text{MnO}_4]^-$, CrO_2Cl_2
 (C) $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{CN})_3]$ (D) $[\text{NiCl}_4]^{2-}$, $[\text{CoCl}_4]^-$
- Q.9 The compound having tetrahedral geometry is **[JEE 2004]**
 (A) $[\text{Ni}(\text{CN})_4]^{2-}$ (B) $[\text{Pd}(\text{CN})_4]^{2-}$ (C) $[\text{PdCl}_4]^{2-}$ (D) $[\text{NiCl}_4]^{2-}$
- Q.10 Spin only magnetic moment of the compound $\text{Hg} [\text{Co}(\text{SCN})_4]$ is **[JEE 2004]**
 (A) $\sqrt{3}$ (B) $\sqrt{15}$ (C) $\sqrt{24}$ (D) $\sqrt{8}$
- Q.11 Which of the following pair is expected to exhibit same colour in solution? **[JEE 2005]**
 (A) VOCl_2 ; FeCl_2 (B) CuCl_2 ; VOCl_2 (C) MnCl_2 ; FeCl_2 (D) FeCl_2 ; CuCl_2
- Q.12 Which type of isomerism is shown by $\text{Co}(\text{NH}_3)_4\text{Br}_2\text{Cl}$? **[JEE 2005]**
 (A) Geometrical and Ionisation (B) Optical and Ionisation
 (C) Geometrical and Optical (D) Geometrical only

Q.13 In the given reaction sequence, identify (A) and (B) [JEE 2005]

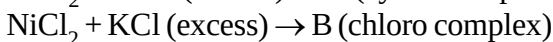
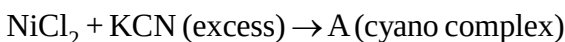


- (a) Write the IUPAC name of (A) and (B)
(b) Find out the spin only magnetic moment of B.

Question No. 14 to 16 (3 questions)

[JEE 2006]

The coordination number of Ni^{2+} is 4.



Q.14 The IUPAC name of A and B are

- (A) Potassium tetracyanonickelate (II), potassium tetrachloridonickelate (II)
(B) Tetracyanidopotassiumnickelate (II), tetrachloridopotassiumnickelate (II)
(C) Tetracyanonickel (II), tetrachloridonickel (II)
(D) Potassium tetracyanonickel (II), potassium tetrachloridonickel (II)

Q.15 Predict the magnetic nature of A and B.

- (A) Both are diamagnetic.
(B) A is diamagnetic and B is paramagnetic with one unpaired electron.
(C) A is diamagnetic and B is paramagnetic with two unpaired electrons.
(D) Both are paramagnetic.

Q.16 The hybridization of A and B are

- (A) dsp^2 , sp^3 (B) sp^3 , sp^3 (C) dsp^2 , dsp^2 (D) sp^3d^2 , d^2sp^3

Q.17 If the bond length of CO bond in carbon monoxide is $1.128\text{\AA}$, then what is the value of CO bond length in $\text{Fe}(\text{CO})_5$?

- (A) $1.15\text{\AA}$ (B) $1.128\text{\AA}$ (C) $1.72\text{\AA}$ (D) $1.118\text{\AA}$ [JEE 2006]

Q.18 Among the following metal carbonyls, the C–O bond order is lowest in

[JEE 2007]

- (A) $[\text{Mn}(\text{CO})_6]^+$ (B) $[\text{Fe}(\text{CO})_5]$ (C) $[\text{Cr}(\text{CO})_6]$ (D) $[\text{V}(\text{CO})_6]^-$

Q.19 Match the complexes in Column I with their properties listed in Column II.

[JEE 2007]

Column I

Column II

- | | |
|---|---------------------------------------|
| (A) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_2$ | (P) Geometrical isomers |
| (B) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ | (Q) Paramagnetic |
| (C) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}$ | (R) Diamagnetic |
| (D) $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$ | (S) Metal ion with 2+ oxidation state |

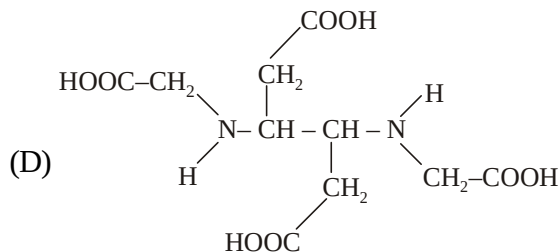
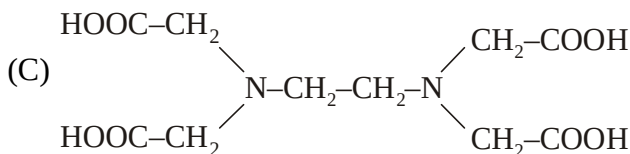
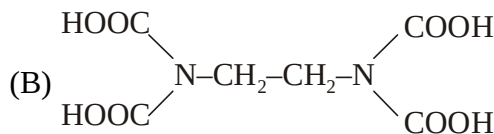
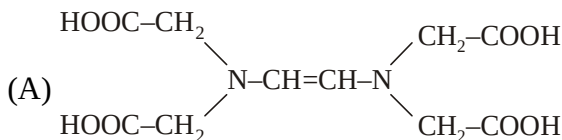
Q.20 Among the following, the coloured compound is

[JEE 2008]

- (A) CuCl (B) $\text{K}_3[\text{Cu}(\text{CN})_4]$ (C) CuF_2 (D) $[\text{Cu}(\text{CH}_3\text{CN})_4]\text{BF}_4$

- Q.21 The IUPAC name of $[\text{Ni}(\text{NH}_3)_4][\text{NiCl}_4]$ is [JEE 2008]
(A) Tetrachloronickel (II)-tetraamminenickel (II)
(B) Tetraamminenickel (II)-tetrachloronickel (II)
(C) Tetraamminenickel (II)-tetrachloronickelate (II)
(D) Tetrachloronickel (II)-tetraamminenickelate (0)
- Q.22 Both $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic. The hybridisations of nickel in these complexes, respectively, are [JEE 2008]
(A) sp^3 , sp^3 (B) sp^3 , dsp^2 (C) dsp^2 , sp^3 (D) dsp^2 , dsp^2
- Q.23 STATEMENT-1 : The geometrical isomers of the complex $[\text{M}(\text{NH}_3)_4\text{Cl}_2]$ are optically inactive.
STATEMENT-2 : Both geometrical isomers of the complex $[\text{M}(\text{NH}_3)_4\text{Cl}_2]$ possess axis of symmetry.
(A) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is a correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True [JEE 2008]
- Q.24 STATEMENT-1 : $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ is paramagnetic [JEE 2008]
STATEMENT-2 : The Fe in $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ has three unpaired electrons.
(A) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is a correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True
- Q.25 The spin only magnetic moment value (in Bohr magneton units) of $\text{Cr}(\text{CO})_6$ is [JEE 2009]
(A) 0 (B) 2.84 (C) 4.90 (D) 5.92
- Q.26 The compound(s) that exhibit(s) geometrical isomerism is (are) : [JEE 2009]
(A) $[\text{Pt}(\text{en})\text{Cl}_2]$ (B) $[\text{Pt}(\text{en})_2]\text{Cl}_2$ (C) $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Cl}_2$ (D) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
- Q.27 The number of water molecule(s) directly bonded to the metal centre in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is [JEE 2009]
- Q.28 The ionization isomer of $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}(\text{NO}_2)]\text{Cl}$ is [JEE 2010]
(A) $[\text{Cr}(\text{H}_2\text{O})_4(\text{O}_2\text{N})]\text{Cl}_2$ (B) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2](\text{NO}_2)$
(C) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}(\text{ONO})]\text{Cl}$ (D) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2(\text{NO}_2)] \cdot \text{H}_2\text{O}$

Q.29 The correct structure of ethylenediaminetetraacetic acid (EDTA) is [JEE 2010]



Q.30 The complex showing a spin-only magnetic moment of 2.82 B.M. is [JEE 2010]

- (A) $\text{Ni}(\text{CO})_4$ (B) $[\text{NiCl}_4]^{2-}$ (C) $\text{Ni}(\text{PPh}_3)_4$ (D) $[\text{Ni}(\text{CN})_4]^{2-}$

Q.31 Total number of geometrical isomers for the complex $[\text{RhCl}(\text{CO})(\text{PPh}_3)(\text{NH}_3)]$ is [JEE 2010]

Q.32 Geometrical shapes of the complexes formed by the reaction of Ni^{2+} with Cl^- , CN^- and H_2O , respectively are [JEE 2011]

- (A) octahedral, tetrahedral and square planar
(B) tetrahedral, square planar and octahedral
(C) square planar, tetrahedral and octahedral
(D) octahedral, square planar and tetrahedral

Q.33 Among the following complexes (K–P) [JEE 2011]

$\text{K}_3[\text{Fe}(\text{CN})_6]$ (K), $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (L), $\text{Na}_3[\text{Co}(\text{oxalate})_3]$ (M), $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$ (N), $\text{K}_2[\text{Pt}(\text{CN})_4]$ (O) and $[\text{Zn}(\text{H}_2\text{O})_6](\text{NO}_3)_2$ (P)
the diamagnetic complexes are

- (A) K, L, M, N (B) K, M, O, P (C) L, M, O, P (D) L, M, N, O

Q.34 As per IUPAC nomenclature, the name of the complex $[\text{Co}(\text{H}_2\text{O})_4(\text{NH}_3)_2]\text{Cl}_3$ is [JEE 2012]

- (A) Tetraaquadiammincobalt (III) chloride (B) Tetraaquadiammincobalt (III) chloride
(C) Diaminetetraaquacobalt (III) chloride (D) Diamminetetraaquacobalt (III) chloride

Q.35 The colour of light absorbed by an aqueous solution of CuSO_4 is [JEE 2012]

- (A) Orange-red (B) Blue-green (C) Yellow (D) Violet

Q.36 $\text{NiCl}_2\{\text{P}(\text{C}_2\text{H}_5)_2(\text{C}_6\text{H}_5)\}_2$ exhibits temperature dependent magnetic behaviour (paramagnetic/diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively [JEE 2012]

- (A) tetrahedral and tetrahedral (B) square planar and square planar
(C) tetrahedral and square planar (D) square planar and tetrahedral

- Q.37 Consider the following complex ions, P, Q and R.
 $P = [FeF_6]^{3-}$, $Q = [V(H_2O)_6]^{2+}$ and $R = [Fe(H_2O)_6]^{2+}$
 the correct order of the complex ions, according to their spin only magnetic moment values (in B.M.) is:
 (A) $R < Q < P$ (B) $Q < R < P$ (C) $R < P < Q$ (D) $Q < P < R$ [JEE (Advanced) 2013]
- Q.38 The pair(s) of coordination complexes/ ions exhibiting the same kind of isomerism is(are)
 (A) $[Cr(NH_3)_5Cl]Cl_2$ and $[Cr(NH_3)_4Cl_2]Cl$ [JEE (Advanced) 2013]
 (B) $[Co(NH_3)_4Cl_2]^+$ and $[Pt(NH_3)_2(H_2O)Cl]^+$
 (C) $[CoBr_2Cl_2]^{2-}$ and $[PtBr_2Cl_2]^{2-}$
 (D) $[Pt(NH_3)_3(NO_3)]Cl$ and $[Pt(NH_3)_3Cl]Br$
- Q.39 $EDTA^{4-}$ is ethylenediaminetetraacetate ion. The total number of N–Co–O bond angles in $[Co(EDTA)]^{1-}$ complex ion is [JEE (Advanced) 2013]
- Q.40 A list of species having the formula XZ_4 is given below. [JEE (Advanced) 2014]
 XeF_4 , SF_4 , SiF_4 , BF_4^- , BrF_4^- , $[Cu(NH_3)_4]^{2+}$, $[FeCl_4]^{2-}$, $[CoCl_4]^{2-}$ and $[PtCl_4]^{2-}$
 Defining shape on the basis of the location of X and Z atoms, the total number of species having a square planar shape is
- Q.41 Match each coordination compound in List-I with an appropriate pair of characteristics from List-II and select the correct answer using the code given below the lists. [JEE (Advanced) 2014]
 $\{en = H_2NCH_2CH_2NH_2; \text{atomic numbers : Ti = 22; Cr = 24; Co = 27; Pt = 78}\}$
- | List-I | | List-II | |
|--------|----------------------------|---------|---|
| P. | $[Cr(NH_3)_4Cl_2]Cl$ | 1. | Paramagnetic and exhibits ionisation isomerism. |
| Q. | $[Ti(H_2O)_5Cl](NO_3)_2$ | 2. | Diamagnetic and exhibits cis-trans isomerism. |
| R. | $[Pt(en)(NH_3)Cl]NO_3$ | 3. | Paramagnetic and exhibits cis-trans isomerism. |
| S. | $[Co(NH_3)_4(NO_3)_2]NO_3$ | 4. | Diamagnetic and exhibits ionisation isomerism. |
- Code: P Q R S
- | | | | | |
|-----|---|---|---|---|
| (A) | 4 | 2 | 3 | 1 |
| (B) | 3 | 1 | 4 | 2 |
| (C) | 2 | 1 | 3 | 4 |
| (D) | 1 | 3 | 4 | 2 |
- Q.42 For the octahedral complexes of Fe^{3+} in SCN^- (thiocyanato - S) and in CN^- ligand environments, the difference between the spin only magnetic moments in Bohr magnetons (when approximated to the nearest integer) is :
 [Atomic number of Fe = 26] [JEE (Advanced) 2015]
- Q.43 In the complex acetyl bromidodicarbonylbis(triethylphosphine)iron(II), the number of Fe–C bond(s) is [JEE (Advanced) 2015]
- Q.44 Among the complex ions, $[Co(NH_2 - CH_2 - CH_2 - NH_2)_2Cl_2]^+$, $[CrCl_2(C_2O_4)_2]^{3-}$, $[Fe(H_2O)_4(OH)_2]^+$, $[Fe(NH_3)_2(CN)_4]^-$, $[Co(NH_2 - CH_2 - CH_2 - NH_2)_2(NH_3)Cl]^{2+}$ and $[Co(NH_3)_4(H_2O)Cl]^{2+}$, the number of complex ion(s) that show(s) *cis-trans* isomerism is : [JEE (Advanced) 2015]

- Q.45 Among $[\text{NiCO}]_4$, $[\text{NiCl}_4]^{2-}$, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$, $\text{Na}_3[\text{CoF}_6]$, Na_2O_2 and CsO_2 , the total number of paramagnetic compounds is
 (A) 2 (B) 3 (C) 4 (D) 5
[JEE (Advanced) 2016]
- Q.46 The number of geometric isomers possible for the complex $[\text{CoL}_2\text{Cl}_2]^-$ ($\text{L} = \text{H}_2\text{NCH}_2\text{CH}_2\text{O}^-$) is
[JEE (Advanced) 2016]
- Q.47 The geometries of the ammonia complexes of Ni^{2+} , Pt^{2+} and Zn^{2+} , respectively, are
 (A) octahedral, square planar and tetrahedral
 (B) square planar, octahedral and tetrahedral
 (C) tetrahedral, square planar and octahedral
 (D) octahedral, tetrahedral and square planar
[JEE (Advanced) 2016]
- Q.48 The correct statement(s) regarding the binary transition metal carbonyl compounds is (are)
 (Atomic numbers: Fe = 26, Ni = 28)
 (A) Total number of valence shell electrons at metal centre in $\text{Fe}(\text{CO})_5$ or $\text{Ni}(\text{CO})_4$ is 16
 (B) These are predominantly low spin in nature
 (C) Metal–carbon bond strengthens when the oxidation state of the metal is lowered
 (D) The carbonyl C–O bond weakens when the oxidation state of the metal is increased
[JEE (Advanced) 2018]
- Q.49 Among the species given below, the total number of diamagnetic species is ____.
 H atom, NO_2 monomer, O_2^- (superoxide), dimeric sulphur in vapour phase, Mn_3O_4 , $(\text{NH}_4)_2[\text{FeCl}_4]$, $(\text{NH}_4)_2[\text{NiCl}_4]$, K_2MnO_4 , K_2CrO_4
[JEE (Advanced) 2018]
- Q.50 The ammonia prepared by treating ammonium sulphate with calcium hydroxide is completely used by $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ to form a stable coordination compound. Assume that both the reactions are 100% complete. If 1584 g of ammonium sulphate and 952 g of $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ are used in the preparation, the combined weight (in grams) of gypsum and the nickel-ammonia coordination compound thus produced is ____.
 (Atomic weights in g mol^{-1} : H = 1, N = 14, O = 16, S = 32, Cl = 35.5, Ca = 40, Ni = 59)
[JEE (Advanced) 2018]
- Q.51 The correct option(s) regarding the complex $[\text{Co}(\text{en})(\text{NH}_3)_3(\text{H}_2\text{O})]^{3+}$ ($\text{en} = \text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$) is (are)
 (A) It has two geometrical isomers
 (B) It will have three geometrical isomers if bidentate 'en' is replaced by two cyanide ligands
 (C) It is paramagnetic
 (D) It absorbs light at longer wavelength as compared to $[\text{Co}(\text{en})(\text{NH}_3)_4]^{3+}$
[JEE (Advanced) 2018]

Q.52 Match each set of hybrid orbitals from LIST-I with complex(es) given in LIST-II.

LIST-I

- P. dsp^2
 Q. sp^3
 R. sp^3d^2
 S. d^2sp^3

LIST-II

1. $[FeF_6]^{4-}$
 2. $[Ti(H_2O)_3Cl_3]$
 3. $[Cr(NH_3)_6]^{3+}$
 4. $[FeCl_4]^{2-}$
 5. $Ni(CO)_4$
 6. $[Ni(CN)_4]^{2-}$

[JEE (Advanced) 2018]

The correct option is

- (A) P $\rightarrow$ 5; Q $\rightarrow$ 4,6; R $\rightarrow$ 2,3; S $\rightarrow$ 1
 (B) P $\rightarrow$ 5,6; Q $\rightarrow$ 4; R $\rightarrow$ 3; S $\rightarrow$ 1, 2
 (C) P $\rightarrow$ 6; Q $\rightarrow$ 4,5; R $\rightarrow$ 1; S $\rightarrow$ 2, 3
 (D) P $\rightarrow$ 4,6; Q $\rightarrow$ 5,6; R $\rightarrow$ 1,2; S $\rightarrow$ 3

Q.53 Total number of cis N–Mn–Cl bond angles (that is, Mn–N and Mn–Cl bonds in cis positions) present in a molecule of cis-[Mn(en)₂Cl₂] complex is ____ (en = $NH_2CH_2CH_2NH_2$)

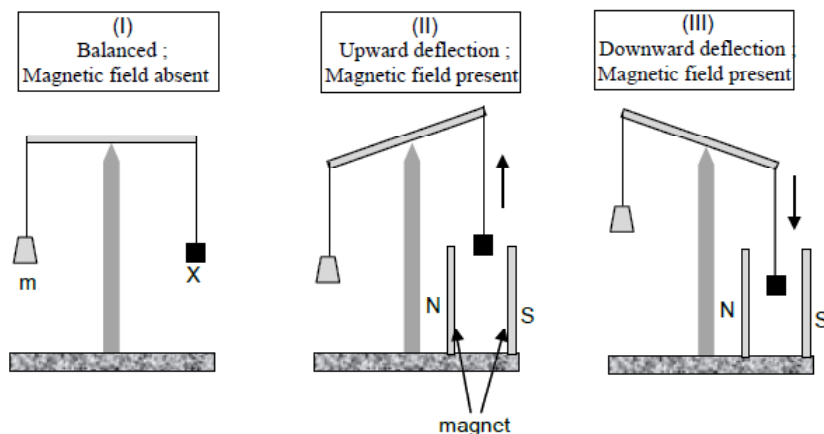
[JEE (Advanced) 2019]

Q.54 Choose the correct statement(s) among the following :

- (A) $[FeCl_4]^-$ has tetrahedral geometry.
 (B) $[Co(en)(NH_3)_2Cl_2]^+$ has 2 geometrical isomers.
 (C) $[FeCl_4]^-$ has higher spin-only magnetic moment than $[Co(en)(NH_3)_2Cl_2]^+$.
 (D) The cobalt ion in $[Co(en)(NH_3)_2Cl_2]^+$ has sp^3d^2 hybridization.

[JEE (Advanced) 2020]

Q.55 In an experiment, m grams of a compound X (gas/liquid/solid) taken in a container is loaded in a balance as shown in figure I below. In the presence of a magnetic field, the pan with X is either deflected upwards (figure II), or deflected downwards (figure III), depending on the compound X. Identify the correct statement(s)



- (A) If X is $H_2O(l)$, deflection of the pan is upwards.
 (B) If X is $K_4[Fe(CN)_6](s)$, deflection of the pan is upwards.
 (C) If X is $O_2(g)$, deflection of the pan is downwards.
 (D) If X is $C_6H_6(l)$, deflection of the pan is downwards.

[JEE (Advanced) 2020]

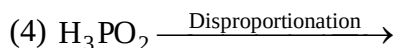
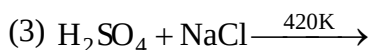
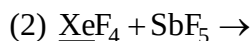
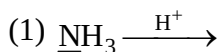
[JEE MAIN]

- Q.1 What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid- [AIEEE 2003]
 (1) Cr^{3+} and $\text{Cr}_2\text{O}_7^{2-}$ are formed
 (2) $\text{Cr}_2\text{O}_7^{2-}$ and H_2O are formed
 (3) $\text{Cr}_2\text{O}_7^{2-}$ is reduced to +3 state of Cr
 (4) $\text{Cr}_2\text{O}_7^{2-}$ is oxidised to +7 state of Cr
- Q.2 Excess of KI reacts with CuSO_4 solution and then $\text{Na}_2\text{S}_2\text{O}_3$ solution is added to it. Which of the statements is incorrect for this reaction - [AIEEE 2004]
 (1) Evolved I_2 is reduced (2) CuI_2 is formed
 (3) $\text{Na}_2\text{S}_2\text{O}_3$ is oxidised (4) Cu_2I_2 is formed
- Q.3 What products are expected from the disproportionation reaction of hypochlorous acid - [AIEEE 2006]
 (1) HClO_2 and HClO_4 (2) HCl and Cl_2O (3) HCl and HClO_3 (4) HClO_3 and Cl_2O
- Q.4 Which one of the following reactions of Xenon compounds is not feasible? [AIEEE 2009]
 (1) $3 \text{XeF}_4 + 6 \text{H}_2\text{O} \rightarrow 2 \text{Xe} + \text{XeO}_3 + 12 \text{HF} + 1.5 \text{O}_2$
 (2) $2 \text{XeF}_2 + 2 \text{H}_2\text{O} \rightarrow 2 \text{Xe} + 4 \text{HF} + \text{O}_2$
 (3) $\text{XeF}_6 + \text{RbF} \rightarrow \text{Rb} [\text{XeF}_7]$
 (4) $\text{XeO}_3 + 6 \text{HF} \rightarrow \text{XeF}_6 + 3 \text{H}_2\text{O}$
- Q.5 In which of the following reactions H_2O_2 acts as a reducing agent? [JEE (Main-Offline) 2014]
 (a) $\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$ (b) $\text{H}_2\text{O}_2 - 2\text{e}^- \rightarrow \text{O}_2 + 2\text{H}^+$
 (c) $\text{H}_2\text{O}_2 + 2\text{e}^- \rightarrow 2\text{OH}^-$ (d) $\text{H}_2\text{O}_2 + 2\text{OH}^- - 2\text{e}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O}$
 (1) (c), (d) (2) (a), (c) (3) (b), (d) (4) (a), (b)
- Q.6 The products obtained when chlorine gas reacts with cold and dilute aqueous NaOH are : [JEE (Main-Offline) 2017]
 (1) ClO_2^- and ClO_3^- (2) Cl^- and ClO^- (3) Cl^- and ClO_2^- (4) ClO^- and ClO_3^-
- Q.7 Which of the following reactions is an example of a redox reaction? [JEE (Main-Offline) 2017]
 (1) $\text{XeF}_2 + \text{PF}_5 \rightarrow [\text{XeF}]^+ \text{PF}_6^-$ (2) $\text{XeF}_6 + \text{H}_2\text{O} \rightarrow \text{XeOF}_4 + 2\text{HF}$
 (3) $\text{XeF}_6 + 2\text{H}_2\text{O} \rightarrow \text{XeO}_2\text{F}_2 + 4\text{HF}$ (4) $\text{XeF}_4 + \text{O}_2\text{F}_2 \rightarrow \text{XeF}_6 + \text{O}_2$
- Q.8 In the following reactions, ZnO is respectively acting as a/an : [JEE (Main-Offline) 2017]
 (a) $\text{ZnO} + \text{Na}_2\text{O} \rightarrow \text{Na}_2\text{ZnO}_2$
 (b) $\text{ZnO} + \text{CO}_2 \rightarrow \text{ZnCO}_3$
 (1) base and base (2) acid and acid (3) acid and base (4) base and acid

- Q.9 When XO_2 is fused with an alkali metal hydroxide in presence of an oxidizing agent such as KNO_3 ; a dark green product is formed which disproportionates in acidic solution to afford a dark purple solution. X is :
 (1) Ti (2) V (3) Cr (4) Mn
[JEE (Main-Online) 2018]
- Q.10 The compound that **does not** produce nitrogen gas by the thermal decomposition is :
 (1) NH_4NO_2 (2) $(\text{NH}_4)_2\text{SO}_4$ (3) $\text{Ba}(\text{N}_3)_2$ (4) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
[JEE (Main-Offline) 2018]
- Q.11 For per gram of reactant, the maximum quantity of N_2 gas is produced in which of the following thermal decomposition reactions? **[Given : Atomic weight: Cr = 52 u, ba = 137 u]**
 (1) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7(\text{s}) \longrightarrow \text{N}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g}) + \text{Cr}_2\text{O}_3(\text{s})$
 (2) $2\text{NH}_3(\text{g}) \longrightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$
 (3) $2\text{NH}_4\text{NO}_3(\text{s}) \longrightarrow 2\text{N}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g}) + \text{O}_2(\text{g})$
 (4) $\text{Na}(\text{N}_3)_2(\text{s}) \longrightarrow \text{Ba}(\text{s}) + 3\text{N}_2(\text{g})$
[JEE (Main-Online) 2018]
- Q.12 Xenon hexafluoride on partial hydrolysis produces compounds X and Y. Compounds X and Y and the oxidation state of Xe are respectively :
 (1) $\text{XeO}_2\text{F}_2(+6)$ and $\text{XeO}_2(+4)$ (2) $\text{XeOF}_4(+6)$ and $\text{XeO}_3(+6)$
 (3) $\text{XeO}_2(+4)$ and $\text{XeO}_3(+6)$ (4) $\text{XeOF}_4(+6)$ and $\text{XeO}_2\text{F}_2(+6)$
[JEE (Main-Online) 2018]
- Q.13 A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the aforementioned solution, a white precipitate is obtained which does not dissolve in dil. nitric acid. The anion is :
 (1) CO_3^{2-} (2) SO_4^{2-} (3) S^{2-} (4) Cl^-
[JEE (Main-Online) 2018]
- Q.14 Iodine reacts with concentrated HNO_3 to yield Y along with other products. The oxidation state of iodine in Y, is :
 (1) 5 (2) 7 (3) 1 (4) 3
[JEE (Main-Online) 2019]
- Q.15 Chlorine on reaction with hot and concentrated sodium hydroxide gives :
 (1) Cl^- and ClO_3^- (2) Cl^- and ClO^- (3) Cl^- and ClO_2^- (4) ClO_3^- and ClO_2^-
[JEE (Main-Online) 2019]
- Q.16 Aqua regia is used for dissolving noble metals (Au, Pt, etc.). The gas evolved in this process is :
 (1) N_2 (2) N_2O_3 (3) NO (4) N_2O_5
[JEE (Main-Online) 2020]

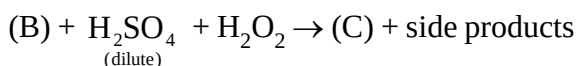
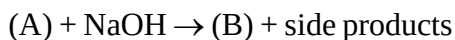
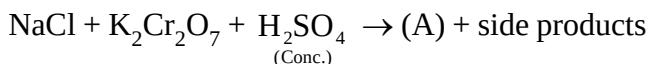
- Q.17 On heating, lead(II) nitrate gives a brown gas (A). The gas (A) on cooling changes to a colourless solid/liquid (B). (B) on heating with NO changes to a blue solid (C). The oxidation number of nitrogen in solid (C) is :
 (1) +5 (2) +2 (3) +4 (4) +3
[JEE (Main-Online) 2020]
- Q.18 Reaction of ammonia with excess Cl_2 gives :
 (1) NH_4Cl and N_2 (2) NCl_3 and NH_4Cl
 (3) NH_4Cl and HCl (4) NCl_3 and HCl
[JEE (Main-Online) 2020]
- Q.19 Reaction of an inorganic sulphite X with dilute H_2SO_4 generates compound Y. Reaction of Y with NaOH gives X. Further, the reaction of X with Y and water affords compound Z. Y and Z, respectively, are:
 (1) S and Na_2SO_3 (2) SO_2 and NaHSO_3
 (3) SO_3 and NaHSO_3 (4) SO_2 and Na_2SO_3
[JEE (Main-Online) 2020]
- Q.20 Chlorine reacts with hot and concentrated NaOH and produces compounds (X) and (Y). Compound (X) gives white precipitate with silver nitrate solution. The average bond order between Cl and O atoms in (Y) is _____.
[JEE (Main-Online) 2020]
- Q.21 The redox reaction among the following is
 (1) formation of ozone from atmospheric oxygen in the presence of sunlight
 (2) reaction of H_2SO_4 with NaOH
 (3) combination of dinitrogen with dioxygen at 2000K
 (4) reaction of $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$ with AgNO_3
[JEE (Main-Online) 2020]
- Q.22 In the following reactions, products (A) and (B), respectively, are:
 $\text{NaOH} + \text{Cl}_2 \rightarrow (\text{A}) + \text{side products}$
 (hot and conc.)
 $\text{Ca}(\text{OH})_2 + \text{Cl}_2 \rightarrow (\text{B}) + \text{side products}$
 (dry)
 (1) NaClO_3 and $\text{Ca}(\text{ClO}_3)_2$ (2) NaOCl and $\text{Ca}(\text{ClO}_3)_2$
 (3) NaOCl and $\text{Ca}(\text{OCl})_2$ (4) NaClO_3 and $\text{Ca}(\text{OCl})_2$
[JEE (Main-Online) 2020]
- Q.23 The reaction of $\text{H}_3\text{N}_3\text{B}_3\text{Cl}_3$ (A) with LiBH_4 in tetrahydrofuran gives inorganic benzene (B). Further, the reaction of (A) with (C) leads to $\text{H}_3\text{N}_3\text{B}_3(\text{Me})_3$. Compounds (B) and (C) respectively, are
 (1) Boron nitride and MeBr (2) Borazine and MeMgBr
 (3) Borazine and BeBr (4) Diborane and MeMgBr
[JEE (Main-Online) 2020]

Q.24 The reaction in which the hybridisation of the underlined atom is affected is :-



[JEE (Main-Online) 2020]

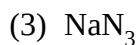
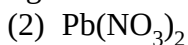
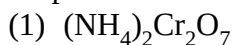
Q.25 Consider the following reactions



The sum of the total number of atoms in one molecule each of (A), (B) and (C) is _____.

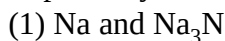
[JEE (Main-Online) 2020]

Q.26 On heating compound (A) gives a gas (B) which is constituent of air. This gas when treated with H_2 in the presence of a catalyst gives another gas (C) which is basic in nature. (A) should not be:



[JEE (Main-Online) 2020]

Q.27 A metal (A) on heating in nitrogen gas gives compound B. B on treatment with H_2O gives a colourless gas which when passed through CuSO_4 solution gives a dark blue-violet coloured solution. A and B respectively, are



[JEE (Main-Online) 2020]

[JEE ADVANCED]

- Q.1 Ammonium dichromate is used in some fireworks. The green colored powder blown in the air is
(A) CrO_3 (B) Cr_2O_3 (C) Cr (D) $\text{CrO}(\text{O}_2)$ **[JEE 1997]**
- Q.2 Which of the following statement(s) is /are correct. When a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with conc. H_2SO_4 ? **[JEE 1998]**
(A) A deep red vapours is evolved.
(B) The vapours when passed into NaOH solution gives a yellow solution of Na_2CrO_4
(C) Chlorine gas is evolved
(D) Chromyl chloride is formed.
- Q.3 An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a : **[JEE 2000]**
(A) Hg_2^+ salt (B) Cr^{2+} salt (C) Ag^+ salt (D) Pb^{2+} salt
- Q.4 The reaction $3\text{ClO}^-(\text{aq}) \longrightarrow \text{ClO}_3^-(\text{aq}) + 2\text{Cl}^-(\text{aq})$ is an example of **[JEE 2001]**
(A) Oxidation reaction (B) Reduction reaction
(C) Disproportionation reaction (D) Decomposition reaction
- Q.5 Anhydrous ferric chloride is prepared by: **[JEE 2002]**
(A) heating hydrated ferric chloride at a high temperature in a stream of air
(B) heating metallic iron in a stream of dry chlorine gas
(C) reaction of ferric oxide with HCl
(D) reaction of metallic iron with HCl
- Q.6 A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas 'Y'. Identify 'X' and 'Y': **[JEE 2002(Mains)]**
(A) $\text{X} = \text{CO}_2$, $\text{Y} = \text{Cl}_2$ (B) $\text{X} = \text{Cl}_2$, $\text{Y} = \text{CO}_2$
(C) $\text{X} = \text{Cl}_2$, $\text{Y} = \text{H}_2$ (D) $\text{X} = \text{H}_2$, $\text{Y} = \text{Cl}_2$
- Q.7 $[\text{X}] + \text{H}_2\text{SO}_4 \rightarrow [\text{Y}]$ a colourless gas with irritating smell **[JEE 2003]**
 $[\text{Y}] + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$ green solution
 $[\text{X}]$ and $[\text{Y}]$ are:
(A) SO_3^{2-} , SO_2 (B) Cl^- , HCl (C) S^{2-} , H_2S (D) CO_3^{2-} , CO_2
- Q.8 When MnO_2 is fused with KOH , a coloured compound is formed, the product and its colour is: **[JEE 2003]**
(A) K_2MnO_4 , green colour (B) KMnO_4 , purple
(C) Mn_2O_3 , brown (D) Mn_3O_4 , black

- Q.9 H_3PO_4 and H_3PO_3 is [JEE 2003]
(A) H_3PO_3 is reducing agent and dibasic acid
(B) H_3PO_4 is reducing agent and dibasic acid
(C) H_3PO_4 is both reducing agent and tribasic acid
(D) H_3PO_3 is not reducing agent and dibasic acid
- Q.10 The product of oxidation of I^- with MnO_4^- in alkaline medium is [JEE 2004]
(A) IO_3^- (B) I_2 (C) IO^- (D) IO_4^-
- Q.11 $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ on heating gives a gas which is also given by: [JEE 2004]
(A) heating NH_4NO_2 (B) heating NH_4NO_3
(C) $\text{Mg}_3\text{N}_2 + \text{H}_2\text{O}$ (D) $\text{Na}(\text{comp.}) + \text{H}_2\text{O}_2$
- Q.12 A metal nitrate reacts with KI to give a black precipitate which on addition of excess of KI convert into orange colour solution. The cation of metal nitrate is: [JEE 2005]
(A) Hg^{2+} (B) Bi^{3+} (C) Pb^{2+} (D) Cu^+
- Q.13 When two gases are mixed at -30°C the blue coloured gas is formed [JEE 2005]
(A) N_2O_3 (B) N_2 (C) NO_2 (D) N_2O_5
- Q.14 When PbO_2 reacts with conc. HNO_3 , the gas evolved is : [JEE 2005]
(A) NO_2 (B) O_2 (C) N_2 (D) N_2O
- Q.15 A solution when diluted with H_2O and boiled, it gives a white precipitate. On addition of excess $\text{NH}_4\text{Cl} / \text{NH}_4\text{OH}$, the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in $\text{NH}_4\text{OH} / \text{NH}_4\text{Cl}$. [JEE 2006]
(A) $\text{Zn}(\text{OH})_2$ (B) $\text{Al}(\text{OH})_3$ (C) $\text{Mg}(\text{OH})_2$ (D) $\text{Ca}(\text{OH})_2$
- Q.16 CuSO_4 decolourises on addition of KCN, the product is: [JEE 2006]
(A) $[\text{Cu}(\text{CN})_4]^{2-}$ (B) Cu^{2+} get reduced to form $[\text{Cu}(\text{CN})_4]^{3-}$
(C) $\text{Cu}(\text{CN})_2$ (D) CuCN
- Q.17 A sodium salt of unknown anion when treated with MgCl_2 gives white precipitate only on boiling. The anion is - [JEE 2006]
(A) SO_4^{2-} (B) HCO_3^- (C) CO_3^{2-} (D) NO_3^-
- Q.18 Sodium fusion extract, obtained from aniline, on treatment with iron (II) sulphate and H_2SO_4 in presence of air gives a Prussian blue precipitate. The blue colour is due to the formation of : [JEE 2007]
(A) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (B) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$ (C) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_2$ (D) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_3$
- Q.19 A solution of a metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is [JEE 2007]
(A) Pb^{2+} (B) Hg^{2+} (C) Cu^{2+} (D) Co^{2+}

- Q.20 A solution of colourless salt H on boiling with excess NaOH produces a non-flammable gas. The gas evolution ceases after sometime. Upon addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt(s) H is (are)- [JEE 2008]
 (A) NH_4NO_3 (B) NH_4NO_2 (C) NH_4Cl (D) $(\text{NH}_4)_2\text{SO}_4$
- Q.21 The compound(s) formed upon combustion of sodium metal in excess air is (are) [JEE 2009]
 (A) Na_2O_2 (B) Na_2O (C) NaO_2 (D) NaOH
- Q.22 The reaction of P_4 with X leads selectively to P_4O_6 . The X is - [JEE 2009]
 (A) Dry O_2 (B) A mixture of O_2 and N_2
 (C) Moist O_2 (D) O_2 in the presence of aqueous NaOH
- Q.23 Match each of reactions given in **Column I** with the corresponding product(s) given in **Column II** - [JEE 2009]
- | Column I | Column II |
|-------------------------------------|--------------------------------|
| (A) $\text{Cu} + \text{dil HNO}_3$ | (p) NO |
| (B) $\text{Cu} + \text{conc HNO}_3$ | (q) NO_2 |
| (C) $\text{Zn} + \text{dil HNO}_3$ | (r) N_2O |
| (D) $\text{Zn} + \text{conc HNO}_3$ | (s) $\text{Cu}(\text{NO}_3)_2$ |
| | (t) $\text{Zn}(\text{NO}_3)_2$ |
- Q.24 In the reaction [JEE 2009]

$$2\text{X} + \text{B}_2\text{H}_6 \rightarrow [\text{BH}_2(\text{X})_2]^+ [\text{BH}_4]^-$$
 the amine(s) X is (are)
 (A) NH_3 (B) CH_3NH_2 (C) $(\text{CH}_3)_2\text{NH}$ (D) $(\text{CH}_3)_3\text{N}$
- Q.25 The oxidation number of Mn in the product of alkaline oxidation fusion of MnO_2 is [JEE 2009]
- Q.26 Passing H_2S gas into a mixture of Mn^{2+} , Ni^{2+} , Cu^{2+} and Hg^{2+} ions in an acidified aqueous solution precipitates [JEE 2011]
 (A) CuS and HgS (B) MnS and CuS (C) MnS and NiS (D) NiS and HgS
- Q.27 Extra pure N_2 can be obtained by heating [JEE 2011]
 (A) NH_3 with CuO (B) NH_4NO_3 (C) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (D) $\text{Ba}(\text{N}_3)_2$
- Q.28 Among the following the number of compounds that can react with PCl_5 to give POCl_3 is [JEE 2011]
 O_2 , CO_2 , SO_2 , H_2O , H_2SO_4 , P_4O_{10}

Paragraph for question nos. 29 to 31

When a metal rod **M** is dipped into an aqueous colourless concentrated solution of compound **N**, the solution turns light blue. Addition of aqueous NaCl to the blue solution gives a white precipitate **O**. Addition of aqueous NH_3 dissolves **O** gives an intense blue solution.

- Q.29 The metal rod **M** is [JEE 2011]
 (A) Fe (B) Cu (C) Ni (D) Co

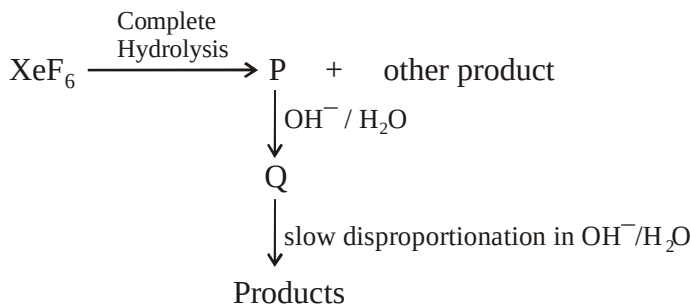
- Q.30 The compound **N** is [JEE 2011]
 (A) AgNO_3 (B) $\text{Zn}(\text{NO}_3)_2$ (C) $\text{Al}(\text{NO}_3)_3$ (D) $\text{Pb}(\text{NO}_3)_2$
- Q.31 The final solution contains [JEE 2011]
 (A) $[\text{Pb}(\text{NH}_3)_4]^{2+}$ and $[\text{CoCl}_4]^{2-}$ (B) $[\text{Al}(\text{NH}_3)_4]^{3+}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 (C) $[\text{Ag}(\text{NH}_3)_2]^+$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (D) $[\text{Ag}(\text{NH}_3)_2]^+$ and $[\text{Ni}(\text{NH}_3)_6]^{2+}$
- Q.32 Which of the following hydrogen halides react(s) with $\text{AgNO}_3(\text{aq})$ to give a precipitate that dissolves in $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$? [JEE 2012]
 (A) HCl (B) HF (C) HBr (D) HI
- Q.33 Upon treatment with ammoniacal H_2S , the metal ion that precipitates as a sulfide is [JEE (Advanced) 2013]
 (A) $\text{Fe}(\text{III})$ (B) $\text{Al}(\text{III})$ (C) $\text{Mg}(\text{II})$ (D) $\text{Zn}(\text{II})$

Paragraph for Questions 34 and 35

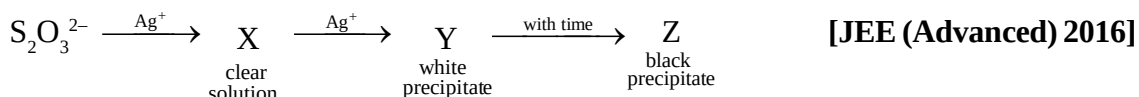
An aqueous solution of a mixture of two inorganic salts, when treated with dilute HCl , gave a precipitate (P) and a filtrate (Q). The precipitate P was found to dissolve in hot water. The filtrate (Q) remained unchanged, when treated with H_2S in a dilute mineral acid medium. However, it gave a precipitate (R) with H_2S in an ammoniacal medium. The precipitate R gave a coloured solution (S), when treated with H_2O_2 in an aqueous NaOH medium.

- Q.34 The coloured solution S contains [JEE (Advanced) 2013]
 (A) $\text{Fe}_2(\text{SO}_4)_3$ (B) CuSO_4 (C) ZnSO_4 (D) Na_2CrO_4
- Q.35 The precipitate P contains [JEE (Advanced) 2013]
 (A) Pb^{2+} (B) Hg_2^{2+} (C) Ag^+ (D) Hg^{2+}
- Q.36 Consider the following list of reagents: [JEE (Advanced) 2014]
 Acidified $\text{K}_2\text{Cr}_2\text{O}_7$, alkaline KMnO_4 , CuSO_4 , H_2O_2 , Cl_2 , O_3 , FeCl_3 , HNO_3 and $\text{Na}_2\text{S}_2\text{O}_3$. The total number of reagents that can oxidise aqueous iodide to iodine is
- Q.37 Hydrogen peroxide in its reaction with KIO_4 and NH_2OH respectively, is acting as a [JEE (Advanced) 2014]
 (A) reducing agent, oxidising agent (B) reducing agent, reducing agent
 (C) oxidising agent, oxidising agent (D) oxidising agent, reducing agent
- Q.38 The product formed in the reaction of SOCl_2 with white phosphorous is [JEE (Advanced) 2014]
 (A) PCl_3 (B) SO_2Cl_2 (C) SCl_2 (D) POCl_3

- Q.39 Under ambient conditions, the total number of gases released as products in the final step of the reaction scheme shown below is **[JEE (Advanced) 2014]**



- (A) 0 (B) 1 (C) 2 (D) 3
- Q.40 Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is : **[JEE (Advanced) 2015]**
- Q.41 Fe^{3+} is reduced to Fe^{2+} by using **[JEE (Advanced) 2015]**
 (A) H_2O_2 in presence of NaOH (B) Na_2O_2 in water
 (C) H_2O_2 in presence of H_2SO_4 (D) Na_2O_2 in presence of H_2SO_4
- Q.42 The pair(s) of ions where BOTH the ions are precipitated upon passing H_2S gas in presence of dilute HCl , is(are) **[JEE (Advanced) 2015]**
 (A) Ba^{2+} , Zn^{2+} (B) Bi^{3+} , Fe^{3+} (C) Cu^{2+} , Pb^{2+} (D) Hg^{2+} , Bi^{3+}
- Q.43 The reagent(s) that can selectively precipitate S^{2-} from a mixture of S^{2-} and SO_4^{2-} in aqueous solution is(are) **[JEE (Advanced) 2016]**
 (A) CuCl_2 (B) BaCl_2 (C) $\text{Pb}(\text{OOCCH}_3)_2$ (D) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$
- Q.44 In the following reaction sequence in aqueous solution, the species X, Y and Z, respectively, are



- (A) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$, $\text{Ag}_2\text{S}_2\text{O}_3$, Ag_2S (B) $[\text{Ag}(\text{S}_2\text{O}_3)_3]^{5-}$, Ag_2SO_3 , Ag_2S
 (C) $[\text{Ag}(\text{SO}_3)_2]^{3-}$, $\text{Ag}_2\text{S}_2\text{O}_3$, Ag (D) $[\text{Ag}(\text{SO}_3)_3]^{3-}$, Ag_2SO_4 , Ag

Paragraph for Question No. 45 & 46

Upon heating KClO_3 in the presence of catalytic amount of MnO_2 , a gas **W** is formed. Excess amount of **W** reacts with white phosphorus to give **X**. The reaction of **X** with pure HNO_3 gives **Y** and **Z**.

[JEE (Advanced) 2017]

- Q.45 **W** and **X** are, respectively

- (A) O_2 and P_4O_{10} (B) O_2 and P_4O_6
 (C) O_3 and P_4O_6 (D) O_3 and P_4O_{10}

- Q.46 **Y** and **Z** are, respectively

- (A) N_2O_5 and HPO_3 (B) N_2O_4 and HPO_3
 (C) N_2O_4 and H_3PO_3 (D) N_2O_3 and H_3PO_4

- Q.47 Which of the following combination will produce H_2 gas? [JEE (Advanced) 2017]
(A) Au metal and NaCN (aq) in the presence of air
(B) Cu metal and conc. HNO_3
(C) Fe metal and conc. HNO_3
(D) Zn metal and NaOH (aq.)
- Q.48 The compound(s) which generate(s) N_2 gas upon thermal decomposition below $300^\circ C$ is (are) : [JEE (Advanced) 2018]
(A) NH_4NO_3 (B) $(NH_4)_2Cr_2O_7$
(C) $Ba(N_3)_2$ (D) Mg_3N_2
- Q.49 The compound(s) which generate(s) N_2 gas upon thermal decomposition below $300^\circ C$ is (are) : [JEE (Advanced) 2018]
(A) NH_4NO_3 (B) $(NH_4)_2Cr_2O_7$
(C) $Ba(N_3)_2$ (D) Mg_3N_2
- Q.50 A tin chloride Q undergoes the following reactions (not balanced) [JEE (Advanced) 2019]
 $Q + Cl^- \rightarrow X$
 $Q + Me_3N \rightarrow Y$
 $Q + CuCl_2 \rightarrow Z + CuCl$
X is a monoanion having pyramidal geometry. Both Y and Z are neutral compounds. Choose the correct option(s).
(1) The oxidation state of the central atom in Z is +2
(2) The central atoms in X is sp^3 hybridized
(3) The central atom in Z has one lone pair of electrons
(4) There is a coordinate bond in Y
- Q.51 At 143 K, the reaction of XeF_4 with O_2F_2 produces a xenon compound Y. The total number of lone pair(s) of electrons present on the whole molecule of Y is _____. [JEE (Advanced) 2019]
- Q.52 With reference to aqua regia, choose the correct option(s): [JEE (Advanced) 2019]
(1) Aqua regia is prepared by mixing conc. HCl and conc. HNO_3 in 3 : 1 (v/v) ratio
(2) The yellow colour of aqua regia is due to presence of NOCl and Cl_2
(3) Reaction of gold with aqua regia produces NO_2 in the absence of air
(4) Reaction of gold with aqua regia produces an anion having Au in +3 oxidation state.
- Q.53 Fusion of MnO_2 with KOH in presence of O_2 produces a salt W. Alkaline solution of W upon electrolytic oxidation yields another salt X. The manganese containing ions present in W and X, respectively, are Y and Z. Correct statement(s) is (are) [JEE (Advanced) 2019]
(1) In both Y and Z, π -bonding occurs between p-orbitals of oxygen and d-orbitals of manganese.
(2) Both Y and Z are coloured and have tetrahedral shape
(3) Y is diamagnetic in nature while Z is paramagnetic
(4) In aqueous acidic solution, Y undergoes disproportionation reaction to give Z and MnO_2 .

- Q.54 Consider the following reactions (unbalanced) [JEE (Advanced) 2019]
 $\text{Zn} + \text{hot conc. H}_2\text{SO}_4 \rightarrow \text{G} + \text{R} + \text{X}$
 $\text{Zn} + \text{conc. NaOH} \rightarrow \text{T} + \text{Q}$
 $\text{G} + \text{H}_2\text{S} + \text{NH}_4\text{OH} \rightarrow \text{Z (a precipitate)} + \text{X} + \text{Y}$
Choose the correct option(s).
(1) Bond order of Q is 1 in its ground state
(2) Z is dirty white in colour
(3) The oxidation state of Zn in T is +1
(4) R is a V-shaped molecule
- Q.55 A colorless aqueous solution contains nitrates of two metals, X and Y. When it was added to an aqueous solution of NaCl, a white precipitate was formed. This precipitate was found to be partly soluble in hot water to give a residue P and a solution Q. The residue P was soluble in aq. NH_3 and also in excess sodium thiosulfate. The hot solution Q gave a yellow precipitate with KI. The metals X and Y, respectively, are
(A) Ag and Pb (B) Ag and Cd (C) Cd and Pb (D) Cd and Zn [JEE (Advanced) 2020]
- Q.56 An acidified solution of potassium chromate was layered with an equal volume of amyl alcohol. When it was shaken after the addition of 1 mL of 3% H_2O_2 , a blue alcohol layer was obtained. The blue color is due to the formation of a chromium (VI) compound 'X'. What is the number of oxygen atoms bonded to chromium through only single bonds in a molecule of X? [JEE (Advanced) 2020]
- Q.57 Which of the following liberates O_2 upon hydrolysis?
(A) Pb_3O_4 (B) KO_2 (C) Na_2O_2 (D) Li_2O_2 [JEE (Advanced) 2020]
- Q.58 Choose the correct statement(s) among the following.
(A) $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ is a reducing agent.
(B) SnO_2 reacts with KOH to form $\text{K}_2[\text{Sn}(\text{OH})_6]$.
(C) A solution of PbCl_2 in HCl contains Pb^{2+} and Cl^- ions.
(D) The reaction of Pb_3O_4 with hot dilute nitric acid to give PbO_2 is a redox reaction. [JEE (Advanced) 2020]

[JEE MAIN]

- Q.1 Which of the following process will produce hard water ? [AIEEE 2003]
(1) Saturation of water with CaCO_3 (2) Saturation of water with MgCO_3
(3) Saturation of water with CaSO_4 (4) Addition of Na_2CO_3 water
- Q.2 The reagent commonly used to determine hardness of water titrimetrically is - [AIEEE 2003]
(1) Oxalic acid (2) Disodium salt of EDTA
(3) Sodium citrate (4) Sodium thiosulphate.
- Q.3 In context with the industrial preparation of hydrogen from water gas ($\text{CO} + \text{H}_2$), which of the following is the correct statement ? [AIEEE 2008]
(1) CO is removed by absorption in aqueous Cu_2Cl_2 solution
(2) H_2 is removed through occlusion with Pd
(3) CO is oxidized to CO_2 with steam in the presence of a catalyst followed by absorption of CO_2 in alkali
(4) CO and H_2 are fractionally separated using differences in their densities
- Q.4 Very pure hydrogen (99.9%) can be made by which of the following processes ? [AIEEE 2012]
(1) Electrolysis of water
(2) Reaction of salt like hydrides with water
(3) Reaction of methane with steam
(4) Mixing natural hydrocarbons of high molecular weight
- Q.5 The numbers of protons, electrons and neutrons in a molecule of heavy water are respectively :
(1) 8, 10, 11 (2) 11, 10, 10 (3) 10, 11, 10 (4) 10, 10, 10 [JEE (Main-Online) 2013]
- Q.6 Which of the following statements about Na_2O_2 is **not** correct ?
(1) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium.
(2) It is diamagnetic in nature.
(3) It is a derivative of H_2O_2
(4) It is the super oxide of sodium. [JEE (Main-Online) 2014]
- Q.7 Hydrogen peroxide acts both as an oxidising and as a reducing agent depending upon the nature of the reacting species. In which of the following cases H_2O_2 acts as a reducing agent in acid medium ?
(1) SO_3^{2-} (2) MnO_4^- (3) KI (4) $\text{Cr}_2\text{O}_7^{2-}$ [JEE (Main-Online) 2014]

- Q.8 Which physical property of dihydrogen is wrong ?
(1) Tasteless gas (2) Non-inflammable gas
(3) Odourless gas (4) Colourless gas
[JEE (Main-Online) 2015]
- Q.9 The concentration of fluoride, lead, nitrate and iron in a water sample from an underground lake was found to be 1000 ppb, 40 ppb, 100 ppm and 0.2 ppm, respectively. This water is unsuitable for drinking due to high concentration of :
(1) Iron (2) Fluoride (3) Lead (4) Nitrate
[JEE (Main-Offline) 2016]
- Q.10 Which one of the following statements about water is **FALSE** ?
(1) Ice formed by heavy water sinks in normal water.
(2) Water is oxidized to oxygen during photosynthesis.
(3) Water can act both as an acid and as a base.
(4) There is extensive intramolecular hydrogen bonding in the condensed phase.
[JEE (Main-Offline) 2016]
- Q.11 Identify the **incorrect** statement regarding heavy water:
(1) It reacts with Al_4C_3 to produce CD_4 and Al(OD)_3 .
(2) It is used as a coolant in nuclear reactors
(3) It reacts with CaC_2 to produce C_2D_2 and Ca(OD)_2
(4) It reacts with SO_3 to form deuterated sulphuric acid (D_2SO_4)
[JEE (Main-Online) 2016]
- Q.12 Identify the reaction which does **not** liberate hydrogen :
(1) Reaction of zinc with aqueous alkali.
(2) Electrolysis of acidified water using Pt electrodes.
(3) Allowing a solution of sodium in liquid ammonia to stand.
(4) Reaction of lithium hydride with B_2H_6 .
[JEE (Main-Online) 2016]
- Q.13 In which of the following reactions, hydrogen peroxide acts as an oxidizing agent ?
(1) $\text{HOCl} + \text{H}_2\text{O}_2 \longrightarrow \text{H}_3\text{O}^+ + \text{Cl}^- + \text{O}_2$
(2) $\text{I}_2 + \text{H}_2\text{O}_2 + 2\text{OH}^- \longrightarrow 2\text{I}^- + 2\text{H}_2\text{O} + \text{O}_2$
(3) $2\text{MnO}_4^- + 3\text{H}_2\text{O}_2 \longrightarrow 2\text{MnO}_2 + 3\text{O}_2 + 2\text{H}_2\text{O} + 2\text{OH}^-$
(4) $\text{PbS} + 4\text{H}_2\text{O}_2 \longrightarrow \text{PbSO}_4 + 4\text{H}_2\text{O}$
[JEE (Main-Online) 2017]
- Q.14 Hydrogen peroxide oxidises $[\text{Fe(CN)}_6]^{4-}$ to $[\text{Fe(CN)}_6]^{3-}$ in acidic medium but reduces $[\text{Fe(CN)}_6]^{3-}$ to $[\text{Fe(CN)}_6]^{4-}$ in alkaline medium. The other products formed are, respectively:
(1) H_2O and $(\text{H}_2\text{O} + \text{O}_2)$ (2) H_2O and $(\text{H}_2\text{O} + \text{OH}^-)$
(3) $(\text{H}_2\text{O} + \text{O}_2)$ and H_2O (4) $(\text{H}_2\text{O} + \text{O}_2)$ and $(\text{H}_2\text{O} + \text{OH}^-)$
[JEE (Main-Offline) 2018]
- Q.15 The isotopes of hydrogen are :
(1) Protium and deuterium only (2) Deuterium and tritium only
(3) Tritium and protium only (4) Protium, deuterium and tritium
[JEE (Main-Online) 2019]

- Q.16 The total number of isotopes of hydrogen and number of radioactive isotopes among them, respectively, are :
 (1) 2 and 0 (2) 3 and 2 (3) 3 and 1 (4) 2 and 1
[JEE (Main-Online) 2019]
- Q.17 The chemical nature of hydrogen peroxide is :
 (1) Reducing agent in basic medium, but not in acidic medium
 (2) Oxidising and reducing agent in both acidic and basic medium
 (3) Oxidising agent in acidic medium, but not in basic medium
 (4) Oxidising and reducing agent in acidic medium, but not in basic medium
[JEE (Main-Online) 2019]
- Q.18 Among the following reactions of hydrogen with halogens, the one that requires a catalyst is :
 (1) $\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$ (2) $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$
 (3) $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ (4) $\text{H}_2 + \text{F}_2 \rightarrow 2\text{HF}$
[JEE (Main-Online) 2019]
- Q.19 The correct statements among (a) to (d) regarding H_2 as a fuel are
 (a) It produces less pollutants than petrol.
 (b) A cylinder of compressed dihydrogen weighs ~30 times more than a petrol tank producing the same amount of energy.
 (c) Dihydrogen is stored in tanks of metal alloys like NaNi_5
 (d) On combustion, values of energy released per gram of liquid dihydrogen and LPG are 50 and 142 kJ, respectively.
 (1) (b) & (d) only (2) (a) & (c) only
 (3) (b), (c) & (d) only (4) (a), (b) & (c) only
[JEE (Main-Online) 2019]
- Q.20 The hydride that is NOT electron deficient is:
 (1) SiH_4 (2) B_2H_6 (3) GaH_3 (4) AlH_3
[JEE (Main-Online) 2019]
- Q.21 The equation that represents the water-gas shift reaction is :
 (1) $\text{CO(g)} + \text{H}_2\text{O(g)} \xrightarrow[\text{Catalyst}]{673\text{K}} \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$
 (2) $\text{CH}_4\text{(g)} + \text{H}_2\text{O(g)} \xrightarrow[\text{Ni}]{1270\text{K}} \text{CO(g)} + 3 \text{H}_2\text{(g)}$
 (3) $\text{C(s)} + \text{H}_2\text{O(g)} \xrightarrow{1270\text{K}} \text{CO(g)} + \text{H}_2\text{(g)}$
 (4) $2\text{C(s)} + \text{O}_2\text{(g)} + 4\text{N}_2\text{(g)} \xrightarrow{1273\text{K}} 2\text{CO(g)} + 4\text{N}_2\text{(g)}$
[JEE (Main-Online) 2020]
- Q.22 The one that is NOT suitable for the removal of permanent hardness of water is :
 (1) Treatment with sodium carbonate (2) Calgon's method
 (3) Clark's method (4) Ion-exchange method
[JEE (Main-Online) 2020]

[JEE (Main-Online) 2020]

[JEE (Main-Online) 2020]

[JEE (Main-Online) 2020]

[JEE (Main-Online) 2020]

[JEE (Main-Online) 2020]

[JEE (Main-Online) 2020]

[JEE ADVANCED]

- Q.1 When zeolite, which is hydrated sodium aluminium silicate is treated with hard water, the sodium ions are exchanged with - **[JEE 1999]**
(A) H^+ ions (B) Ca^{2+} ions (C) SO_4^{2-} (D) OH^-
- Q.2 Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is - **[JEE 2000]**
(A) H_2O , because of hydrogen bonding (B) H_2Te , because of higher molecular weight
(C) H_2S , because of hydrogen bonding (D) H_2Se , because of lower molecular weight
- Q.3 Polyphosphates are used as water softening agents because they - **[JEE 2002]**
(A) Form soluble complexes with anionic species
(B) Precipitate anionic species
(C) Form soluble complexes with cationic species
(D) Precipitate cationic species
- Q.4 The reagent(s) used for softening the temporary hardness of water is/are **[JEE 2010]**
(A) $\text{Ca}_3(\text{PO}_4)_2$ (B) $\text{Ca}(\text{OH})_2$ (C) Na_2CO_3 (D) NaOCl

pmssir

[JEE MAIN]

- Q.1 A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is-[AIEEE 2002]
(1) Mg (2) Ba (3) Ca (4) Be
- Q.2 KO_2 is used in space and submarines because it - [AIEEE 2002]
(1) Absorbs CO_2 and increase O_2 concentration
(2) Absorbs moisture
(3) Absorbs CO_2
(4) Produces ozone
- Q.3 In current cement plasters, water is sprinkled from time to time. This helps in - [AIEEE 2003]
(1) Hydrating sand and gravel mixed with cement
(2) Converting sand into silicate
(3) Developing interlocking needle like crystals of hydrated silicates
(4) Keeping it cool
- Q.4 The solubilities of carbonates decreases down the magnesium group due to decrease in- [AIEEE 2003]
(1) Inter-ionic attraction (2) Entropy of solution formation
(3) Lattice energy of solids (4) Hydration energy of cations
- Q.5 The substance not likely to contain CaCO_3 is - [AIEEE 2003]
(1) Sea shells (2) Dolomite (3) A marble statue (4) Calcined gypsum
- Q.6 One mole of magnesium nitride on reaction with excess of water gives - [AIEEE 2004]
(1) Two mole of HNO_3 (2) Two mole of NH_3
(3) 1 mole of NH_3 (4) 1 mole of HNO_3
- Q.7 Beryllium and aluminium exhibit many properties which are similar. But the two elements differ in [AIEEE 2004]
(1) Exhibiting maximum covalency in compounds
(2) Forming polymeric hydrides
(3) Forming covalent halides
(4) Exhibiting amphoteric nature in their oxides
- Q.8 The ionic mobility of alkali metal ions in aqueous solution is maximum for - [AIEEE 2006]
(1) Rb^+ (2) Li^+ (3) Na^+ (4) K^+

- Q.9 Which one of the following is the correct statement ? [AIEEE 2008]
(1) Beryllium exhibits coordination number of six
(2) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase
(3) $B_2H_6 \cdot 2NH_3$ is known as inorganic benzene
(4) Boric acid is a protonic acid
- Q.10 The set representing the correct order of ionic radius is : [AIEEE 2009]
(1) $Li^+ > Be^{2+} > Na^+ > Mg^{2+}$ (2) $Na^+ > Li^+ > Mg^{2+} > Be^{2+}$
(3) $Li^{+2} > Na^+ > Mg^{2+} > Be^{2+}$ (4) $Mg^{2+} > Be^{2+} > Li^+ > Na^+$
- Q.11 Which of the following on thermal decomposition yields a basic as well as an acidic oxide ? [AIEEE 2012]
(1) $CaCO_3$ (2) NH_4NO_3 (3) $NaNO_3$ (4) $KClO_3$
- Q.12 Equimolar solutions of the following compounds are prepared separately in water. Which will have the lowest pH value ? [JEE (Main-Online) 2013]
(1) $CaCl_2$ (2) $BeCl_2$ (3) $MgCl_2$ (4) $SrCl_2$
- Q.13 Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy ? [JEE (Main-Offline) 2015]
(1) $BaSO_4$ (2) $SrSO_4$ (3) $CaSO_4$ (4) $BeSO_4$
- Q.14 The main oxides formed on combustion of Li, Na and K in excess of air are, respectively: [JEE (Main-Offline) 2016]
(1) Li_2O , Na_2O_2 and KO_2 (2) Li_2O , Na_2O and KO_2
(3) LiO_2 , Na_2O_2 and K_2O (4) Li_2O_2 , Na_2O_2 and KO_2
- Q.15 The non-metal that does **not** exhibit positive oxidation state is: [JEE (Main-Online) 2016]
(1) Oxygen (2) Iodine (3) Chlorine (4) Fluorine
- Q.16 Aqueous solution of which salt will **not** contain ions with the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6$? [JEE (Main-Online) 2016]
(1) NaF (2) NaCl (3) KBr (4) CaI_2
- Q.17 The commercial name for calcium oxide is : [JEE (Main-Online) 2016]
(1) Milk of lime (2) Slaked lime (3) Limestone (4) Quick lime
- Q.18 Both lithium and magnesium display several similar properties due to the diagonal relationship; however, the one which is incorrect is- [JEE (Main-Offline) 2017]
(1) both form soluble bicarbonates
(2) both form nitrides
(3) nitrates of both Li and Mg yield NO_2 and O_2 on heating
(4) both form basic carbonates

- Q.19 A metal 'M' reacts with nitrogen gas to afford ' M_3N '. ' M_3N ' on heating at high temperature gives back 'M' and on reaction with water produces a gas 'B'. Gas 'B' reacts with aqueous solution of $CuSO_4$ to form a deep blue compound. 'M' and 'B' respectively are :
(1) Li and NH_3 (2) Ba and N_2 (3) Na and NH_3 (4) Al and N_2
[JEE (Main-Online) 2017]
- Q.20 Which one of the following is an oxide ?
(1) KO_2 (2) BaO_2 (3) SiO_2 (4) CsO_2
[JEE (Main-Online) 2017]
- Q.21 In KO_2 , the nature of oxygen species and the oxidation state of oxygen atom are, respectively:
(1) Peroxide and $-1/2$ (2) Superoxide and -1
(3) Superoxide and $-1/2$ (4) Oxide and -2
[JEE (Main-Online) 2018]
- Q.22 The alkaline earth metal nitrate that does not crystallise with water molecules, is :
(1) $Ca(NO_3)_2$ (2) $Mg(NO_3)_2$ (3) $Ba(NO_3)_2$ (4) $Sr(NO_3)_2$
[JEE (Main-Online) 2019]
- Q.23 The metal that forms nitride by reacting directly with N_2 of air, is :
(1) Rb (2) Li (3) Cs (4) K
[JEE (Main-Online) 2019]
- Q.24 The temporary hardness of water is due to :
(1) Na_2SO_4 (2) $CaCl_2$ (3) NaCl (4) $Ca(HCO_3)_2$
[JEE (Main-Online) 2019]
- Q.25 The metal used for making X-ray tube window is :
(1) Na (2) Be (3) Ca (4) Mg
[JEE (Main-Online) 2019]
- Q.26 Sodium metal on dissolution in liquid ammonia gives a deep blue solution due to the formation of :
(1) sodamide (2) sodium-ammonia complex
(3) ammoniated electrons (4) sodium ion-ammonia complex
[JEE (Main-Online) 2019]
- Q.27 NaH is an example of
(1) electron rich hydride (2) metallic hydride
(3) saline hydride (4) molecular hydride
[JEE (Main-Online) 2019]
- Q.28 The amphoteric hydroxide is
(1) $Be(OH)_2$ (2) $Ca(OH)_2$ (3) $Mg(OH)_2$ (4) $Sr(OH)_2$
[JEE (Main-Online) 2019]

- Q.29 A metal on combustion in excess air forms X. X upon hydrolysis with water yields H_2O_2 and O_2 along with another product. The metal is :
 (1) Na (2) Mg (3) Rb (4) Li
[JEE (Main-Online) 2019]
- Q.30 The metal mainly used in devising photoelectric cells is:
 (1) Na (2) Rb (3) Li (4) Cs
[JEE (Main-Online) 2020]
- Q.31 Two elements A and B have similar chemical properties. They don't form solid hydrogencarbonates, but react with nitrogen to form nitrides. A and B, respectively, are :
 (1) Na and Ca (2) Li and Mg (3) Cs and Ba (4) Na and Rb
[JEE (Main-Online) 2020]
- Q.32 Among the statements (I-IV), the correct ones are :
 (I) Be has smaller atomic radius compared to Mg.
 (II) Be has higher ionization enthalpy than Al.
 (III) Charge/radius ratio of Be is greater than that of Al.
 (IV) Both Be and Al form mainly covalent compounds.
 (1) (I), (II) and (IV) (2) (II), (III) and (IV)
 (3) (I), (II) and (III) (4) (I), (III) and (IV)
[JEE (Main-Online) 2020]
- Q.33 On combustion Li, Na and K in excess of air, the major oxides formed, respectively, are :
 (1) Li_2O , Na_2O and K_2O_2 (2) Li_2O , Na_2O_2 and K_2O
 (3) Li_2O , Na_2O_2 and KO_2 (4) Li_2O_2 , Na_2O_2 and K_2O_2
[JEE (Main-Online) 2020]
- Q.34 Match the following compounds (Column-I) with their uses (Column-II) :
- | S.No. | Column-I | S.No. | Column-II |
|-------|---|-------|--------------------------|
| (I) | $\text{Ca}(\text{OH})_2$ | (A) | Casts of statues |
| (II) | NaCl | (B) | White wash |
| (III) | $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ | (C) | Antacid |
| (IV) | CaCO_3 | (D) | Washing soda preparation |
- (1) (I)-(D), (II)-(A), (III)-(C), (IV)-(B) (2) (I)-(B), (II)-(C), (III)-(D), (IV)-(A)
 (3) (I)-(C), (II)-(D), (III)-(B), (IV)-(A) (4) (I)-(B), (II)-(D), (III)-(A), (IV)-(C)
[JEE (Main-Online) 2020]
- Q.35 Oxidation number of potassium in K_2O , K_2O_2 and KO_2 , respectively, is
 (1) +2, +1 and $+\frac{1}{2}$ (2) +1, +1 and +1 (3) +1, +4 and +2 (4) +1, +2 and +4
[JEE (Main-Online) 2020]

Q.36 When gypsum is heated to 393 K, it forms

- (1) $\text{CaSO}_4 \cdot 0.5 \text{H}_2\text{O}$ (2) Dead burnt plaster
(3) $\text{CaSO}_4 \cdot 5 \text{H}_2\text{O}$ (4) Anhydrous CaSO_4

[JEE (Main-Online) 2020]

Q.37 The first and second ionisation enthalpies of a metal are 496 and 4560 kJ mol⁻¹, respectively. How many moles of HCl and H₂SO₄, respectively, will be needed to react completely with 1 mole of the metal hydroxide?

- (1) 1 and 0.5 (2) 2 and 0.5 (3) 1 and 1 (4) 1 and 2

[JEE (Main-Online) 2020]

Q.38 Among the statements (a)-(d) the correct ones are

- (a) Lithium has the highest hydration enthalpy among the alkali metals.
(b) Lithium chloride is insoluble in pyridine.
(c) Lithium cannot form ethynide upon its reaction with ethyne.
(d) Both lithium and magnesium react slowly with H₂O.

- (1) (a), (b) and (d) only (2) (b) and (c) only
(3) (a), (c) and (d) only (4) (a) and (d) only

[JEE (Main-Online) 2020]

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[JEE ADVANCED]

- Q.1 The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order - **[JEE 1996]**
K₂CO₃ (I) MgCO₃(II) CaCO₃(III) BeCO₃(IV)
(A) I < II < III < IV (B) IV < II < III < I (C) IV < II < I < III (D) II < IV < III < I
- Q.2 Amongst the following, the total number of compounds whose aqueous solution turns red litmus paper blue is **[JEE 2010]**
KCN K₂SO₄ (NH₄)₂C₂O₄ NaCl Zn(NO₃)₂
FeCl₃ K₂CO₃ NH₄NO₃ LiCN
- Q.3 The pair(s) of reagents that yield paramagnetic species is/are **[JEE (Advanced) 2014]**
(A) Na and excess of NH₃ (B) K and excess of O₂
(C) Cu and dilute HNO₃ (D) O₂ and 2-ethylantraquinol

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[JEE MAIN]

- Q.1 The correct no. of lone pairs on the central atom of compounds XeF_2 , XeF_4 & XeF_6 are respectively- [AIEEE 2002]
(1) 3, 2, 1 (2) 4, 3, 2 (3) 1, 3, 4 (4) 1, 2, 3
- Q.2 The no. of σ bonds in the compound P_4O_{10} is - [AIEEE 2002]
(1) 1 (2) 4 (3) 3 (4) 16
- Q.3 Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite : [AIEEE 2003]
(1) Has molecules of variable molecular masses like polymers
(2) Has carbon atoms arranged in large plates of rings of strongly bonded carbon atoms with weak interplate bonds
(3) Is a non crystalline substance
(4) Is an allotropic form of diamond
- Q.4 For making good quality mirrors, plates of float glass are used. These are obtained by floating molten glass over liquid metal which does not solidify before glass. The metal used can be: [AIEEE 2003]
(1) Sodium (2) Magnesium (3) Mercury (4) Tin
- Q.5 The Soldiers of Napoleon army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White Metallic tin buttons get converted to grey powder. This transformation is related to: [AIEEE 2004]
(1) An interaction with water vapour contained in humid air
(2) A change in crystalline structure of tin
(3) A change in the partial pressure of O_2 in air
(4) An interaction with N_2 of air at low temperature
- Q.6 The states of hybridisation of boron and oxygen atoms in boric acid (H_3BO_3) are respectively [AIEEE 2004]
(1) sp^2 , sp^2 (2) sp^3 , sp^3 (3) sp^3 , sp^2 (4) sp^2 , sp^3
- Q.7 Aluminium chloride exists as dimer, Al_2Cl_6 in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives- [AIEEE 2004]
(1) $\text{Al}^{3+} + 3\text{Cl}^-$ (2) $[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 3\text{Cl}^-$
(3) $[\text{Al}(\text{OH})_6]^{3-} + 3\text{HCl}$ (4) $\text{Al}_2\text{O}_3 + \text{HCl}$

- Q.8 The structure of diborane contains [AIEEE 2005]
(1) Four 2c-2e bonds and four 3c-2e bonds
(2) Two 2c-2e bonds and two 3c-2e bonds
(3) Two 2c-2e bonds and four 3c-2e bonds
(4) Four 2c-2e bonds and two 3c-2e bonds
- Q.9 Heating an aqueous solution of aluminium chloride to dryness will give :- [AIEEE 2005]
(1) AlCl_3 (2) Al_2Cl_6 (3) Al_2O_3 (4) $\text{Al}(\text{OH})\text{Cl}_2$
- Q.10 The number of hydrogen atoms (s) attached to phosphorus atom in hypophosphorous acid is – [AIEEE 2005]
(1) two (2) zero (3) three (4) one
- Q.11 In silicon dioxide : [AIEEE 2005]
(1) Each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms
(2) Each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms
(3) Silicon atom is bonded to two oxygen atoms
(4) There are double bonds between silicon and oxygen atoms
- Q.12 Which of the following oxide is amphoteric: [AIEEE 2005]
(1) CaO (2) CO_2 (3) SiO_2 (4) SnO_2
- Q.13 The stability of dihalides of Si, Ge, Sn and Pb increases steadily in the sequence: [AIEEE 2007]
(1) $\text{GeX}_2 \ll \text{SiX}_2 \ll \text{SnX}_2 \ll \text{PbX}_2$ (2) $\text{SiX}_2 \ll \text{GeX}_2 \ll \text{PbX}_2 \ll \text{SnX}_2$
(3) $\text{SiX}_2 \ll \text{GeX}_2 \ll \text{SnX}_2 \ll \text{PbX}_2$ (4) $\text{PbX}_2 \ll \text{SnX}_2 \ll \text{GeX}_2 \ll \text{SiX}_2$
- Q.14 Which one of the following statements regarding helium is incorrect [AIEEE 2007]
(1) It is used to produce and sustain powerful superconducting magnets
(2) It is used as a cryogenic agent for carrying out experiments at low temperatures
(3) It is used to fill gas balloons instead of hydrogen because it is lighter than hydrogen and non-inflammable
(4) It is used in gas-cooled nuclear reactors
- Q.15 Which one of the following is the correct statement [AIEEE 2008]
(1) Boric acid is a protonic acid
(2) Beryllium exhibits coordination number of six
(3) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase
(4) B_2H_6 , 2NH_3 is known as 'inorganic benzene'

- Q.16 In which of the following arrangements, the sequence is not strictly according to the property written against it ? [AIEEE 2009]
- (1) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: increasing acid strength
 - (2) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: increasing basic strength
 - (3) $\text{B} < \text{C} < \text{O} < \text{N}$: increasing first ionization enthalpy
 - (4) $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: increasing oxidising power
- Q.17 Which of the following statements is wrong ? [AIEEE 2011]
- (1) The stability of hydrides increases from NH_3 to BiH_3 in group 15 of the periodic table.
 - (2) Nitrogen cannot form $d\pi - p\pi$ bond.
 - (3) Single N – N bond is weaker than the single P – P bond.
 - (4) N_2O_4 has two resonance structures
- Q.18 Boron cannot form which one of the following anions ? [AIEEE 2011]
- (1) BF_6^{3-}
 - (2) BH_4^-
 - (3) $\text{B}(\text{OH})_4^-$
 - (4) BO_2^-
- Q.19 Identify the incorrect statement :
- (1) $(\text{Si}_3\text{O}_9)^{6-}$ has cyclic structure
 - (2) Trialkylchlorosilane on hydrolysis gives R_3SiOH
 - (3) In $(\text{Si}_3\text{O}_9)^{6-}$, tetrahedral SiO_4 units share two oxygen atoms
 - (4) SiCl_4 undergoes hydrolysis to give H_4SiO_4
- [JEE (Main-Online) 2013]
- Q.20 The catenation tendency of C, Si and Ge is in the order $\text{Ge} < \text{Si} < \text{C}$. The bond energies (in kJ mol^{-1}) of C–C, Si – Si and Ge–Ge bonds are respectively :
- (1) 348, 297, 260
 - (2) 297, 348, 260
 - (3) 260, 297, 248
 - (4) 348, 260, 297
- [JEE (Main-Online) 2013]
- Q.21 Among the following oxoacids, the correct decreasing order of acid strength is : [JEE (Main-Offline) 2014]
- (1) $\text{HClO}_4 > \text{HOCl} > \text{HClO}_2 > \text{HClO}_3$
 - (2) $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HOCl}$
 - (3) $\text{HClO}_2 > \text{HClO}_4 > \text{HClO}_3 > \text{HOCl}$
 - (4) $\text{HOCl} > \text{HClO}_2 > \text{HClO}_3 > \text{HClO}_4$
- Q.22 In the following sets of reactants which two sets best exhibit the amphoteric character of $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$?
- Set 1 : $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}(\text{s})$ and $\text{OH}^-(\text{aq})$
- Set 2 : $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}(\text{s})$ and $\text{H}_2\text{O}(\text{l})$
- Set 3 : $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}(\text{s})$ and $\text{H}^+(\text{aq})$
- Set 4 : $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}(\text{s})$ and $\text{NH}_3(\text{aq})$
- (1) 3 and 4
 - (2) 1 and 3
 - (3) 2 and 4
 - (4) 1 and 2
- [JEE (Main-Online) 2014]

Q.23 The gas evolved on heating CaF_2 and SiO_2 with concentrated H_2SO_4 , on hydrolysis gives a white gelatinous precipitate. The precipitate is :

- (1) silica gel (2) hydrofluosilicic acid
(3) calciumfluorosilicate (4) silicic acid

[JEE (Main-Online) 2014]

Q.24 Which of the following xenon-OXO compounds may not be obtained by hydrolysis of xenon fluorides ?

- (1) XeO_3 (2) XeO_4 (3) XeOF_4 (4) XeO_2F_2

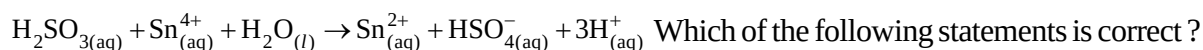
[JEE (Main-Online) 2014]

Q.25 The conjugate base of hydrazoic acid is :

- (1) N_3^- (2) HN_3^- (3) N^{3-} (4) N_2^-

[JEE (Main-Online) 2014]

Q.26 Consider the reaction :



Which of the following statements is correct ?

- (1) Sn^{4+} is the reducing agent because it undergoes oxidation
(2) H_2SO_3 is the reducing agent because it undergoes reducing
(3) Sn^{4+} is the oxidizing agent because it undergoes oxidation
(4) H_2SO_3 is the reducing agent because it undergoes oxidation

[JEE (Main-Online) 2014]

Q.27 Which of these statements is not true ?

- (1) B is always covalent in its compounds
(2) LiAlH_4 is versatile reducing agent in organic synthesis.
(3) In aqueous solution, the Ti^+ ion is much more stable than Ti(III)
(4) NO^+ is not isoelectronic with O_2

[JEE (Main-Online) 2014]

Q.28 Match the catalysts to the correct processes:

[JEE (Main-Offline) 2015]

Catalyst

Process

- (A) TiCl_3 (i) Wacker process
(B) PdCl_2 (ii) Ziegler-Natta Polymerization
(C) CuCl_2 (iii) Contact process
(D) V_2O_5 (iv) Deacon's process

- (1) (A) -(ii), (B) -(iii), (C) -(iv), (D) -(i)
(2) (A) -(iii), (B) -(i), (C) -(ii), (D) -(iv)
(3) (A) -(iii), (B) -(ii), (C) -(iv), (D) -(i)
(4) (A) -(ii), (B) -(i), (C) -(iv), (D) -(iii)

- Q.29 Which among the following is the most reactive? [JEE (Main-Offline) 2015]
 (1) I_2 (2) ICl (3) Cl_2 (4) Br_2
- Q.30 **Assertion:** Nitrogen and Oxygen are the main components in the atmosphere but these do not react to form oxides of nitrogen. [JEE (Main-Offline) 2015]
Reason: The reaction between nitrogen and oxygen requires high temperature.
 (1) The assertion is incorrect, but the reason is correct
 (2) Both the assertion and reason are incorrect
 (3) Both assertion and reason are correct, and the reason is the correct explanation for the assertion
 (4) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion
- Q.31 Which one has the highest boiling point? [JEE (Main-Offline) 2015]
 (1) Kr (2) Xe (3) He (4) Ne
- Q.32 From the following statements regarding H_2O_2 , choose the incorrect statement: [JEE (Main-Offline) 2015]
 (1) It has to be stored in plastic or wax lined glass bottles in dark
 (2) It has to be kept away from dust
 (3) It can act only as an oxidizing agent
 (4) It decomposes on exposure to light
- Q.33 The least number of oxyacids are formed by : [JEE (Main-Online) 2015]
 (1) Sulphur (2) Fluorine (3) Chlorine (4) Nitrogen
- Q.34 Chlorine water on standing loses its colour and forms : [JEE (Main-Online) 2015]
 (1) HCl and $HOCl$ (2) HCl only (3) $HOCl$ and $HOCl_2$ (4) HCl and $HClO_2$
- Q.35 The reaction of zinc with dilute and concentrated nitric acid, respectively, produces [JEE (Main-Offline) 2016]
 (1) NO_2 and N_2O (2) N_2O and NO_2 (3) NO_2 and NO (4) NO and N_2O

- Q.36 Match the items in **column I** with its main use listed in **column II** :

Column I	Column II
(A) Silica gel	(i) Transistor
(B) Silicon	(ii) Ion-exchanger
(C) Silicone	(iii) Drying agent
(D) Silicate	(iv) Sealant

- (1) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii) (2) (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)
 (3) (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii) (4) (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)

[JEE (Main-Online) 2016]

Q.37 **Assertion:** Among the carbon allotropes, diamond is an insulator, whereas, graphite is a good conductor of electricity.

Reason : Hybridization of carbon in diamond and graphite are sp^3 and sp^2 , respectively.

- (1) Both assertion and reason are correct, and the reason is the correct explanation for the assertion.
- (2) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion.
- (3) Assertion is incorrect statement, but the reason is correct.
- (4) Both assertion and reason are incorrect.

[JEE (Main-Online) 2016]

Q.38 Identify the **incorrect** statement :

- (1) S_2 is paramagnetic like oxygen.
- (2) Rhombic and monoclinic sulphur have S_8 molecules.
- (3) S_8 ring has a crown shape.
- (4) The S-S-S bond angles in the S_8 and S_6 rings are the same.

[JEE (Main-Online) 2016]

Q.39 Consider the following standard electrode potentials (E° in volts) in aqueous solution :

Element	M^{3+} / M	M^+ / M
Al	-1.66	+0.55
Tl	+1.26	-0.34

Based on these data, which of the following statements is correct ?

- (1) Tl^+ is more stable than Al^{3+}
- (2) Al^+ is more stable than Al^{3+}
- (3) Tl^+ is more stable than Al^+
- (4) Tl^{3+} is more stable than Al^{3+}

[JEE (Main-Online) 2017]

Q.40 XeF_6 on partial hydrolysis with water produces a compound 'X'. The same compound 'X' is formed when XeF_6 reacts with silica. The compound 'X' is :

- (1) XeF_2
- (2) XeF_4
- (3) $XeOF_4$
- (4) XeO_3

[JEE (Main-Online) 2017]

Q.41 The number of P – OH bonds and the oxidation state of phosphorus atom in pyrophosphoric acid ($H_4P_2O_7$) respectively are :

- (1) four and four
- (2) five and four
- (3) five and five
- (4) four and five

[JEE (Main-Online) 2017]

Q.42 The correct sequence of decreasing number of π -bonds in the structures of H_2SO_3 , H_2SO_4 and $H_2S_2O_7$ is :

- (1) $H_2SO_3 > H_2SO_4 > H_2S_2O_7$
- (2) $H_2SO_4 > H_2S_2O_7 > H_2SO_3$
- (3) $H_2S_2O_7 > H_2SO_4 > H_2SO_3$
- (4) $H_2S_2O_7 > H_2SO_3 > H_2SO_4$

[JEE (Main-Online) 2017]

Q.43 Lithium aluminium hydride reacts with silicon tetrachloride to form:

- (1) LiH , $AlCl_3$ and $SiCl_2$
- (2) $LiCl$, AlH_3 and SiH_4
- (3) $LiGH$, AlH_3 and SiH_4
- (4) $LiCl$, $AlCl_3$ and SiH_4

[JEE (Main-Online) 2018]

- Q.44 The number of P – O bonds in P_4O_6 is :
 (1) 12 (2) 6 (3) 9 (4) 18
[JEE (Main-Online) 2018]
- Q.45 In graphite and diamond, the percentage of p-characters of the hybrid orbitals in hybridisation are respectively:
 (1) 50 and 75 (2) 33 and 25 (3) 67 and 75 (4) 3 and 75
[JEE (Main-Online) 2018]
- Q.46 A group 13 element 'X' reacts with chlorine gas to produce a compound XCl_3 . XCl_3 is electron deficient and easily reacts with NH_3 to form $Cl_3X \leftarrow NH_3$ adduct; however, XCl_3 does not dimerize. X is :
 (1) B (2) Al (3) Ga (4) In
[JEE (Main-Online) 2018]
- Q.47 The one that is extensively used as a piezoelectric material is :
 (1) Quartz (2) Amorphous silica (3) Mica (4) Tridymite
[JEE (Main-Online) 2019]
- Q.48 Aluminium is usually found in +3 oxidation state. In contrast, thallium exists in +1 and +3 oxidation states. This is due to :
 (1) Diagonal relationship (2) Inert pair effect
 (3) Lanthanoid contraction (4) Lattice effect
[JEE (Main-Online) 2019]
- Q.49 Correct statements among a to d regarding silicones are :
 (a) They are polymers with hydrophobic character.
 (b) They are biocompatible
 (c) In general, they have high thermal stability and low dielectric strength.
 (d) Usually, they are resistant to oxidation and used as greases.
 (1) (a), (b), (c) and (d) (2) (a), (b) and (d) only
 (3) (a), (b) and (c) only (4) (a) and (b) only
[JEE (Main-Online) 2019]
- Q.50 The number of 2-centre-2-electron and 3-centre-2-electron bonds in B_2H_6 , respectively, are :
 (1) 2 and 2 (2) 4 and 2 (3) 2 and 4 (4) 2 and 1
[JEE (Main-Online) 2019]
- Q.51 The pair that contains two P – H bonds in each of the oxoacids is :
 (1) H_3PO_2 and $H_4P_2O_5$ (2) H_3PO_3 and H_3PO_2
 (3) $H_4P_2O_5$ and H_3PO_3 (4) $H_4P_2O_5$ and $H_4P_2O_6$
[JEE (Main-Online) 2019]
- Q.52 The chloride that CANNOT get hydrolysed is:
 (1) $PbCl_4$ (2) CCl_4 (3) $SnCl_4$ (4) $SiCl_4$
[JEE (Main-Online) 2019]

- Q.53 The relative stability of +1 oxidation state of group 13 elements follows the order:
(1) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$ (2) $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$ (3) $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$ (4) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
[JEE (Main-Online) 2019]
- Q.54 The element that does NOT show catenation is:
(1) Ge (2) Si (3) Sn (4) Pb
[JEE (Main-Online) 2019]
- Q.55 The correct statement with respect to dinitrogen is :
(1) liquid dinitrogen is not used in cryosurgery.
(2) it can be used as an inert diluent for reactive chemicals.
(3) it can combine with dioxygen at 25°C
(4) N_2 is paramagnetic in nature.
[JEE (Main-Online) 2020]
- Q.56 The reaction of NO with N_2O_4 at 250 K gives :
(1) N_2O_5 (2) NO_2 (3) N_2O (4) N_2O_3
[JEE (Main-Online) 2020]
- Q.57 The number of bonds between sulphur and oxygen atoms in $\text{S}_2\text{O}_8^{2-}$ and the number of bonds between sulphur and sulphur atoms in rhombic sulphur, respectively are
(1) 8 and 6 (2) 4 and 6 (3) 8 and 8 (4) 4 and 8
[JEE (Main-Online) 2020]
- Q.58 The first ionization energy (in kJ/mol) of Na, Mg, Al and Si respectively, are
(1) 496, 737, 577, 786 (2) 496, 577, 737, 786
(3) 496, 577, 786, 737 (4) 786, 737, 577, 496
[JEE (Main-Online) 2020]
- Q.59 White phosphorus on reaction with concentrated NaOH solution in an inert atmosphere of CO_2 gives phosphine and compound (X). (X) on acidification with HCl gives compound (Y). The basicity of compound (Y) is
(1) 4 (2) 3 (3) 2 (4) 1
[JEE (Main-Online) 2020]
- Q.60 If enthalpy of atomisation for $\text{Br}_{2(l)}$ is x kJ/mol and bond enthalpy for Br_2 is y kJ/mol, the relation between them
(1) does not exist (2) is $x < y$
(3) is $x = y$ (4) is $x > y$
[JEE (Main-Online) 2020]

[JEE ADVANCED]

- Q.1 Ammonia can be dried by - [JEE 2000]
 (A) Conc. H_2SO_4 (B) P_4O_{10} (C) CaO (D) Anhydrous CaCl_2
- Q.2 The number of P – O – P bonds in cyclic trimetaphosphoric acid is - [JEE 2000]
 (A) Zero (B) Two (C) Three (D) Four
- Q.3 Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is – [JEE 2000]
 (A) H_2O because of hydrogen bonding
 (B) H_2Te because of higher molecular weight
 (C) H_2S because of hydrogen bonding
 (D) H_2Se because of lower molecular weight
- Q.4 The number of S – S bonds in sulphur trioxide trimer (S_3O_9) is – [JEE 2001]
 (A) Three (B) Two (C) One (D) Zero
- Q.5 Read the following statement and explanation and answer as per the option given below : [JEE 2001]
Statement-I : F atom has a less negative electron gain enthalpy affinity than Cl atom.
Statement-II : Additional electrons are repelled more effectively by 3 p electrons in Cl atom than by 2p electron in F atom
 (A) If both Statement-I and Statement-II are correct, and Statement-II is the correct explanation of the Statement-I
 (B) If both Statement-I and Statement-II are correct, but Statement-II is not correct explanation of the Statement-I
 (C) If Statement-I is correct but Statement-II is incorrect
 (D) If Statement-I is incorrect but Statement-II is correct
- Q.6 The set with correct order of acidity is [JEE 2001]
 (A) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$ (B) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$
 (C) $\text{HClO} < \text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2$ (D) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}$
- Q.7 The acid having O–O bond is - [JEE 2004]
 (A) $\text{H}_2\text{S}_2\text{O}_3$ (B) $\text{H}_2\text{S}_2\text{O}_6$ (C) $\text{H}_2\text{S}_2\text{O}_8$ (D) $\text{H}_2\text{S}_4\text{O}_6$
- Q.8 Which phosphorus is thermodynamically most stable ? [JEE 2005]
 (A) White (B) Red (C) Black (D) Yellow
- Q.9 $\text{B(OH)}_3 + \text{NaOH} \rightleftharpoons \text{NaBO}_2 + \text{Na[B(OH)}_4] + \text{H}_2\text{O}$
 How can this reaction is made to proceed in forward direction? [JEE 2006]
 (A) addition of cis 1,2 diol (B) addition of borax
 (C) addition of trans 1,2 diol (D) addition of Na_2HPO_4

Q.10 Match the following:

[JEE 2006]

Column I

- (A) $\text{Bi}^{3+} \rightarrow (\text{BiO})^+$
 (B) $[\text{AlO}_2]^- \rightarrow \text{Al}(\text{OH})_3$
 (C) $\text{SiO}_4^{4-} \rightarrow \text{Si}_2\text{O}_7^{6-}$
 (D) $(\text{B}_4\text{O}_7^{2-}) \rightarrow [\text{B}(\text{OH})_3]$

Column II

- (P) Heat
 (Q) Hydrolysis
 (R) Acidification
 (S) Dilution by water

Paragraph for Question No. 11 to 13

The noble gases have closed-shell electronic configuration and are monoatomic gases under normal conditions. The low boiling points of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions.

The direct reaction of xenon with fluorine leads to a series of compounds with oxidation numbers +2, +4 and +6. XeF_4 reacts violently with water to give XeO_3 . The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valence shell.

Q.11 Argon is used in arc welding because of its-

[JEE 2007]

- (A) low reactivity with metal
 (B) ability to lower the melting point of metal
 (C) flammability
 (D) high calorific value

Q.12 The structure of XeO_3 is -

- (A) linear (B) planar (C) pyramidal (D) T-shaped

Q.13 XeF_4 and XeF_6 are expected to be -

- (A) oxidizing (B) reducing (C) unreactive (D) strongly basic

Paragraph for Question No. 14 to 16

There are some deposits of nitrates and phosphates in earth's crust. Nitrates are more soluble in water. Nitrates are difficult to reduce under the laboratory conditions but microbes do it easily. Ammonia forms large number of complexes with transition metal ions. Hybridization easily explains the ease of sigma donation capability of NH_3 and PH_3 . Phosphine is a flammable gas and is prepared from white phosphorous.

[JEE 2008]

Q.14 Among the following, the correct statement is-

- (A) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional
 (B) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
 (C) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
 (D) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

- Q.15 White phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a -
(A) dimerization reaction (B) disproportionation reaction
(C) condensation reaction (D) precipitation reaction
- Q.16 Among the following, the correct statement is
(A) Phosphates have no biological significance in humans
(B) Between nitrates and phosphates, phosphates are less abundant in earth's crust
(C) Between nitrates and phosphates, nitrates are less abundant in earth's crust
(D) Oxidation of nitrates is possible in soil
- Q.17 The nitrogen oxide(s) that contain(s) N-N bond(s) is (are) - [JEE 2009]
(A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5
- Q.18 The value of n in the molecular formula $\text{Be}_n\text{Al}_2\text{Si}_6\text{O}_{18}$ is [JEE 2010]
- Q.19 Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen? [JEE 2012]
(A) HNO_3 , NO, NH_4Cl , N_2 (B) HNO_3 , NO, N_2 , NH_4Cl
(C) HNO_3 , NH_4Cl , NO, N_2 (D) NO, HNO_3 , NH_4Cl , N_2
- Q.20 The correct statement(s) for orthoboric acid is/are [JEE (Advanced) 2014]
(A) It behaves as a weak acid in water due to self ionization.
(B) Acidity of its aqueous solution increases upon addition of ethylene glycol
(C) It has a three dimensional structure due to hydrogen bonding
(D) It is a weak electrolyte in water.
- Q.21 Under hydrolytic conditions, the compounds used for preparation of linear polymer and for chain termination, respectively, are [JEE (Advanced) 2015]
(A) CH_3SiCl_3 and $\text{Si}(\text{CH}_3)_4$ (B) $(\text{CH}_3)_2\text{SiCl}_2$ and $(\text{CH}_3)_3\text{SiCl}$
(C) $(\text{CH}_3)_2\text{SiCl}_2$ and CH_3SiCl_3 (D) SiCl_4 and $(\text{CH}_3)_3\text{SiCl}$
- Q.22 The crystalline form of borax has [JEE (Advanced) 2016]
(A) tetranuclear $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ unit
(B) all boron atoms in the same plane
(C) equal number of sp^2 and sp^3 hybridized boron atoms
(D) one terminal hydroxide per boron atom.
- Q.23 The nitrogen containing compound produced in the reaction of HNO_3 with P_4O_{10} [JEE (Advanced) 2016]
(A) can also be prepared by reaction of P_4 and HNO_3
(B) is diamagnetic
(C) contains one N-N bond
(D) reacts with Na metal producing a brown gas

- Q.24 The correct statement(s) about the oxoacids, HClO_4 and HClO , is(are) [JEE (Advanced) 2017]
(A) The conjugate base of HClO_4 is weaker base than H_2O
(B) The central atom in both HClO_4 and HClO is sp^3 hybridized
(C) HClO_4 is formed in the reaction between Cl_2 and H_2O
(D) HClO_4 is more acidic than HClO because of the resonance stabilization of its anion
- Q.25 Among the following, the correct statement(s) is(are) [JEE (Advanced) 2017]
(A) BH_3 has the three-centre two-electron bonds in its dimeric structure.
(B) $\text{Al}(\text{CH}_3)_3$ has the three-centre two-electron bonds in its dimeric structure.
(C) AlCl_3 has the three-centre two-electron bonds in its dimeric structure.
(D) The Lewis acidity of BCl_3 is greater than that of AlCl_3 .
- Q.26 The green colour produced in the borax bead test of a chromium(III) salt is due to - [JEE (Advanced) 2019]
(1) $\text{Cr}(\text{BO}_2)_3$ (2) $\text{Cr}_2(\text{B}_4\text{O}_7)_3$ (3) CrB (4) Cr_2O_3
- Q.27 The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc. HNO_3 to a compound with the highest oxidation state of sulphur is ____ [JEE (Advanced) 2019]
(Given data : Molar mass of water = 18 g mol^{-1})
- Q.28 A tin chloride Q undergoes the following reactions (not balanced)
 $\text{Q} + \text{Cl}^- \rightarrow \text{X}$
 $\text{Q} + \text{Me}_3\text{N} \rightarrow \text{Y}$
 $\text{Q} + \text{CuCl}_2 \rightarrow \text{Z} + \text{CuCl}$
X is a monoanion having pyramidal geometry. Both Y and Z are neutral compounds. Choose the correct option(s).
(1) The oxidation state of the central atom in Z is +2
(2) The central atoms in X is sp^3 hybridized
(3) The central atom in Z has one lone pair of electrons
(4) There is a coordinate bond in Y [JEE (Advanced) 2019]
- Q.29 With respect to hypochlorite, chlorate and perchlorate ions, choose the correct statement(s).
(A) The hypochlorite ion is the strongest conjugate base.
(B) The molecular shape of only chlorate ion is influenced by the lone pair of electrons of Cl.
(C) The hypochlorite and chlorate ions disproportionate to give rise to identical set of ions.
(D) The hypochlorite ion oxidizes the sulfite ion. [JEE (Advanced) 2020]

[JEE MAIN]

- Q.1 Arrange Ce^{3+} , La^{3+} , Pm^{3+} , and Yb^{3+} in increasing order of their ionic radius - [AIEEE 2002]
 (1) $\text{Yb}^{3+} < \text{Pm}^{3+} < \text{Ce}^{3+} < \text{La}^{3+}$ (2) $\text{Ce}^{3+} > \text{Yb}^{3+} < \text{Pm}^{3+} < \text{La}^{3+}$
 (3) $\text{Yb}^{3+} > \text{Pm}^{3+} < \text{La}^{3+} < \text{Ce}^{3+}$ (4) $\text{Pm}^{3+} < \text{La}^{3+} < \text{Ce}^{3+} < \text{Yb}^{3+}$
- Q.2 The atomic numbers of vanadium (V), Chromium (Cr), manganese (Mn) and iron (Fe) respectively 23, 24, 25 and 26. Which one of these may be expected to have the higher second ionization enthalpy? [AIEEE 2003]
 (1) Cr (2) Mn (3) Fe (4) V
- Q.3 The radius of La^{3+} is $1.06 \text{ \AA}$, which of the following given values will be closest to the radius of Lu^{3+} (At no. of Lu = 71, La = 57) - [AIEEE 2003]
 (1) $1.6 \text{ \AA}$ (2) $1.4 \text{ \AA}$ (3) $1.06 \text{ \AA}$ (4) $0.85 \text{ \AA}$
- Q.4 Cerium ($Z = 58$) is an important member of the lanthanoids. Which of the following statement about cerium is incorrect - [AIEEE 2004]
 (1) Cerium (IV) acts as an oxidising agent
 (2) The +3 oxidation state of cerium is more stable than the +4 oxidation state
 (3) The +4 oxidation state of cerium is not known in solutions
 (4) The common oxidation states of cerium are +3 and +4
- Q.5 Calomel on reaction with NH_4OH gives - [AIEEE 2004]
 (1) HgNH_2Cl (2) $\text{NH}_2\text{-Hg-Hg-Cl}$ (3) Hg_2O (4) HgO
- Q.6 The lanthanoid contraction is responsible for the fact that - [AIEEE 2005]
 (1) Zr and Y have about the same radius (2) Zr and Nb have similar oxidation state
 (3) Zr and Hf have about the same radius (4) Zr and Zn have similar oxidation state
- Q.7 Lanthanoid contraction is caused due to - [AIEEE 2006]
 (1) the same effective nuclear charge from Ce to Lu
 (2) the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
 (3) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
 (4) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge

- Q.8 Identify the incorrect statement among the following - [AIEEE 2007]
(1) d-block elements show irregular and erratic chemical properties among themselves
(2) La and Lu have partially filled d-orbitals and no other have partially filled orbitals
(3) The chemistry of various lanthanoids is very similar
(4) 4f and 5f-orbitals are equally shielded
- Q.9 The actinoids exhibits more number of oxidation states in general than the lanthanoids. This is because - [AIEEE 2007]
(1) The 5f-orbitals are more buried than the 4f-orbitals
(2) There is a similarity between 4f-and-5f in their angular part of the wave function
(3) The actinoids are more reactive than the lanthanoids
(4) The 5f-orbitals extend further from the nucleus than the 4f-orbitals
- Q.10 In context with the transition elements, which of the following statements is incorrect ? [AIEEE 2009]
(1) In the highest oxidation states, the transition metal show basic character and form cationic complexes.
(2) In the highest oxidation states of the first five transition elements (Sc to Mn), all the 4s and 3d electrons are used for bonding.
(3) Once the d^5 configuration is exceeded, the tendency to involve all the 3d electrons in bonding decreases.
(4) In addition to the normal oxidation states, the zero oxidation state is also shown by these elements in complexes.
- Q.11 Knowing that the Chemistry of lanthanoids (Ln) is dominated by its +3 oxidation state, which of the following statements is incorrect ? [AIEEE 2009]
(1) The ionic sizes of Ln (III) decrease in general with increasing atomic number.
(2) Ln (III) compounds are generally colourless.
(3) Ln (III) hydroxides are mainly basic in character.
(4) Because of the large size of the Ln (III) ions the bonding in its compounds is predominantly ionic in character.
- Q.12 In context of the lanthanoids, which of the following statements is not correct ? [AIEEE 2011]
(1) There is gradual decrease in the radii of the members with increasing atomic number in the series.
(2) All the members exhibit + 3 oxidation state.
(3) Because of similar properties the separation of lanthanoids is not easy.
(4) Availability of 4f electrons results in the formation of compounds in + 4 state for all the members of the series
- Q.13 Iron exhibit + 2 and + 3 oxidation states. Which of the following statements about iron is incorrect? [AIEEE 2012]
(1) Ferrous compounds are less volatile than the corresponding ferric compounds.
(2) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.
(3) Ferrous oxide is more basic in nature than the ferric oxide.
(4) Ferrous compounds are relatively more ionic than the corresponding ferric compounds.

- Q.14 Four successive members of the first row transition elements are listed below with atomic numbers.
Which one of them is expected to have the highest $E_{M^{3+}/M^{2+}}^0$ value? [JEE (Main-Offline) 2013]
(1) Mn (Z = 25) (2) Fe (Z = 26) (3) Co (Z = 27) (4) Cr (Z = 24)
- Q.15 Which of the following arrangements does not represent the correct order of the property stated against it? [JEE (Main-Offline) 2013]
(1) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size
(2) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: stability in aqueous solution
(3) $Sc < Ti < Cr < Mn$: number of oxidation states
(4) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: paramagnetic behaviour
- Q.16 Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown-red vapours of :
(1) CrO_3 (2) CrO_2Cl_2 (3) Cr_2O_3 (4) $CrCl_3$ [JEE (Main-Online) 2013]
- Q.17 The element with which of the following outer electron configuration may exhibit the largest number of oxidation states in its compounds :
(1) $3d^6 4s^2$ (2) $3d^8 4s^2$ (3) $3d^7 4s^2$ (4) $3d^5 4s^2$ [JEE (Main-Online) 2013]
- Q.18 When a small amount of $KMnO_4$ is added to concentrated H_2SO_4 , a green oily compound is obtained which is highly explosive in nature. Compound may be :
(1) MnO_2 (2) Mn_2O_7 (3) $MnSO_4$ (4) Mn_2O_3 [JEE (Main-Online) 2013]
- Q.19 Identify incorrect statement :
(1) Copper (I) compound are diamagnetic
(2) Copper(I) compound are colourless except where colour results from charge transfer
(3) Cu_2O is colourless
(4) Cu_2S is black [JEE (Main-Online) 2013]
- Q.20 Which one of the following cannot function as an oxidising agent?
(1) I^- (2) $Cr_2O_7^{2-}$ (3) NO_3^- (aq) (4) $S_{(s)}$ [JEE (Main-Online) 2013]
- Q.21 Which one of the following exhibits the largest number of oxidation states ?
(1) V(23) (2) Mn(25) (3) Ti(22) (4) Cr(24) [JEE (Main-Online) 2014]

- Q.22 Copper becomes green when exposed to moist air for a long period. This is due to :
 (1) the formation of basic copper sulphate layer on the surface of the metal.
 (2) the formation of a layer of cupric oxide on the surface of copper.
 (3) the formation of a layer of basic carbonate of copper on the surface of copper.
 (4) the formation of a layer of cupric hydroxide on the surface of copper.
[JEE (Main-Online) 2014]
- Q.23 The color of KMnO_4 is due to : **[JEE (Main-Offline) 2015]**
 (1) $\text{L} \rightarrow \text{M}$ charge transfer transition (2) $\sigma - \sigma^*$ transition
 (3) $\text{M} \rightarrow \text{L}$ charge transfer transition (4) d – d transition
- Q.24 Which of the following statements is false ?
 (1) CrO_4^{2-} is tetrahedral in shape (2) $\text{Na}_2\text{Cr}_2\text{O}_7$ is less soluble than $\text{K}_2\text{Cr}_2\text{O}_7$
 (3) $\text{Cr}_2\text{O}_7^{2-}$ has a Cr–O–Cr bond (4) $\text{Na}_2\text{Cr}_2\text{O}_7$ is a primary standard in volumetry
[JEE (Main-Online) 2015]
- Q.25 What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M ZnSO_4 ?
 (1) The copper metal will dissolve and zinc metal will be deposited
 (2) The copper metal will dissolve with evolution of hydrogen gas.
 (3) The copper metal will dissolve with evolution of oxygen gas.
 (4) No reaction will occur.
[JEE (Main-Online) 2016]
- Q.26 Which one of the following species is stable in aqueous solution?
 (1) Cr^{2+} (2) Cu^+ (3) MnO_4^{3-} (4) MnO_4^{2-}
[JEE (Main-Online) 2016]
- Q.27 Given **[JEE (Main-Offline) 2017]**
 $E^\circ_{\text{Cl}_2/\text{Cl}^-} = 1.36\text{V}$, $E^\circ_{\text{Cr}^{3+}/\text{Cr}} = -0.74\text{V}$
 $E^\circ_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}} = 1.33\text{V}$, $E^\circ_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51\text{V}$.
 Among the following, the strongest reducing agent is :
 (1) Mn^{2+} (2) Cr^{3+} (3) Cl^- (4) Cr
- Q.28 The pair of compounds having metals in their highest oxidation state is :
 (1) $[\text{MnO}_4]^-$ and CrO_2Cl_2 (2) $[\text{NiCl}_4]^{2-}$ and $[\text{CoCl}_4]^{2-}$
 (3) $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Cu}(\text{CN})_4]^{2-}$ (4) $[\text{FeCl}_4]^-$ and Co_2O_3
[JEE (Main-Online) 2017]
- Q.29 Which of the following ions does not liberate hydrogen gas on reaction with dilute acids ?
 (1) Ti^{2+} (2) V^{2+} (3) Cr^{2+} (4) Mn^{2+}
[JEE (Main-Online) 2017]

- Q.30 The highest value of the calculated spin only magnetic moment (in BM) among all the transition metal complexes is :
 (1) 4.90 (2) 3.87 (3) 5.92 (4) 6.93
[JEE (Main-Online) 2019]
- Q.31 The transition element that has lowest enthalpy of atomisation, is -
 (1) Fe (2) Zn (3) V (4) Cu
[JEE (Main-Online) 2019]
- Q.32 Consider the following reduction processes:
 $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn(s)} ; E^\circ = -0.76 \text{ V}$
 $\text{Ca}^{2+} + 2\text{e}^- \rightarrow \text{Ca(s)} ; E^\circ = -2.87 \text{ V}$
 $\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg(s)} ; E^\circ = -2.36 \text{ V}$
 $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni(s)} ; E^\circ = -0.25 \text{ V}$
 The reducing power of the metals increases in the order :
 (1) $\text{Ca} < \text{Mg} < \text{Zn} < \text{Ni}$ (2) $\text{Ni} < \text{Zn} < \text{Mg} < \text{Ca}$
 (3) $\text{Ca} < \text{Zn} < \text{Mg} < \text{Ni}$ (4) $\text{Zn} < \text{Mg} < \text{Ni} < \text{Ca}$
[JEE (Main-Online) 2019]
- Q.33 The effect of lanthanoid contraction in the lanthanoid series of elements by and large means:
 (1) increase in both atomic and ionic radii
 (2) increase in atomic radii and decrease in ionic radii
 (3) decrease in atomic radii and increase in ionic radii
 (4) decrease in both atomic and ionic radii
[JEE (Main-Online) 2019]
- Q.34 The element that usually does **not** show variable oxidation states is:
 (1) Cu (2) Ti (3) Sc (4) V
[JEE (Main-Online) 2019]
- Q.35 Match the metals (Column I) with the coordination compound (s) / enzymes (s) (column II)
- | Column (I) | Column (II) |
|--|--|
| Metals | Coordination compound(s) / enzyme(s) |
| (A) Co | (i) Wilkinson catalyst |
| (B) Zn | (ii) Chlorophyll |
| (C) Rh | (iii) Vitamin B ₁₂ |
| (D) Mg | (iv) Carbonic anhydrase |
| (1) (A)→(iii); (B)→(iv); (C)→(i); (D)→(ii) | (2) (A)→(i); (B)→(ii); (C)→(iii); (D)→(iv) |
| (3) (A)→(ii); (B)→(i); (C)→(iv); (D)→(iii) | (4) (A)→(iv); (B)→(iii); (C)→(i); (D)→(ii) |
- [JEE (Main-Online) 2019]**
- Q.36 The incorret statement is :
 (1) In manganate and permanganate ions, the π -bonding takes place by overlap of p-orbitals of oxygen and d-orbitals of manganese
 (2) Manganate ion is green in colour and permanganate ion in purple in colour
 (3) Manganate and permanganate ions are paramagnetic
 (4) Manganate and permanganate ions are tetrahedral
[JEE (Main-Online) 2020]

- Q.37 The incorrect statement(s) among (a) - (c) is (are) :-
(a) W(VI) is more stable than Cr(VI).
(b) in the presence of HCL, permanganate titrations provide satisfactory results.
(c) some lanthanoid oxides can be used as phosphors.
(1) (a) and (b) only (2) (a) only (3) (b) and (c) only (4) (b) only
[JEE (Main-Online) 2020]
- Q.38 The set that contains atomic number of only transition element is -
(1) 21, 32, 53, 64 (2) 21, 25, 42, 72 (3) 9, 17, 34, 38 (4) 37, 42, 50, 64
[JEE (Main-Online) 2020]
- Q.39 The correct electronic configuration and spin-only magnetic moment (BM) of Gd^{3+} ($Z = 64$), respectively, are
(1) $[\text{Xe}]5f^7$ and 8.9 (2) $[\text{Xe}]4f^7$ and 7.9 (3) $[\text{Xe}]5f^7$ and 7.9 (4) $[\text{Xe}]4f^7$ and 8.9
[JEE (Main-Online) 2020]
- Q.40 The lanthanoid that does NOT show +4 oxidation state is
(1) Dy (2) Eu (3) Ce (4) Tb
[JEE (Main-Online) 2020]
- Q.41 Mischmetal is an alloy consisting mainly of:
(1) lanthanoid metals (2) actinoid metals
(3) actinoid and transition metals (4) lanthanoid and actinoid metals
[JEE (Main-Online) 2020]
- Q.42 The atomic radius of Ag is closest to
(1) Ni (2) Hg (3) Cu (4) Au
[JEE (Main-Online) 2020]
- Q.43 The third ionization enthalpy is minimum for
(1) Mn (2) Co (3) Ni (4) Fe
[JEE (Main-Online) 2020]
- Q.44 The sum of the total number of bonds between chromium and oxygen atoms in chromate and dichromate ions is _____.
[JEE (Main-Online) 2020]
- Q.45 The electronic configurations of bivalent europium and trivalent cerium are (atomic number : Xe = 54, Ce = 58, Eu = 63)
(1) $[\text{Xe}] 4f^7$ and $[\text{Xe}] 4f^1$ (2) $[\text{Xe}] 4f^7 6s^2$ and $[\text{Xe}] 4f^2 6s^2$
(3) $[\text{Xe}] 4f^4$ and $[\text{Xe}] 4f^9$ (4) $[\text{Xe}] 4f^2$ and $[\text{Xe}] 4f^7$
[JEE (Main-Online) 2020]

[JEE ADVANCED]

- Q.1 Which of the following is an organometallic compound ? [JEE 1997]
(A) lithium methoxide (B) lithium acetate
(C) lithium dimethylamide (D) methyl lithium
- Q.2 Which of the following compounds is expected to be colored ? [JEE 1997]
(A) Ag_2SO_4 (B) CuF_2 (C) MgF_2 (D) CuCl
- Q.3 In the standardization of $\text{Na}_2\text{S}_2\text{O}_3$ using $\text{K}_2\text{Cr}_2\text{O}_7$ by iodometry, the equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ is [JEE 1999]
(A) (molecular weight) / 2 (B) (molecular weight) / 6
(C) (molecular weight) / 3 (D) same as molecular weight
- Q.4 Amongst the following, identify the species with an atom in +6 oxidation state. [JEE 2000]
(A) MnO_4^- (B) $\text{Cr}(\text{CN})_6^{3-}$ (C) NiF_6^{2-} (D) CrO_2Cl_2
- Q.5 The complex ion which has no "d" electrons in the central atom is (at. no Cr=24, Mn = 25, Fe = 26, Co = 27) [JEE 2001]
(A) $[\text{MnO}_4]^-$ (B) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (C) $[\text{Fe}(\text{CN})_6]^{3-}$ (D) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
- Q.6 The pair of the compounds in which both the metals are in the highest possible oxidation state is [JEE 2004]
(A) $[\text{Fe}(\text{N})_6]^{3-}$, $[\text{Co}(\text{CN})_6]^{3-}$ (B) CrO_2Cl_2 , MnO_4^-
(C) TiO_3 , MnO_2 (D) $[\text{Co}(\text{CN})_6]^{3-}$, MnO_2
- Q.7 The correct statement(s) about Cr^{2+} and Mn^{3+} is (are) [JEE (Advanced) 2015]
[Atomic numbers of Cr = 24 and Mn = 25]
(A) Cr^{2+} is a reducing agent
(B) Mn^{3+} is an oxidizing agent
(C) Both Cr^{2+} and Mn^{3+} exhibit d^4 electronic configuration
(D) When Cr^{2+} is used as a reducing agent, the chromium ion attains d^5 electronic configuration.
- Q.8 In the chemical reaction between stoichiometric quantities of KMnO_4 and KI in weakly basic solution, what is the number of moles of I_2 released for 4 moles of KMnO_4 consumed ? [JEE (Advanced) 2020]

[JEE MAIN]

- Q.1 HgCl_2 on passing H_2S gives - [AIEEE 2002]
 (1) HgS (2) Hg_2S (3) $\text{Hg} + \text{HgS}$ (4) $\text{HgS} + \text{H}_2\text{S}$
- Q.2 How do we differentiate between Fe^{3+} and Cr^{3+} in qualitative analysis gp. III : [AIEEE 2002]
 (1) By taking excess of NH_4OH
 (2) By increasing NH_4^+ ion concentration
 (3) By decreasing OH^- ion concentration
 (4) Both (2) and (3)
- Q.3 Which statement is **correct** : [AIEEE 2003]
 (1) Fe^{3+} ions give deep green precipitate with $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution
 (2) On heating K^+ , Ca^{2+} and HCO_3^- ions, we get a precipitate of $\text{K}_2[\text{Ca}(\text{CO}_3)_2]$
 (3) Manganese salts give a violet borax bead test in the reducing flame
 (4) From a mixed precipitate of AgCl and AgI ammonia solution dissolves only AgCl
- Q.4 Sodium carboate cannot be used in place of $(\text{NH}_4)_2\text{CO}_3$ for the identification of Ca^{2+} , Ba^{2+} and Sr^{2+} ions (in group V) during mixture analysis because :
 (1) Mg^{2+} ions will also be precipitated
 (2) sodium ions will react with acid radicals
 (3) Concentration of CO_3^{2-} ions is very low
 (4) Na^+ ions will interfere with the detection of Ca^{2+} , Ba^{2+} , Sr^{2+} ions [JEE (Main-Online) 2013]
- Q.5 Which of the following statements is incorrect?
 (1) Fe^{3+} ion gives blood red colour with SCN^- ion
 (2) Fe^{2+} ion also gives blood red colour with SCN^- ion.
 (3) On passing H_2S into Na_2ZnO_2 solution, a white ppt of ZnCl is formed.
 (4) Cupic ion react with excess of ammonia soln to give deep blue colour of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ion [JEE (Main-Online) 2013]
- Q.6 Which of the following is **not** formed when H_2S reacts with acidic $\text{K}_2\text{Cr}_2\text{O}_7$ solution ?
 (1) S (2) $\text{Cr}_2(\text{SO}_4)_3$ (3) K_2SO_4 (4) CrSO_4 [JEE (Main-Online) 2014]
- Q.7 Consider the coordination compound, $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$. In the formation of this complex, the species which acts as the Lewis acid is :
 (1) Cl^- (2) NH_3 (3) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (4) Co^{3+} [JEE (Main-Online) 2014]

- Q.8 Consider the following equilibrium
 $\text{AgCl} \downarrow + 2\text{NH}_3 \rightleftharpoons [\text{Ag}(\text{NH}_3)_2]^+ + \text{Cl}^-$
 White precipitate of AgCl appears on adding which of the following ?
 (1) aqueous NaCl (2) aqueous HNO_3 (3) aqueous NH_4Cl (4) NH_3
[JEE (Main-Online) 2014]

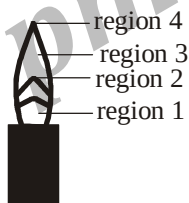
- Q.9 Which of the following compounds is not colored yellow ?
 (1) $(\text{NH}_4)_3[\text{As}(\text{Mo}_3\text{O}_{10})_4]$ (2) BaCrO_4
 (3) $\text{Zn}_2[\text{Fe}(\text{CN})_6]$ (4) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$
[JEE (Main-Offline) 2015]

- Q.10 An aqueous solution of a salt X turns blood red on treatment with SCN^- and blue on treatment with $\text{K}_4[\text{Fe}(\text{CN})_6]$. X also gives a positive chromyl chloride test. The salt X is :
 (1) $\text{Cu}(\text{NO}_3)_2$ (2) FeCl_3 (3) $\text{Fe}(\text{NO}_3)_3$ (4) CuCl_2
[JEE (Main-Online) 2015]

- Q.11 The cation that will not be precipitated by H_2S in the presence of dil HCl is :
 (1) Pb^{2+} (2) Co^{2+} (3) Cu^{2+} (4) As^{3+}
[JEE (Main-Online) 2015]

- Q.12 A pink coloured salt turns blue on heating. The presence of which cation is most likely ?
 (1) Zn^{2+} (2) Fe^{2+} (3) Cu^{2+} (4) Co^{2+}
[JEE (Main-Online) 2015]

- Q.13 The hottest region of Bunsen flame shown in the figure below is : **[JEE (Main-Offline) 2016]**



- (1) region 4 (2) region 1 (3) region 2 (4) region 3
- Q.14 Extraction of copper by smelting uses silica as an additive to remove :
 (1) Cu_2S (2) FeO (3) FeS (4) Cu_2O
[JEE (Main-Online) 2016]

- Q.15 Sodium extract is heated with concentrated HNO_3 before testing for halogens because :
 (1) Silver halides are totally insoluble in nitric acid.
 (2) Ag_2S and AgCN are soluble in acidic medium.
 (3) S^{2-} and CN^- , if present, are decomposed by conc. HNO_3 and hence do not interfere in the test.
 (4) Ag reacts faster with halides in acidic medium.
[JEE (Main-Online) 2016]

- Q.16 Sodium salt of an organic acid 'X' produces effervescence with conc. H_2SO_4 , 'X' reacts with the acidified aqueous CaCl_2 solution to give a white precipitate which decolourises acidic solution of KMnO_4 . 'X' is:
(1) HCOONa (2) CH_3COONa [JEE (Main-Offline) 2017]
(3) $\text{Na}_2\text{C}_2\text{O}_4$ (4) $\text{C}_6\text{H}_5\text{COONa}$
- Q.17 A solution containing a group-IV cation gives a precipitate on passing H_2S . A solution of this precipitate in dil. HCl produces a white precipitate with NaOH solution and bluish-white precipitate with basic potassium ferrocyanide. The cation is :
(1) Co^{2+} (2) Ni^{2+} (3) Mn^{2+} (4) Zn^{2+}
[JEE (Main-Online) 2017]
- Q.18 The **incorrect** statement is :
(1) Cu^{2+} salts give red coloured borax bead test in reducing flame.
(2) Cu^{2+} and Ni^{2+} ions give black precipitate with H_2S in presence of HCl solution.
(3) Ferric ion gives blood red colour with potassium thiocyanate.
(4) Cu^{2+} ion gives chocolate coloured precipitate with potassium ferrocyanide solution.
[JEE (Main-Online) 2018]
- Q.19 For the following Assertion and Reason, the correct option is
Assertion (A) : When Cu (II) and sulphide ions are mixed, they react together extremely quickly to give a solid.
Reason (R) : The equilibrium constant of $\text{Cu}^{2+}(\text{aq}) + \text{S}^{2-}(\text{aq}) \rightleftharpoons \text{CuS}(\text{s})$ is high because the solubility product is low.
(1) Both (A) and (R) are true and (R) is the explanation for (A)
(2) Both (A) and (R) are false
(3) (A) is false and (R) is true
(4) Both (A) and (R) are true but (R) is not the explanation for (A)
[JEE (Main-Online) 2020]

[JEE ADVANCED]

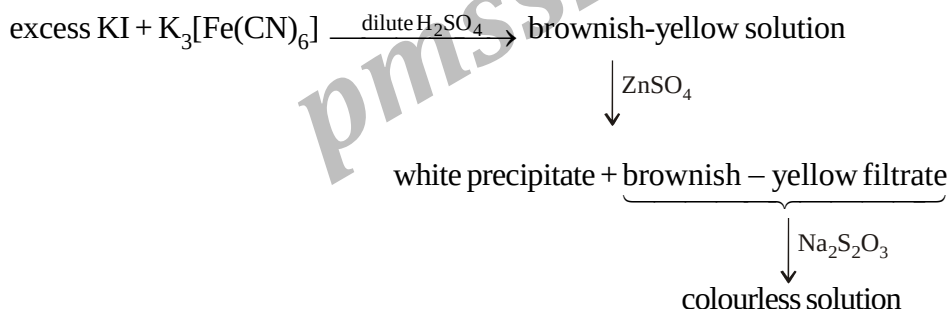
- Q.1 In nitroprusside ion the iron and NO exist. NO exist as Fe^{II} and NO^+ rather than Fe^{III} and NO. These forms can be differentiated by: [JEE 1998]
 (A) estimating the concentration of iron
 (B) measuring the concentration of CN
 (C) measuring the solid state magnetic moment
 (D) thermally decomposing the compound.
- Q.2 **Statement-1** : Sulphate is estimated as BaSO_4 and not as MgSO_4 . [JEE 1998]
Statement-2 : Ionic radius of Mg^{2+} is smaller than that of Ba^{2+} .
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- Q.3 Which of the following statement(s) is (are) correct with reference to the ferrous and ferric ions: [JEE 1998]
 (A) Fe^{3+} gives brown colour with potassium ferricyanide
 (B) Fe^{2+} gives blue precipitate with potassium ferricyanide
 (C) Fe^{3+} give red colour with potassium thiocyanate
 (D) Fe^{2+} gives brown colour with ammonium thiocyanate
- Q.4 A sodium salt of an unknown anion when treated with MgCl_2 give white precipitate only on boiling. The anion is: [JEE 2004]
 (A) SO_4^{2-} (B) HCO_3^- (C) CO_3^{2-} (D) NO_3^-
- Q.5 Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The number of moles of Mohr's salt required per mole of dichromate is: [JEE 2007]
 (A) 3 (B) 4 (C) 5 (D) 6
- Q.6 The species present in solution when CO_2 is dissolved in water are [JEE 2007]
 (A) $\text{CO}_2, \text{H}_2\text{CO}_3, \text{HCO}_3^-, \text{CO}_3^{2-}$ (B) $\text{H}_2\text{CO}_3, \text{CO}_3^{2-}$
 (C) $\text{CO}_3^{2-}, \text{HCO}_3^-$ (D) $\text{CO}_2, \text{H}_2\text{CO}_3$
- Q.7 **Column I** [JEE 2007]
 (A) $\text{O}_2^- \rightarrow \text{O}_2 + \text{O}_2^{2-}$
 (B) $\text{CrO}_4^{2-} + \text{H}^+ \rightarrow$
 (C) $\text{MnO}_4^- + \text{NO}_2^- + \text{H}^+ \rightarrow$
 (D) $\text{NO}_3^- + \text{H}_2\text{SO}_4 + \text{Fe}^{2+} \rightarrow$
- Column II**
 (P) Redox reaction
 (Q) One of the products has trigonal planar structure
 (R) Dimeric bridged tetrahedral metal ion
 (S) Disproportionation

- Q.8 A solution of colourless salt H on boiling with excess NaOH produces a non-flammable gas. The gas evolution ceases after sometime. Upon addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt(s) H is (are) [JEE 2008]
- (A) NH_4NO_3 (B) NH_4NO_2 (C) NH_4Cl (D) $(\text{NH}_4)_2\text{SO}_4$

Paragraph for question nos. 9 to 11

p-Amino-N, N-dimethylaniline is added to a strongly acidic solution of **X**. The resulting solution is treated with a few drops of aqueous solution of **Y** to yield blue coloration due to the formation of methylene blue. Treatment of the aqueous solution of **Y** with the reagent potassium hexacyanoferrate(II) leads to the formation of an intense blue precipitate. The precipitate dissolves on excess addition of the reagent. Similarly, treatment of the solution of **Y** with the solution of potassium hexacyanoferrate(III) leads to a brown coloration due to the formation of **Z**. [JEE 2009]

- Q.9 The compound **X** is
(A) NaNO_3 (B) NaCl (C) Na_2SO_4 (D) Na_2S
- Q.10 The compound **Y** is
(A) MgCl_2 (B) FeCl_2 (C) FeCl_3 (D) ZnCl_2
- Q.11 The compound **Z** is
(A) $\text{Mg}_2[\text{Fe}(\text{CN})_6]$ (B) $\text{Fe}[\text{Fe}(\text{CN})_6]$ (C) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (D) $\text{K}_2\text{Zn}_3[\text{Fe}(\text{CN})_6]_2$
- Q.12 For the given aqueous reactions, which of the statement(s) is (are) true?



- (A) The first reaction is a redox reaction
(B) White precipitate is $\text{Zn}_3[\text{Fe}(\text{CN})_6]_2$
(C) Addition of filtrate to starch solution gives blue colour
(D) White precipitate is soluble in NaOH solution.

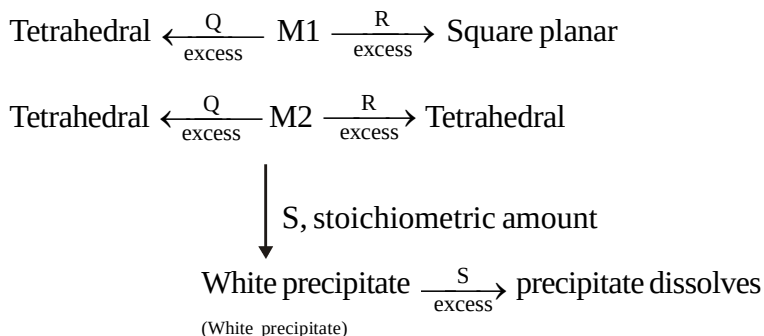
[JEE 2012]

- Q.13 Among PbS , CuS , HgS , MnS , Ag_2S , NiS , CoS , Bi_2S_3 and SnS_2 , the total number of BLACK coloured sulphides is [JEE (Advanced) 2014]

Paragraph for Questions 14 and 15

An aqueous solution of metal ion **M1** reacts separately with reagents **Q** and **R** in excess to give tetrahedral and square planar complexes, respectively. An aqueous solution of another metal ion **M2** always forms tetrahedral complexes with these reagents. Aqueous solution of **M2** on reaction with reagent **S** gives white precipitate which dissolves in excess of **S**. The reactions are summarized in the scheme given below :

SCHEME :



- Q.14 M1, Q and R respectively are [JEE (Advanced) 2014]
 (A) Zn^{2+} , KCN and HCl (B) Ni^{2+} , HCl and KCN
 (C) Cd^{2+} , KCN and HCl (D) Co^{2+} , HCl and KCN
- Q.15 Reagent S is [JEE (Advanced) 2014]
 (A) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (B) Na_2HPO_4 (C) K_2CrO_4 (D) KOH
- Q.16 Addition of excess aqueous ammonia to a pink coloured aqueous solution of $\text{MCl}_2 \cdot 6\text{H}_2\text{O}$ (**X**) and NH_4Cl gives an octahedral complex **Y** in the presence of air. In aqueous solution, complex **Y** behaves as 1 : 3 electrolyte. The reaction of **X** with excess HCl at room temperature results in the formation of a blue coloured complex **Z**. The calculated spin only magnetic moment of **X** and **Z** is 3.87 B.M., whereas it is zero for complex **Y**. Among the following options, which statement(s) is(are) correct? [JEE (Advanced) 2017]
 (A) The hybridization of the central metal ion in **Y** is d^2sp^3
 (B) When **X** and **Z** are in equilibrium at 0°C , the colour of the solution is pink
 (C) **Z** is a tetrahedral complex
 (D) Addition of silver nitrate to **Y** gives only two equivalents of silver chloride
- Q.17 The correct option(s) to distinguish nitrate salts of Mn^{2+} and Cu^{2+} taken separately is (are) [JEE (Advanced) 2018]
 (A) Mn^{2+} shows the characteristic green colour in the flame test
 (B) Only Cu^{2+} shows the formation of precipitate by passing H_2S in acidic medium
 (C) Only Mn^{2+} shows the formation of precipitate by passing H_2S in faintly basic medium
 (D) Cu^{2+}/Cu has higher reduction potential than Mn^{2+}/Mn (measured under similar conditions)
- Q.18 The correct statement(s) regarding the binary transition metal carbonyl compounds is (are) [JEE (Advanced) 2018]
 (Atomic numbers: Fe = 26, Ni = 28)
 (A) Total number of valence shell electrons at metal centre in $\text{Fe}(\text{CO})_5$ or $\text{Ni}(\text{CO})_4$ is 16
 (B) These are predominantly low spin in nature
 (C) Metal–carbon bond strengthens when the oxidation state of the metal is lowered
 (D) The carbonyl C–O bond weakens when the oxidation state of the metal is increased

- Q.19 Among the species given below, the total number of diamagnetic species is ____.
H atom, NO_2 monomer, O_2^- (superoxide), dimeric sulphur in vapour phase,
 Mn_3O_4 , $(\text{NH}_4)_2[\text{FeCl}_4]$, $(\text{NH}_4)_2[\text{NiCl}_4]$, K_2MnO_4 , K_2CrO_4 [JEE (Advanced) 2018]
- Q.20 The ammonia prepared by treating ammonium sulphate with calcium hydroxide is completely used by $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ to form a stable coordination compound. Assume that both the reactions are 100% complete. If 1584 g of ammonium sulphate and 952 g of $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ are used in the preparation, the combined weight (in grams) of gypsum and the nickel-ammonia coordination compound thus produced is ____.
(Atomic weights in g mol^{-1} : H = 1, N = 14, O = 16, S = 32, Cl = 35.5, Ca = 40, Ni = 59) [JEE (Advanced) 2018]
- Q.21 The correct option(s) regarding the complex $[\text{Co}(\text{en})(\text{NH}_3)_3(\text{H}_2\text{O})]^{3+}$ [JEE (Advanced) 2018]
(en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$) is (are)
(A) It has two geometrical isomers
(B) It will have three geometrical isomers if bidentate 'en' is replaced by two cyanide ligands
(C) It is paramagnetic
(D) It absorbs light at longer wavelength as compared to $[\text{Co}(\text{en})(\text{NH}_3)_4]^{3+}$
- Q.22 Match each set of hybrid orbitals from LIST-I with complex(es) given in LIST-II. [JEE (Advanced) 2018]
- | LIST-I | | LIST-II | |
|--------|-------------------------|---------|--|
| P. | dsp^2 | 1. | $[\text{FeF}_6]^{4-}$ |
| Q. | sp^3 | 2. | $[\text{Ti}(\text{H}_2\text{O})_3\text{Cl}_3]$ |
| R. | sp^3d^2 | 3. | $[\text{Cr}(\text{NH}_3)_6]^{3+}$ |
| S. | d^2sp^3 | 4. | $[\text{FeCl}_4]^{2-}$ |
| | | 5. | $\text{Ni}(\text{CO})_4$ |
| | | 6. | $[\text{Ni}(\text{CN})_4]^{2-}$ |
- The correct option is
(A) $\text{P} \rightarrow 5$; $\text{Q} \rightarrow 4,6$; $\text{R} \rightarrow 2,3$; $\text{S} \rightarrow 1$
(B) $\text{P} \rightarrow 5,6$; $\text{Q} \rightarrow 4$; $\text{R} \rightarrow 3$; $\text{S} \rightarrow 1, 2$
(C) $\text{P} \rightarrow 6$; $\text{Q} \rightarrow 4,5$; $\text{R} \rightarrow 1$; $\text{S} \rightarrow 2, 3$
(D) $\text{P} \rightarrow 4,6$; $\text{Q} \rightarrow 5,6$; $\text{R} \rightarrow 1,2$; $\text{S} \rightarrow 3$
- Q.23 The green colour produced in the borax bead test of a chromium(III) salt is due to -
(1) $\text{Cr}(\text{BO}_2)_3$ (2) $\text{Cr}_2(\text{B}_4\text{O}_7)_3$ (3) CrB (4) Cr_2O_3 [JEE (Advanced) 2019]

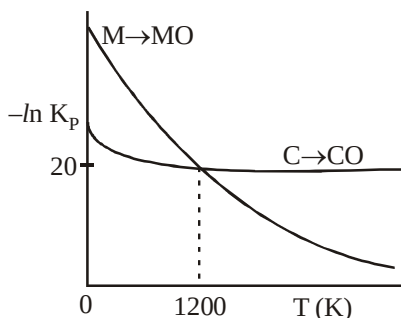
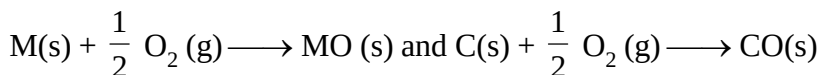
[JEE MAIN]

- Q.1 Aluminium is industrially prepared by : [AIEEE 2002]
 (1) Fused cryolite (2) Bauxite ore (3) Alumina (4) Borax
- Q.2 Which one of the following ores is best concentrated by froath-flotation method ? [AIEEE 2004]
 (1) Magnetite (2) Cassiterite (3) Galena (4) Malachite
- Q.3 During the process of electrolytic refining of copper, some metals present as impurity settle as 'anode mud'. These are – [AIEEE 2005]
 (1) Pb and Zn (2) Sn and Ag (3) Fe and Ni (4) Ag and Au
- Q.4 Heating mixture of Cu_2O and Cu_2S will give – [AIEEE 2005]
 (1) $\text{Cu} + \text{SO}_3$ (2) $\text{Cu} + \text{SO}_2$ (3) Cu_2SO_3 (4) $\text{CuO} + \text{CuS}$
- Q.5 Which of the following factors is of no significance for roasting sulphide ores to the oxides and not subjecting the sulphide ores to carbon reduction directly [AIEEE 2008]
 (1) Metal sulphides are thermodynamically more stable than CS_2
 (2) CO_2 is thermodynamically more stable than CS_2
 (3) Metal sulphides are less stable than the corresponding oxides
 (4) CO_2 is more volatile than CS_2
- Q.6 Which method of purification is represented by the following equation : [AIEEE 2012]

$$\text{Ti(s)} + 2\text{I}_2(\text{g}) \xrightarrow{523\text{K}} \text{TiI}_4(\text{g}) \xrightarrow{1700\text{K}} \text{Ti(s)} + 2\text{I}_2(\text{g})$$
 (1) Poling (2) Van Arkel (3) Zone refining (4) Cupellation
- Q.7 Calcination is the process in which :
 (1) Ore is heated strongly below its melting point in the presence of excess of air and is used for the conversion of carbonates and hydrated oxide ores to their respective oxides.
 (2) Ore is heated strongly above its melting point in the limited supply of air to convert sulphide ores to their respective oxides.
 (3) Ore is heated strongly below its melting point in the absence or limited supply of air and is used for the conversion of sulphide ores to their respective oxides.
 (4) Ore is heated strongly below its melting point either in the limited or absence of air and is used to convert carbonates and hydrated oxide ores to their respective oxides.
[JEE (Main-Online) 2013]
- Q.8 In Goldschmidt aluminothermite process which of the following reducing agents is used :
 (1) Al-powder (2) sodium (3) coke (4) calcium
[JEE (Main-Online) 2013]

- Q.9 The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is:
(1) Ca (2) Cu (3) Cr (4) Ag
[JEE (Main-Offline) 2014]
- Q.10 Which series of reactions correctly represents chemical relations related to iron and its compound ?
(1) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{FeO} \xrightarrow{\text{dil H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{heat}} \text{Fe}$ [JEE (Main-Offline) 2014]
(2) $\text{Fe} \xrightarrow{\text{Cl}_2, \text{heat}} \text{FeCl}_3 \xrightarrow{\text{heat, air}} \text{FeCl}_2 \xrightarrow{\text{Zn}} \text{Fe}$
(3) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{Fe}_3\text{O}_4 \xrightarrow{\text{CO}, 600^\circ\text{C}} \text{FeO} \xrightarrow{\text{CO}, 700^\circ\text{C}} \text{Fe}$
(4) $\text{Fe} \xrightarrow{\text{dil H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{H}_2\text{SO}_4, \text{O}_2} \text{Fe}_2(\text{SO}_4)_3 \xrightarrow{\text{heat}} \text{Fe}$
- Q.11 The form of iron obtained from blast furnace is
(1) Pig iron (2) Cast iron (3) Wrought Iron (4) Steel
[JEE (Main-Online) 2014]
- Q.12 Which one of the following ores is known as Malachite :
(1) Cu_2O (2) CuFeS_2 (3) $\text{Cu}(\text{OH})_2 \cdot \text{CuCO}_3$ (4) Cu_2S
[JEE (Main-Online) 2014]
- Q.13 In the context of the Hall-Heroult process for the extraction of Al, which of the following statements is false?
(1) Al^{3+} is reduced at the cathode to form Al
(2) Na_3AlF_6 serves as the electrolyte
(3) CO and CO_2 are produced in this process
(4) Al_2O_3 is mixed with CaF_2 which lowers the melting point of the mixture and brings conductivity
[JEE (Main-Offline) 2015]
- Q.14 In the isolation of metals, calcination process usually results in :
(1) metal oxide (2) metal sulphide (3) metal hydroxide (4) metal carbonate
[JEE (Main-Online) 2015]
- Q.15 Calamine is an ore of :
(1) Iron (2) Zinc (3) Aluminium (4) Copper
[JEE (Main-Online) 2015]
- Q.16 Galvanization is applying a coating of :
(1) Zn (2) Pb (3) Cr (4) Cu
[JEE (Main-Offline) 2016]
- Q.17 Which one of the following ores is best concentrated by froth floatation method?
(1) Malachite (2) Magnetite (3) Siderite (4) Galena
[JEE (Main-Offline) 2016]

Q.18 The plot shows the variation of $-\ln K_p$ versus temperature for the two reactions.



Identify the correct statement:

- (1) At $T > 1200$ K, carbon will reduce $MO(s)$ to $M(s)$.
- (2) At $T < 1200$ K, the reaction $MO(s) + C(s) \longrightarrow M(s) + CO(g)$ is spontaneous
- (3) At $T < 1200$ K, oxidation of carbon is unfavourable
- (4) Oxidation of carbon is favourable at all temperatures.

[JEE (Main-Online) 2016]

Q.19 When metal 'M' is treated with NaOH, a white gelatinous precipitate 'X' is obtained which is soluble in excess of NaOH. Compound 'X' when heated strongly gives an oxide which is used in chromatography as an adsorbent. The metal 'M' is :

- (1) Al
- (2) Fe
- (3) Zn
- (4) Ca

[JEE (Main-Offline) 2018]

Q.20 In the leaching method, bauxite ore is digested with a concentrated solution of NaOH that produces 'X'. When CO_2 gas is passed through the aqueous solution of 'X', a hydrated compound 'Y' is precipitated. 'X' and 'Y' respectively are :

- (1) $NaAlO_2$ and $Al_2(CO_3)_3 \cdot xH_2O$
- (2) $Al(OH)_3$ and $Al_2O_3 \cdot xH_2O$
- (3) $Na[Al(OH)_4]$ and $Al_2(CO_3)_3 \cdot xH_2O$
- (4) $Na[Al(OH)_4]$ and $Al_2O_3 \cdot xH_2O$

[JEE (Main-Online) 2018]

Q.21 Two compound I and II are eluted by column chromatography (adsorption of $I > II$). Which of the following is a **correct** statement?

- (1) II moves faster and has higher R_f value than I
- (2) I moves faster and has higher R_f value than II
- (3) II moves slower and has higher R_f value than I
- (4) I moves slower and has higher R_f value than II

[JEE (Main-Online) 2018]

Q.22 In the extraction of copper from its sulphide ore, metal is finally obtained by the oxidation of cuprous sulphide with

- (1) Fe_2O_3
- (2) Cu_2O
- (3) SO_2
- (4) CO

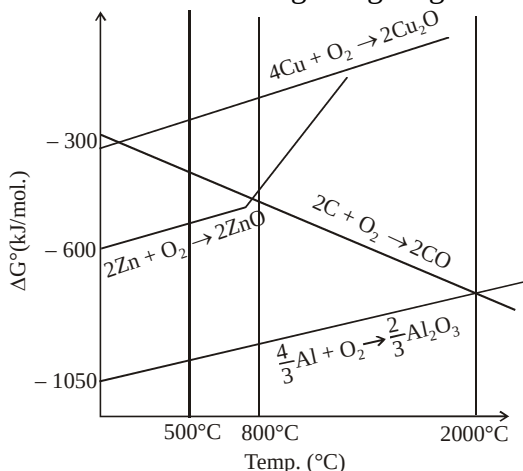
[JEE (Main-Online) 2018]

Q.23 The ore that contains both iron and copper is :

- (1) azurite (2) dolomite (3) malachite (4) copper pyrites

[JEE (Main-Online) 2019]

Q.24 The correct statement regarding the given Ellingham diagram is :



- (1) At 800°C, Cu can be used for the extraction of Zn from ZnO.
 (2) At 500°C, coke can be used for the extraction of Zn from ZnO.
 (3) At 1400°C, Al can be used for the extraction of Zn from ZnO.
 (4) Coke cannot be used for the extraction of Cu from Cu₂O.

[JEE (Main-Online) 2019]

Q.25 Hall-Heroult's process is given by :

- (1) $\text{ZnO} + \text{C} \xrightarrow{\text{Coke, 1673K}} \text{Zn} + \text{CO}$
 (2) $\text{Cu}^{2+}(\text{aq}) + \text{H}_2(\text{g}) \rightarrow \text{Cu}(\text{s}) + 2\text{H}^+(\text{aq})$
 (3) $2\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Al} + 3\text{CO}_2$
 (4) $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$

[JEE (Main-Online) 2019]

Q.26 Match the ores (Column A) with the metals (column B)

- | Column (A)
Ores | Column (B)
Metals |
|--------------------|----------------------|
| (I) Siderite | (a) Zinc |
| (II) Kaolinite | (b) Copper |
| (III) Malachite | (c) Iron |
| (IV) Calamine | (d) Aluminium |
- (1) (I)→(a); (II)→(b); (III)→(c); (IV)→(d)
 (2) (I)→(c); (II)→(d); (III)→(b); (IV)→(a)
 (3) (I)→(c); (II)→(d); (III)→(a); (IV)→(b)
 (4) (I)→(b); (II)→(c); (III)→(d); (IV)→(a)

[JEE (Main-Online) 2019]

Q.27 In the Hall-Heroult process, aluminium is formed at the cathode. The cathode is made out of :

- (1) Platinum
- (2) Copper
- (3) Carbon
- (4) Pure aluminium

[JEE (Main-Online) 2019]

Q.28 The pair that does NOT require calcination is :

- (1) Fe_2O_3 and $\text{CaCO}_3 \cdot \text{MgCO}_3$
- (2) ZnO and $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- (3) ZnO and MgO
- (4) ZnCO_3 and CaO

[JEE (Main-Online) 2019]

Q.29 Cast iron is used for the manufacture of :

- (1) wrought iron and pig iron
- (2) wrought iron and steel
- (3) wrought iron, pig iron and steel
- (4) pig iron, scrap iron and steel

[JEE (Main-Online) , 2020]

Q.30 Among statements (a) -(d), the correct ones are :

- (a) Lime stone is decomposed to CaO during the extraction of iron from its oxides.
- (b) In the extraction of silver, silver is extracted as an anionic complex.
- (c) Nickel is purified by Mond's process.
- (d) Zr and Ti are purified by Van Arkel method.

- (1) (c) and (d) only
- (2) (a), (c) and (d) only
- (3) (b), (c) and (d) only
- (4) (a), (b), (c) and (d)

[JEE (Main-Online) 2020]

Q.31 An Ellingham diagram provides information about :

- (1) the pressure dependence of the standard electrode potentials of reduction reactions involved in the extraction of metals.
- (2) the kinetics of the reduction process.
- (3) the temperature dependence of the standard Gibbs energies of formation of some metal oxides.
- (4) the conditions of pH and potential under which a species is thermodynamically stable.

[JEE (Main-Online) 2020]

Q.32 Boron and silicon of very high purity can be obtained through :

- (1) vapour phase refining
- (2) electrolytic refining
- (3) liquation
- (4) zone refining

[JEE (Main-Online) 2020]

Q.33 The INCORRECT statement is :

- (1) bronze is an alloy of copper and tin.
- (2) brass is an alloy of copper and nickel
- (3) cast iron is used to manufacture wrought iron.
- (4) german silver is an alloy of zinc, copper and nickel

[JEE (Main-Online) 2020]

- Q.34 For a reaction,
 $4M(s) + nO_2(g) \rightarrow 2M_2O_n(s)$,
the free energy change is plotted as a function of temperature. The temperature below which the oxide is stable could be inferred from the plot as the point at which :
- (1) the slope changes from positive to zero
 - (2) the free energy change shows a change from negative to positive value
 - (3) the slope changes from negative to positive
 - (4) the slope changes from positive to negative

[JEE (Main-Online) 2020]

- Q.35 The element that can be refined by distillation is :
- (1) nickel (2) zinc (3) gallium (4) tin

[JEE (Main-Online) 2020]

- Q.36 The purest form of commercial iron is
- (1) cast iron (2) wrought iron
(3) scrap iron and pig iron (4) pig iron

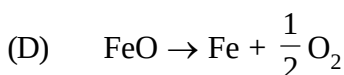
[JEE (Main-Online) 2020]

- Q.37 The refining method used when the metal and the impurities have low and high melting temperatures, respectively, is
- (1) vapour phase refining (2) distillation
(3) liquation (4) zone refining

[JEE (Main-Online) 2020]

- Q.38 Among the reactions (a) - (d), the reaction(s) that does/do not occur in the blast furnace during the extraction of iron is/are

- (A) $CaO + SiO_2 \rightarrow CaSiO_3$
(B) $3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$
(C) $FeO + SiO_2 \rightarrow FeSiO_3$



- (1) (c) and (d) (2) (a) (3) (a) and (d) (4) (d)

[JEE (Main-Online) 2020]

[JEE ADVANCED]

- Q.1 Answer the following questions briefly: **[JEE 1987]**
- (i) What is the actual reducing agent of haematite in blast furnace?
 - (ii) Give the equations for the recovery of lead from galena by air reduction.
 - (iii) Why is sodium chloride added during electrolysis of fused anhydrous magnesium chloride?
 - (iv) Why copper metal is not used for the recovery of metallic silver from complex $[\text{Ag}(\text{CN})_2]^-$ explain.
 - (v) Why is chalcocite roasted and not calcinated during recovery of copper?
- Q.2 In extractive metallurgy of zinc partial fusion of ZnO with coke is called _____ and reduction of the ore to the molten metal is called _____ (smelting, calcining, roasting, sintering). **[JEE 1988]**
- Q.3 Write balanced equation for "the extraction of copper from copper pyrites by self reduction." **[JEE 1990]**
- Q.4 Give briefly the isolation of magnesium from sea water by the Dow's process. Give equations for the steps involved. **[JEE 1993]**
- Q.5 Give reasons for the following: **[JEE 1994]**
"Although aluminium is above hydrogen in the electrochemical series, it is stable in air and water."
- Q.6 Give balanced equations for the following: **[JEE 1998]**
"Extraction of silver from silver glance by cyanide process."
- Q.7 In the commercial electrochemical process for aluminium extraction, the electrolyte used as:
(A) $\text{Al}(\text{OH})_3$ in NaOH solution (B) an aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
(C) a molten mixture of Al_2O_3 and Na_3AlF_6 (D) a molten mixture of $\text{AlO}(\text{OH})$ and $\text{Al}(\text{OH})_3$ **[JEE 1999]**
- Q.8 The chemical process in the production of steel from haematite ore involve: **[JEE 2000 Qualifying]**
(A) reduction (B) oxidation
(C) reduction followed by oxidation (D) oxidation followed by reduction
- Q.9 Electrolytic reduction of alumina to aluminium by Hall-Heravlt process is carried out:-**[JEE 2000]**
(A) In the presence of NaCl
(B) In the presence of fluorite
(C) In the presence of cryolite which forms a melt with lower melting temperature
(D) In the presence of cryolite which forms a melt with higher melting temperature
- Q.10 The chemical composition of "slag" formed during the smelting process in the extraction of copper is: **[JEE 2001 Qualifying]**
(A) $\text{Cu}_2\text{O} + \text{FeS}$ (B) FeSiO_3 (C) CuFeS_2 (D) $\text{Cu}_2\text{S} + \text{FeO}$

- Q.11 Which of the following processes is used in extractive metallurgy of magnesium?
 (A) Fused salt electrolysis (B) Self reduction [JEE 2002 Qualifying]
 (C) Aqueous solution electrolysis (D) Thermite reduction
- Q.12 In the process of extraction of gold,

$$\text{Roasted gold ore} + \text{CN}^- + \text{H}_2\text{O} \xrightarrow{\text{O}_2} [\text{X}] + \text{OH}^-$$

$$[\text{X}] + \text{Zn} \longrightarrow [\text{Y}] + \text{Au}$$
 Identify the complexes [X] and [Y]: [JEE 2003 Qualifying]
 (A) $\text{X} = [\text{Au}(\text{CN})_2]^-$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$ (B) $\text{X} = [\text{Au}(\text{CN})_4]^{3-}$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
 (C) $\text{X} = [\text{Au}(\text{CN})_2]^-$, $\text{Y} = [\text{Zn}(\text{CN})_6]^{4-}$ (D) $\text{X} = [\text{Au}(\text{CN})_4]^-$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
- Q.13 The methods chiefly used for the extraction of lead and tin from their ores are respectively: [JEE 2004]
 (A) self reduction and carbon reduction (B) self reduction and electrolytic reduction
 (C) carbon reduction and self reduction (D) cyanide process and carbon reduction
- Q.14 Which ore contains both iron and copper? [JEE 2004]
 (A) Cuprite (B) Chalcocite (C) Chalcopyrite (D) Malachite
- Q.15 Match the extraction processes listed in column I with metals listed in column II. [JEE 2006]
- | Column I | Column II |
|---|------------|
| (A) Self reduction | (P) Lead |
| (B) Carbon reduction | (Q) Silver |
| (C) Complex formation and displacement by metal | (R) Copper |
| (D) Decomposition of iodide | (S) Boron |
- Q.16 Extraction for zinc from zinc blende is achieved by: [JEE 2007]
 (A) electrolytic reduction
 (B) roasting followed by reduction with carbon
 (C) roasting followed by reduction with another metal
 (D) roasting followed by self-reduction
- Q.17 Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of
 (A) nitrogen (B) oxygen (C) carbon dioxide (D) argon [JEE 2008]
- Q.18 Match the conversions in **Column I** with the type(s) of reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE 2008]
- | Column I | Column II |
|---|----------------------|
| (A) $\text{PbS} \rightarrow \text{PbO}$ | (P) Roasting |
| (B) $\text{CaCO}_3 \rightarrow \text{CaO}$ | (Q) Calcination |
| (C) $\text{ZnS} \rightarrow \text{Zn}$ | (R) Carbon reduction |
| (D) $\text{Cu}_2\text{S} \rightarrow \text{Cu}$ | (S) Self reduction |

Paragraph for Questions 19 to 21

Copper is the most noble of the first row transition metals and occurs in small deposits in several countries. Ores of copper include chalcantite ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), atacamite ($\text{Cu}_2\text{Cl}(\text{OH})_3$), cuprite (Cu_2O), copper glance (Cu_2S) and malachite ($\text{Cu}_2(\text{OH})_2\text{CO}_3$). However, 80% of the world copper production comes from the ore chalcopyrite (CuFeS_2). The extraction of copper from chalcopyrite involved partial roasting, removal of iron and self-reduction.

- Q.19 Partial roasting of chalcopyrite produces [JEE 2010]
 (A) Cu_2S and FeO (B) Cu_2O and FeO
 (C) CuS and Fe_2O_3 (D) Cu_2O and Fe_2O_3
- Q.20 Iron is removed from chalcopyrite as [JEE 2010]
 (A) FeO (B) FeS (C) Fe_2O_3 (D) FeSiO_3
- Q.21 In self-reduction, the reducing species is [JEE 2010]
 (A) S (B) O^{2-} (C) S^{2-} (D) SO_2
- Q.22 Extraction of metal from the ore **cassiterite** involves [JEE 2011]
 (A) carbon reduction of an oxide ore (B) self-reduction of a sulphide ore
 (C) removal of copper impurity (D) removal of iron impurity
- Q.23 Oxidation states of the metal in the minerals haematite and magnetite, respectively, are [JEE 2011]
 (A) II, III in haematite and III in magnetite (B) II, III in haematite and II in magnetite
 (C) II in haematite and II, III in magnetite (D) III in haematite and II, III in magnetite
- Q.24 In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are: [JEE 2012]
 (A) O_2 and CO respectively (B) O_2 and Zn dust respectively
 (C) HNO_3 and Zn dust respectively (D) HNO_3 and CO respectively
- Q.25 Sulfide ores are common for the metals : [JEE (Advanced) 2013]
 (A) Ag , Cu and Pb (B) Ag , Cu and Sn
 (C) Ag , Mg and Pb (D) Al , Cu and Pb
- Q.26 The carbon-based reduction method is **NOT** used for the extraction of [JEE (Advanced) 2013]
 (A) tin from SnO_2 (B) iron from Fe_2O_3
 (C) aluminium from Al_2O_3 (D) magnesium from $\text{MgCO}_3 \cdot \text{CaCO}_3$
- Q.27 Upon heating with Cu_2S , the reagent(s) that give copper metal is/are [JEE (Advanced) 2014]
 (A) CuFeS_2 (B) CuO
 (C) Cu_2O (D) CuSO_4
- Q.28 Copper is purified by electrolytic refining of blister copper. The correct statement(s) about this process is(are): [JEE (Advanced) 2015]
 (A) Impure Cu strip is used as cathode
 (B) Acidified aqueous CuSO_4 is used as electrolyte
 (C) Pure Cu deposits at cathode
 (D) Impurities settle as anode - mud

Q.29 Match the anionic species given in **Column I** that are present in the ore(s) given in **Column II**

Column I		Column II	
(A)	Carbonate	(P)	Siderite
(B)	Sulphide	(Q)	Malachite
(C)	Hydroxide	(R)	Bauxite
(D)	Oxide	(S)	Calamine
		(T)	Argentite

[JEE (Advanced) 2015]

Q.30 Extraction of copper from copper pyrite (CuFeS_2) involves

[JEE (Advanced) 2016]

- (A) crushing followed by concentration of the ore by froth-flotation
- (B) removal of iron as slag
- (C) self-reduction step to produce 'blister copper' following evolution of SO_2
- (D) refining of 'blister copper' by carbon reduction

Q.31 Galena (an ore) is partially oxidized by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnace such that the contents undergo self-reduction. The weight (in kg) of Pb produced per kg of O_2 consumed is ____.

(Atomic weights in g mol^{-1} : O = 16, S = 32, Pb = 207)

[JEE (Advanced) 2018]

Q.32 The cyanide process of gold extraction involves leaching out gold from its ore with CN^- in the presence of **Q** in water to form **R**. Subsequently, **R** is treated with **T** to obtain Au and **Z**. Choose the correct option(s).

[JEE (Advanced) 2019]

- (1) Z is $[\text{Zn}(\text{CN})_4]^{2-}$
- (2) R is $[\text{Au}(\text{CN})_4]^-$
- (3) Q is O_2
- (4) T is Zn

Q.33 Calamine, malachite, magnetite and cryolite, respectively are

[JEE (Advanced) 2019]

- (1) ZnCO_3 , CuCO_3 , Fe_2O_3 , Na_3AlF_6
- (2) ZnCO_3 , $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$, Fe_3O_4 , Na_3AlF_6
- (3) ZnSO_4 , $\text{Cu}(\text{OH})_2$, Fe_3O_4 , Na_3AlF_6
- (4) ZnSO_4 , CuCO_3 , Fe_2O_3 , AlF_3

Q.34 Which among the following statement(s) is(are) true for the extraction of aluminium from bauxite?

- (A) Hydrated Al_2O_3 precipitates, when CO_2 is bubbled through a solution of sodium aluminate.
- (B) Addition of Na_3AlF_6 lowers the melting point of alumina.
- (C) CO_2 is evolved at the anode during electrolysis.
- (D) The cathode is a steel vessel with a lining of carbon.

[JEE (Advanced) 2020]

[JEE MAIN]

- Q.1 Which of the following acts as rocket propellants [AIEEE 2003]
(1) Liq. H_2 + Liq. O_2 (2) Liq. N_2 + Liq. O_2
(3) Liq. H_2 + Liq. N_2 (4) Liq. O_2 + Liq. Argon
- Q.2 When rain is accompanied by a thunderstorm, the collected rain water will have pH? [AIEEE 2003]
(1) slightly lower than that of rain water without thunderstorm
(2) slightly higher than that of rain water without thunderstorm
(3) uninfluenced by occurrence of thunderstorm
(4) which depends on amount of dust in air
- Q.3 The smog is essentially caused by the presence of [AIEEE 2004]
(1) O_3 and N_2 (2) O_2 and N_2
(3) Oxides of sulphur and N_2 (4) O_2 and O_3
- Q.4 Ozone in stratosphere is depleted by [AIEEE 2004]
(1) CF_2Cl_2 (2) C_7F_{16} (3) $C_6H_6Cl_6$ (4) C_6F_6
- Q.5 Which of the following is responsible for depletion of ozone layer in upper strata of the atmosphere? [AIEEE 2004]
(1) Polyhalogens (2) Ferrocene (3) Fullerenes (4) Freons
- Q.6 Identify the wrong statements in the following [AIEEE 2008]
(1) Chlorofluorocarbons are responsible for ozone layer depletion
(2) Green house effect is responsible for global warming
(3) Ozone layer does not permit infrared radiation from the sun to reach the earth
(4) Acid rain is mostly because of oxides of nitrogen and sulphur
- Q.7 Identify the incorrect statement from the following [AIEEE 2011]
(1) Oxides of nitrogen in the atmosphere can cause the depletion of ozone layer
(2) Ozone absorbs the intense ultraviolet radiation of the sun
(3) Depletion of ozone layer is because of its chemical reactions with chlorofluoro alkanes
(4) Ozone absorbs infra red radiations
- Q.8 What is DDT among the following? [AIEEE 2012]
(1) Green house gas (2) A fertilizer
(3) Biodegradable pollutant (4) Non-biodegradable pollutant

- Q.9 Which of the following statements about the depletion of ozone layer is correct ? :
- (1) The problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone by $\text{Cl}^\bullet$ and $\text{ClO}^\bullet$ radicals.
 - (2) Freons, chlorofluorocarbons, are inert chemically, they do not react with ozone in stratosphere.
 - (3) The problem of ozone depletion is less serious at poles because NO_2 solidifies and is not available for consuming $\text{ClO}^\bullet$ radicals.
 - (4) Oxides of nitrogen also do not react with ozone in stratosphere.

[JEE (Main-Online) 2014]

- Q.10 Global warming is due to increase of :
- (1) methane and O_3 in atmosphere
 - (2) methane and nitrous oxide in atmosphere
 - (3) methane and CO in atmosphere
 - (4) methane and CO_2 in atmosphere

[JEE (Main-Online) 2014]

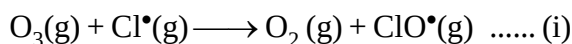
- Q.11 Photochemical smog consists of excessive amount of X, in addition to aldehydes, ketones, peroxy acetyl nitrile (PAN), and so forth. X is :
- (1) CO
 - (2) CO_2
 - (3) O_3
 - (4) CH_4

[JEE (Main-Online) 2015]

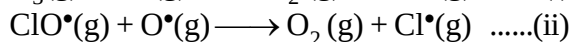
- Q.12 Addition of phosphate fertilisers to water bodies causes :
- (1) enhanced growth of algae
 - (2) increase in amount of dissolved oxygen in water
 - (3) increase in fish population
 - (4) deposition of calcium phosphate

[JEE (Main-Online) 2015]

- Q.13 The reaction of ozone with oxygen atoms in the presence of chlorine atoms can occur by a two step process shown below:



$$k_{\text{i}} = 5.2 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$$



$$k_{\text{ii}} = 2.6 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1}$$

The closest rate constant for the overall reaction $\text{O}_3(\text{g}) + \text{O}^\bullet(\text{g}) \longrightarrow 2\text{O}_2(\text{g})$ is:

- (1) $5.2 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$
- (2) $2.6 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1}$
- (3) $3.1 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1}$
- (4) $1.4 \times 10^{20} \text{ L mol}^{-1} \text{ s}^{-1}$

[JEE (Main-Online) 2016]

- Q.14 BOD stands for :
- (1) Biological Oxygen Demand
 - (2) Bacterial Oxidation Demand
 - (3) Biochemical Oxygen Demand
 - (4) Biochemical Oxidation Demand

[JEE (Main-Online) 2016]

- Q.15 Identify the pollutant gases largely responsible for the discoloured and lustreless nature of marble of the Taj Mahal.

- (1) O_3 and CO_2
- (2) CO_2 and NO_2
- (3) SO_2 and NO_2
- (4) SO_2 and O_3

[JEE (Main-Online) 2017]

- Q.16 Which of the following is a set of green house gases ?
(1) CH_4 , O_3 , N_2 , SO_2 (2) O_3 , N_2 , CO_2 , NO_2
(3) O_3 , NO_2 , SO_2 , Cl_2 (4) CO_2 , CH_4 , N_2O , O_3
[JEE (Main-Online) 2017]
- Q.17 The recommended concentration of fluoride ion in drinking water is up to 1 ppm as fluoride ion is required to make teeth enamel harder by converting $[\text{3Ca}_3(\text{PO}_4)_2 \cdot \text{Ca}(\text{OH})_2]$ to
(1) $[\text{3Ca}_3(\text{PO}_4)_2 \cdot \text{CaF}_2]$ (2) $[\text{3}\{\text{Ca}(\text{OH})_2\} \cdot \text{CaF}_2]$
(3) $[\text{CaF}_2]$ (4) $[\text{3}(\text{CaF}_2) \cdot \text{Ca}(\text{OH})_2]$
[JEE (Main-Offline) 2018]
- Q.18 Biochemical Oxygen Demand (BOD) value can be a measure of water pollution caused by the organic matter. Which of the following statements is **correct**?
(1) Anaerobic bacteria increase the BOD value
(2) Aerobic bacteria decrease the BOD value
(3) Polluted water has BOD value higher than 10 ppm.
(4) Clean water has BOD value higher than 10 ppm.
[JEE (Main-Online) 2018]
- Q.19 The statement that is not true about ozone is :
(1) in the stratosphere, it forms a protective shield against UV radiation.
(2) it is at oxig gas and its reaction with NO gives NO_2 .
(3) in the atmosphere, it is depleted by CFCs.
(4) in the stratophere, CFCs release chlorine free radicals ($\text{Cl}^\cdot$) which reacts with O_3 to give chlorine dioxide radicals.
[JEE (Main-Online) 2020]
- Q.20 Thermal power plants can lead to :
(1) Ozone layer depletion (2) Eutrophication
(3) Acid rain (4) Blue baby syndrome
[JEE (Main-Online) 2020]
- Q.21 The incorrect statement(s) among (a) - (d) regarding acid rain is (are) :
(a) It can corrode water pipes.
(b) It can damage structures made up of stone.
(c) It cannot cause respiratory ailments in animals.
(d) It is not harmful for trees
(1) (c) and (d) (2) (a), (b) and (d) (3) (c) only (4) (a), (c) and (d)
[JEE (Main-Online) 2020]
- Q.22 The processes of calcination and roasting in metallurgical industries, respectively, can lead to :-
(1) Global warming and acid rain
(2) Photochemical smog and ozone layer depletion
(3) Global warming and photochemical smog
(4) Photochemical smog and global warming
[JEE (Main-Online) 2020]

Q.23 The condition that indicates a polluted environment is

- (1) BOD value of 5 ppm (2) eutrophication
(3) 0.03% of CO_2 in the atmosphere (4) pH of rain water to be 5.6

[JEE (Main-Online) 2020]

Q.24 The presence of soluble fluoride ion upto 1 ppm concentration in drinking water, is :

- (1) harmful to bones (2) harmful for teeth (3) safe for teeth (4) harmful to skin

[JEE (Main-Online) 2020]

pmssir

ANSWER KEY

CLASSIFICATION OF ELEMENTS & PERIODIC PROPERTIES

[JEE MAIN]

Q.1	1	Q.2	2	Q.3	2	Q.4	2	Q.5	1
Q.6	1	Q.7	3	Q.8	1	Q.9	4	Q.10	1
Q.11	2	Q.12	1	Q.13	3	Q.14	4	Q.15	3
Q.16	3	Q.17	2	Q.18	4	Q.19	1	Q.20	1
Q.21	1	Q.22	2	Q.23	4	Q.24	3	Q.25	3
Q.26	3	Q.27	4	Q.28	4	Q.29	1	Q.30	3
Q.31	1	Q.32	1	Q.33	1	Q.34	1	Q.35	3
Q.36	4	Q.37	3	Q.38	3	Q.39	1	Q.40	2
Q.41	4	Q.42	3	Q.43	1	Q.44	3	Q.45	1
Q.46	1	Q.47	2	Q.48	4	Q.49	3	Q.50	2
Q.51	4	Q.52	3	Q.53	4	Q.54	2	Q.55	2
Q.56	4	Q.57	2	Q.58	4	Q.59	3	Q.60	101.00
Q.61	2	Q.62	2	Q.63	2	Q.64	3	Q.65	3
Q.66	19.00								

[JEE ADVANCED]

Q.1	A	Q.2	C	Q.3	C	Q.4	A	Q.5	D
Q.6	A	Q.7	D	Q.8	A	Q.9	B	Q.10	A
Q.11	$\text{Al}^{+3} < \text{Mg}^{2+} < \text{Li}^{+} < \text{K}^{+}$			Q.12	ABC	Q.13	C	Q.14	C
Q.15	B	Q.16	A	Q.17	True	Q.18	B	Q.19	B
Q.20	A	Q.21	2	Q.22	B	Q.23	AC		

BASIC CHEMICAL BONDING

[JEE MAIN]

Q.1	2	Q.2	3	Q.3	2	Q.4	2	Q.5	1
Q.6	1	Q.7	4	Q.8	4	Q.9	3	Q.10	3
Q.11	4	Q.12	2	Q.13	4	Q.14	1	Q.15	1
Q.16	4	Q.17	2	Q.18	3	Q.19	1	Q.20	3
Q.21	2	Q.22	2	Q.23	2	Q.24	1	Q.25	3
Q.26	1	Q.27	3	Q.28	4	Q.29	4	Q.30	1
Q.31	3	Q.32	2	Q.33	1	Q.34	1	Q.35	4
Q.36	2	Q.37	1	Q.38	1	Q.39	4	Q.40	4
Q.41	2	Q.42	3	Q.43	3	Q.44	4	Q.45	3
Q.46	2	Q.47	4	Q.48	2	Q.49	1	Q.50	4
Q.51	1	Q.52	1	Q.53	1	Q.54	3	Q.55	2
Q.56	2								

[JEE ADVANCED]

Q.1	B	Q.2	D	Q.3	A	Q.4	B	Q.5	A
Q.6	B	Q.7	A	Q.8	A	Q.9	A	Q.10	D
Q.11	D	Q.12	C	Q.13	D	Q.14	0	Q.15	B
Q.16	D	Q.17	4	Q.18	6	Q.19	3,4	Q.20	6

ADVANCE CHEMICAL BONDING**[JEE MAIN]**

Q.1	2	Q.2	2	Q.3	3	Q.4	3	Q.5	2
Q.6	2	Q.7	3	Q.8	4	Q.9	1	Q.10	1
Q.11	1	Q.12	4	Q.13	2	Q.14	1	Q.15	1
Q.16	4	Q.17	1	Q.18	1,4	Q.19	2	Q.20	4
Q.21	3	Q.22	2	Q.23	4	Q.24	4	Q.25	2
Q.26	2	Q.27	2	Q.28	4	Q.29	1	Q.30	4
Q.31	2	Q.32	1	Q.33	2	Q.34	4	Q.35	3
Q.36	4	Q.37	1	Q.38	4	Q.39	4	Q.40	4
Q.41	2,4	Q.42	2	Q.43	4	Q.44	1	Q.45	3
Q.46	3	Q.47	4	Q.48	1	Q.49	2	Q.50	2
Q.51	4	Q.52	1	Q.53	2	Q.54	4	Q.55	3,4
Q.56	4.00	Q.57	1	Q.58	2	Q.59	2	Q.60	4
Q.61	1	Q.62	2	Q.63	4	Q.64	3,4	Q.65	4.00
Q.66	3	Q.67	2	Q.68	4	Q.69	4	Q.70	2
Q.71	1	Q.72	3.00	Q.73	3	Q.74	3	Q.75	4
Q.76	2								

[JEE ADVANCED]

Q.1	B	Q.2	A	Q.3	A	Q.4	C	Q.5	B
Q.7	D	Q.8	A	Q.9	D	Q.10	4		
Q.11	(A) P,Q,R, T (B) Q, R, S, T (C) P, Q, R (D) P,Q,R, S					Q.12	ABC	Q.13	A
Q.14	6	Q.15	5	Q.16	BD	Q.17	ABD	Q.18	C
Q.19	C	Q.20	BC	Q.21	8	Q.22	AC	Q.23	BC
Q.24	Bonus	Q.25	D	Q.26	ABC	Q.27	6	Q.28	4.00

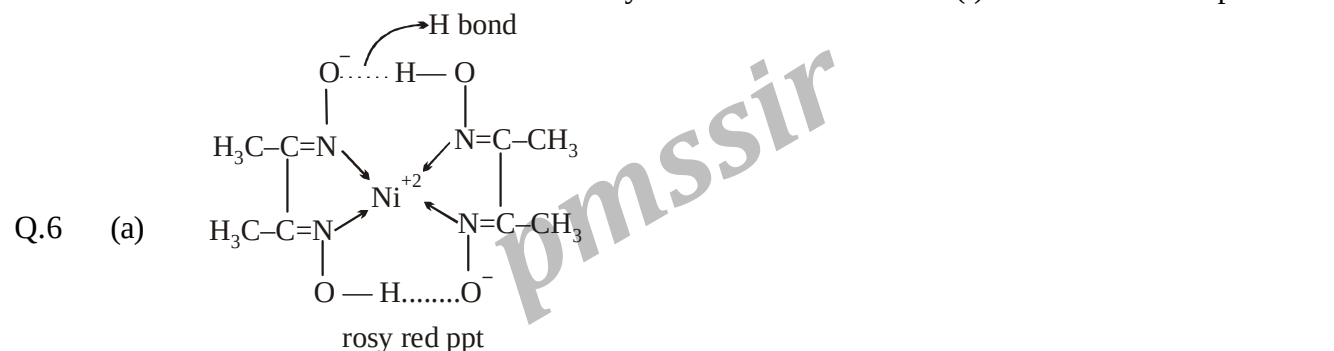
COORDINATION COMPOUND**[JEE MAIN]**

Q.1	2	Q.2	2	Q.3	4	Q.4	1	Q.5	3
Q.6	2	Q.7	1	Q.8	3	Q.9	4	Q.10	3
Q.11	3	Q.12	1	Q.13	2	Q.14	4	Q.15	4
Q.16	1	Q.17	3	Q.18	4	Q.19	4	Q.20	1
Q.21	3	Q.22	1	Q.23	1	Q.24	2	Q.25	3
Q.26	3	Q.27	4	Q.28	2	Q.29	2	Q.30	1
Q.31	1	Q.32	3	Q.33	3	Q.34	1	Q.35	1
Q.36	3	Q.37	3	Q.38	3	Q.39	4	Q.40	4
Q.41	4	Q.42	1	Q.43	2	Q.44	4	Q.45	4
Q.46	4	Q.47	1	Q.48	3	Q.49	3	Q.50	3
Q.51	3	Q.52	3	Q.53	1	Q.54	4	Q.55	4
Q.56	1	Q.57	4	Q.58	1	Q.59	2	Q.60	2

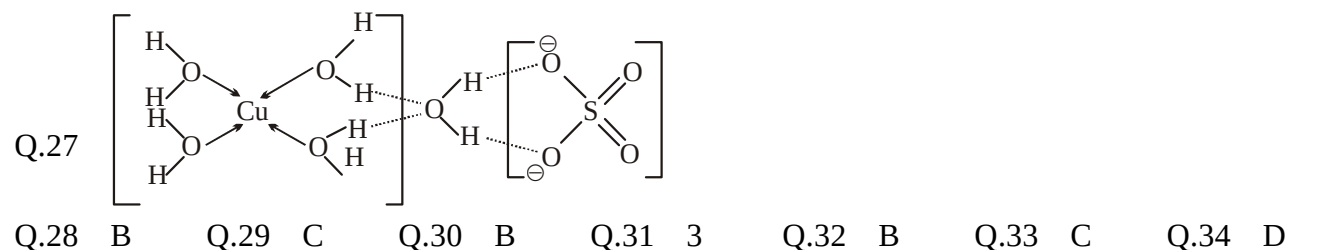
PREVIOUS YEARS QUESTIONS			ENVIRONMENTAL CHEMISTRY		
Q.61	1	Q.62	3	Q.63	1
Q.66	3	Q.67	1	Q.68	4
Q.71	1	Q.72	4	Q.73	4
Q.76	3	Q.77	3	Q.78	2
Q.81	3	Q.82	4	Q.83	3
Q.86	3	Q.87	4	Q.88	3
Q.91	6.00	Q.92	4	Q.93	0000
Q.96	4	Q.97	3	Q.98	3
Q.101	1	Q.102	2	Q.103	20.00
Q.106	2	Q.107	4	Q.104	1
				Q.65	4
				Q.70	2
				Q.75	1
				Q.80	2
				Q.85	1
				Q.90	3
				Q.95	3
				Q.100	3
				Q.105	3

[JEE ADVANCED]

- Q.1 [Co(NH₃)₆]³⁺ d²sp³ Octahedral
 [Ni(CN)₄]²⁻ dsp² Square planar
 [Ni(CO)₄] sp³ Tetrahedral
- Q.2 A Q.3 B
- Q.4 Tetrahedral, $\sqrt{8}$ B.M. Square planar, $\mu = 0$ B.M.
- Q.5 K₂[Cr(NO)(NH₃)(CN)₄], $\mu_{\text{eff}} = 1.73$ BM. Chromium is +1 oxidation state and hybridization is d²sp³ and IUPAC name is Potassium aminetetrahydridonitrosylchromate(I) and Octahedral shape.



- (b) dsp² hybridization square planar structure and oxidation state of Ni = 2+
- (c) Diamagnetic
- Q.7 D Q.8 B Q.9 D Q.10 B Q.11 B Q.12 A
- Q.13 (a) $\text{Fe}^{3+} + \xrightarrow[\text{(Excess)}]{\text{SCN}^-} [\text{Fe}(\text{H}_2\text{O})_5\text{SCN}]^{2+} \xrightarrow{\text{F}^- (\text{Excess})} \text{colouless (B)} [\text{FeF}_6]^{3-}$
 Hexafluoridoferrate (III) ion
- (b) The magnetic moment value of B is 5.93 B.M.
- Q.14 A Q.15 C Q.16 A Q.17 A Q.18 D
- Q.19 (A) P, Q, S; (B) P, R, S; (C) Q, S; (D) Q, S Q.20 C Q.21 C Q.22 B
- Q.23 B Q.24 A Q.25 A Q.26 CD



PREVIOUS YEARS QUESTIONS								ENVIRONMENTAL CHEMISTRY					
Q.35	A	Q.36	C	Q.37	B	Q.38	BD	Q.39	8	Q.40	4	Q.41	B
Q.42	4	Q.43	3	Q.44	6	Q.45	B	Q.46	5	Q.47	A	Q.48	BC
Q.49	1	Q.50	2992	Q.51	ABD	Q.52	C	Q.53	6.00	Q.54	AC	Q.55	ABC

TYPES OF REACTION

[JEE MAIN]

Q.1	2	Q.2	2	Q.3	3	Q.4	4	Q.5	3
Q.6	2	Q.7	4	Q.8	3	Q.9	4	Q.10	2
Q.11	2	Q.12	4	Q.13	4	Q.14	1	Q.15	1
Q.16	3	Q.17	4	Q.18	4	Q.19	2		
Q.20	1.66 to 1.67	Q.21	3	Q.22	4	Q.23	2	Q.24	2
Q.25	18.00	Q.26	2	Q.27	2				

[JEE ADVANCED]

Q.1	B	Q.2	ABD	Q.3	D	Q.4	C	Q.5	B
Q.6	C	Q.7	A	Q.8	A	Q.9	A	Q.10	A
Q.11	A	Q.12	B	Q.13	A	Q.14	B	Q.15	A
Q.16	D	Q.17	B	Q.18	A	Q.19	B	Q.20	AB
Q.21	AB	Q.22	B	Q.23	(A) p, s ; (B) q, s (C) r, t ; (D) q, t				
Q.24	BC	Q.25	6	Q.26	A	Q.27	D	Q.28	4
Q.29	B	Q.30	A	Q.31	C	Q.32	ACD	Q.33	D
Q.34	D	Q.35	A	Q.36	7	Q.37	A	Q.38	A
Q.39	C	Q.40	6	Q.41	AB	Q.42	CD	Q.43	AC
Q.44	A	Q.45	A	Q.46	A	Q.47	D	Q.48	BC
Q.49	BC	Q.50	2,4	Q.51	19.00	Q.52	1,2,4	Q.53	1,2,4
Q.54	1,2,4	Q.55	A	Q.56	4	Q.57	B	Q.58	AB

HYDROGEN

[JEE MAIN]

Q.1	3	Q.2	2	Q.3	3	Q.4	1	Q.5	4
Q.6	4	Q.7	2	Q.8	2	Q.9	4	Q.10	4
Q.11	2	Q.12	4	Q.13	4	Q.14	1	Q.15	4
Q.16	3	Q.17	2	Q.18	1	Q.19	4	Q.20	4
Q.21	1	Q.22	3	Q.23	1	Q.24	2	Q.25	1
Q.26	2	Q.27	3	Q.28	1				

[JEE ADVANCED]

Q.1	B	Q.2	A	Q.3	C	Q.4	BC
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s-BLOCK ELEMENTS

[JEE MAIN]

Q.1	4	Q.2	1	Q.3	3	Q.4	4	Q.5	4
Q.6	2	Q.7	1,2	Q.8	1	Q.9	2	Q.10	2
Q.11	1	Q.12	2	Q.13	4	Q.14	1	Q.15	4
Q.16	1	Q.17	4	Q.18	4	Q.19	1	Q.20	3

PREVIOUS YEARS QUESTIONS				ENVIRONMENTAL CHEMISTRY					
Q.21	3	Q.22	3	Q.23	2	Q.24	4	Q.25	2
Q.26	3	Q.27	3	Q.28	1	Q.29	3	Q.30	4
Q.31	2	Q.32	3	Q.33	3	Q.34	4	Q.35	2
Q.36	1	Q.37	1	Q.38	3				

[JEE ADVANCED]

Q.1	B	Q.2	3	Q.3	ABC
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p-BLOCK ELEMENTS

[JEE MAIN]

Q.1	1	Q.2	4	Q.3	2	Q.4	3	Q.5	2
Q.6	4	Q.7	2	Q.8	4	Q.9	3	Q.10	1
Q.11	1	Q.12	4	Q.13	3	Q.14	3	Q.15	3
Q.16	2	Q.17	1	Q.18	1	Q.19	3	Q.20	1
Q.21	2	Q.22	2	Q.23	4	Q.24	2	Q.25	1
Q.26	4	Q.27	1	Q.28	4	Q.29	2	Q.30	3
Q.31	2	Q.32	3	Q.33	2	Q.34	1	Q.35	2
Q.36	1	Q.37	2	Q.38	4	Q.39	4	Q.40	3
Q.41	4	Q.42	4	Q.43	4	Q.44	1	Q.45	3
Q.46	1	Q.47	1	Q.48	2	Q.49	2	Q.50	2
Q.51	1	Q.52	2	Q.53	4	Q.54	4	Q.55	2
Q.56	4	Q.57	3	Q.58	1	Q.59	4	Q.60	4

[JEE ADVANCED]

Q.1	C	Q.2	C	Q.3	A	Q.4	D	Q.5	C
Q.6	A	Q.7	C	Q.8	C	Q.9	A		
Q.10	(A) Q,S (B) R (C) P (D) Q,R			Q.11	A	Q.12	C	Q.13	A
Q.14	C	Q.15	B	Q.16	C	Q.17	ABC	Q.18	3
Q.19	B	Q.20	BD	Q.21	B	Q.22	ACD	Q.23	BD
Q.24	ABD	Q.25	ABD	Q.26	1	Q.27	288 (288.00 – 288.30)		
Q.28	2,4	Q.29	ABD						

d&f-BLOCK ELEMENTS

[JEE MAIN]

Q.1	1	Q.2	1	Q.3	4	Q.4	3	Q.5	1
Q.6	3	Q.7	2	Q.8	4	Q.9	4	Q.10	1
Q.11	2	Q.12	4	Q.13	2	Q.14	3	Q.15	2,4
Q.16	2	Q.17	4	Q.18	2	Q.19	3	Q.20	1
Q.21	2	Q.22	3	Q.23	1	Q.24	2	Q.25	4
Q.26	4	Q.27	4	Q.28	1	Q.29	4	Q.30	3
Q.31	4	Q.32	2	Q.33	4	Q.34	3	Q.35	1
Q.36	3	Q.37	4	Q.38	2	Q.39	2	Q.40	2
Q.41	1	Q.42	4	Q.43	4	Q.44	12.00	Q.45	1

[JEE ADVANCED]

Q.1	D	Q.2	B	Q.3	B	Q.4	D	Q.5	A
Q.6	B	Q.7	ABC	Q.8	6				

SALT ANALYSIS**[JEE MAIN]**

Q.1	1	Q.2	4	Q.3	4	Q.4	4	Q.5	2
Q.6	4	Q.7	4	Q.8	2	Q.9	3	Q.10	2
Q.11	2	Q.12	4	Q.13	3	Q.14	2	Q.15	2
Q.16	3	Q.17	4	Q.18	2	Q.19	4		

[JEE ADVANCED]

Q.1	C	Q.2	B	Q.3	ABC	Q.4	B	Q.5	D
Q.6	A	Q.7	(A) P,S (B) R (C) P,Q (D) P	Q.8	AB	Q.9	D	Q.10	D
Q.10	C	Q.11	B	Q.12	ACD	Q.13	6 or 7	Q.14	B
Q.15	D	Q.16	ABC	Q.17	BD	Q.18	BC	Q.19	1
Q.20	2992	Q.21	ABD	Q.22	C	Q.23	1		

METALLURGY**[JEE MAIN]**

Q.1	2	Q.2	3	Q.3	4	Q.4	2	Q.5	3
Q.6	2	Q.7	4	Q.8	1	Q.9	1	Q.10	3
Q.11	1	Q.12	3	Q.13	2	Q.14	1	Q.15	2
Q.16	1	Q.17	4	Q.18	4	Q.19	1	Q.20	4
Q.21	1	Q.22	2	Q.23	4	Q.24	3	Q.25	3
Q.26	2	Q.27	3	Q.28	3	Q.29	2	Q.30	4
Q.31	3	Q.32	4	Q.33	2	Q.34	2	Q.35	2
Q.36	2	Q.37	3	Q.38	1				

[JEE ADVANCED]

Q.2	sintering ,smelting	Q.7	C	Q.8	C	Q.9	C		
Q.10	B	Q.11	A	Q.12	A	Q.13	A	Q.14	C
Q.15	(A) P, R ; (B) P ; (C) Q ;(D) S	Q.16	B	Q.17	B				
Q.18	(A) P; (B) Q; (C) P,R (D) P,S	Q.19	A	Q.20	D	Q.21	C		
Q.22	ACD	Q.23	D	Q.24	B	Q.25	A	Q.26	CD
Q.27	BCD	Q.28	BCD	Q.29	(A) PQS (B) T (C) QR (D) R	Q.30	ABC		
Q.31	6.47	Q.32	1,3,4	Q.33	2	Q.34	ABCD		

ENVIRONMENTAL CHEMISTRY**[JEE MAIN]**

Q.1	1	Q.2	1	Q.3	3	Q.4	1	Q.5	4
Q.6	3	Q.7	4	Q.8	4	Q.9	1	Q.10	4
Q.11	3	Q.12	1	Q.13	1	Q.14	3	Q.15	3
Q.16	4	Q.17	1	Q.18	3	Q.19	4	Q.20	3
Q.21	2	Q.22	1	Q.23	2	Q.24	3		